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(54) **OPTICAL IMAGING SYSTEM**

(30) **Foreign Application Priority Data**

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(52) **U.S. Cl.**

CPC **G02B 9/64** (2013.01); **G02B 3/02** (2013.01); **G02B 13/0045** (2013.01)

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Suwon-si (KR)

(21) Appl. No.: **19/220,436**

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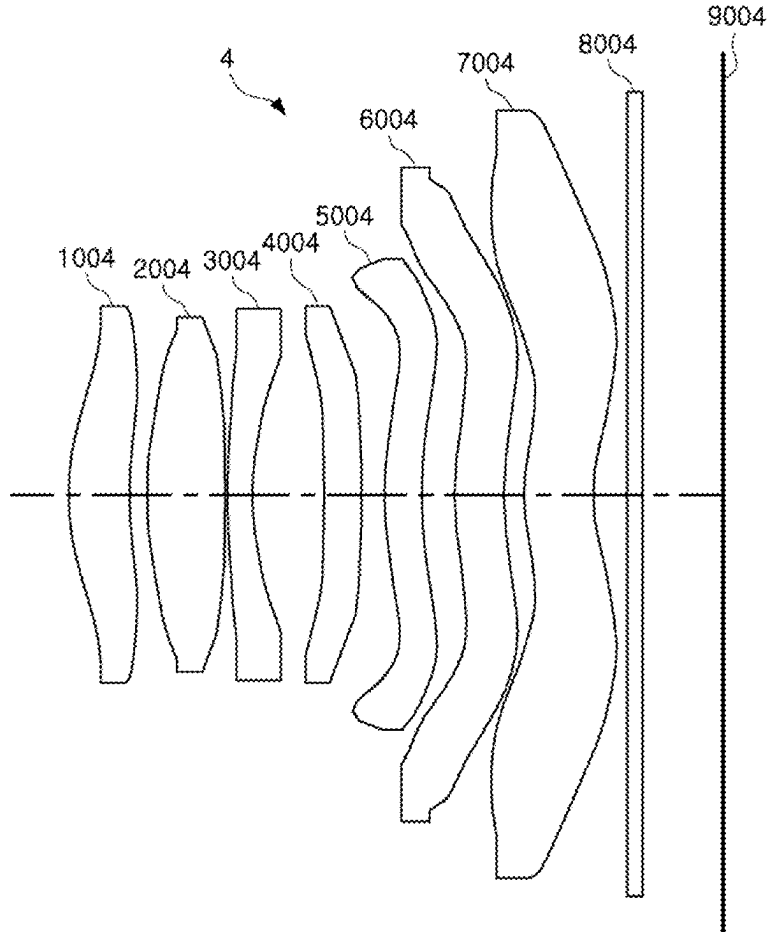
Related U.S. Application Data

(63) Continuation of application No. 18/670,871, filed on May 22, 2024, now Pat. No. 12,353,059, which is a continuation of application No. 18/330,645, filed on Jun. 7, 2023, now Pat. No. 12,032,226, which is a continuation of application No. 17/088,665, filed on Nov. 4, 2020, now Pat. No. 11,714,263, which is a continuation of application No. 16/424,842, filed on May 29, 2019, now Pat. No. 11,016,271.

(57)

ABSTRACT

An optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $0.5 < L12345TR_{avg}/L7TR < 0.9$, where $L12345TR_{avg}$ is an average value of overall outer diameters of the first to fifth lenses, $L7TR$ is an overall outer diameter of the seventh lens, and $L12345TR_{avg}$ and $L7TR$ are expressed in a same unit of measurement.



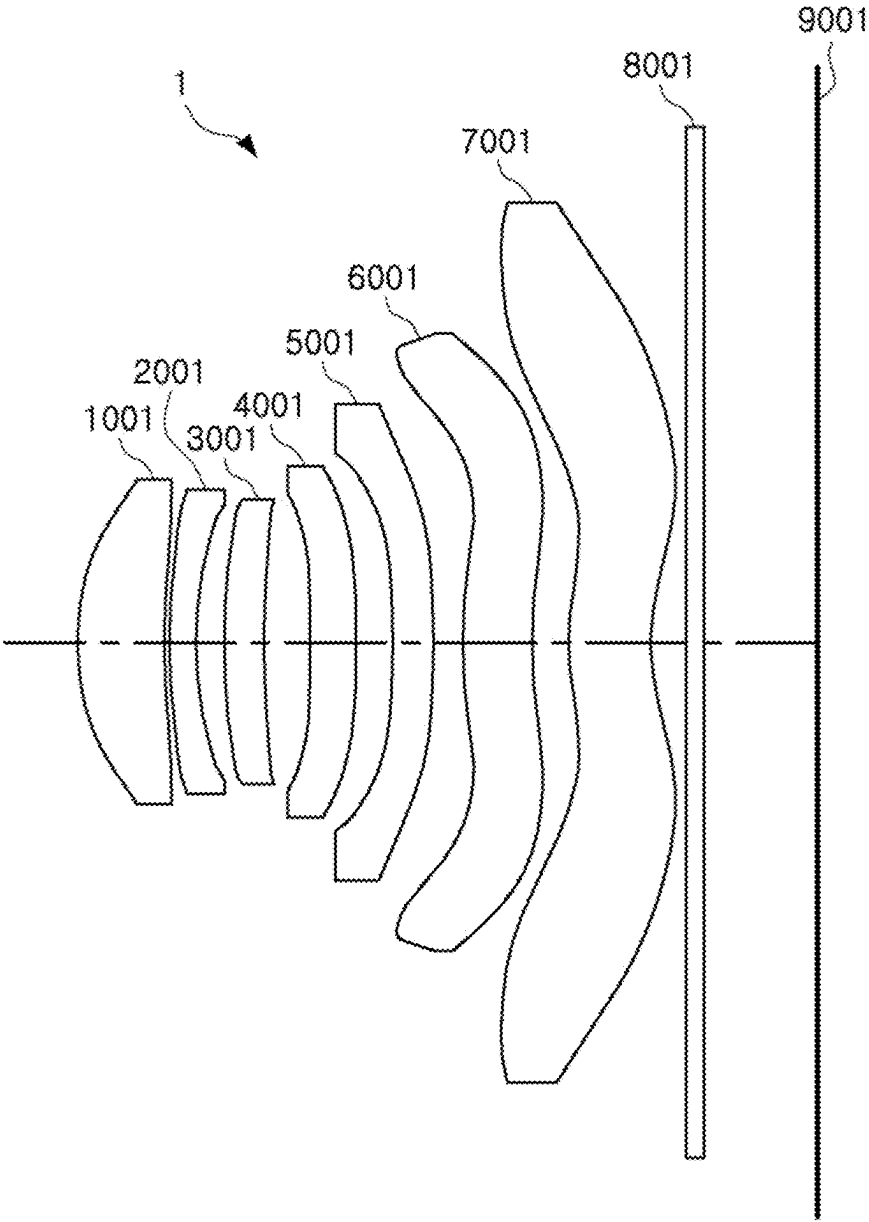


FIG. 1

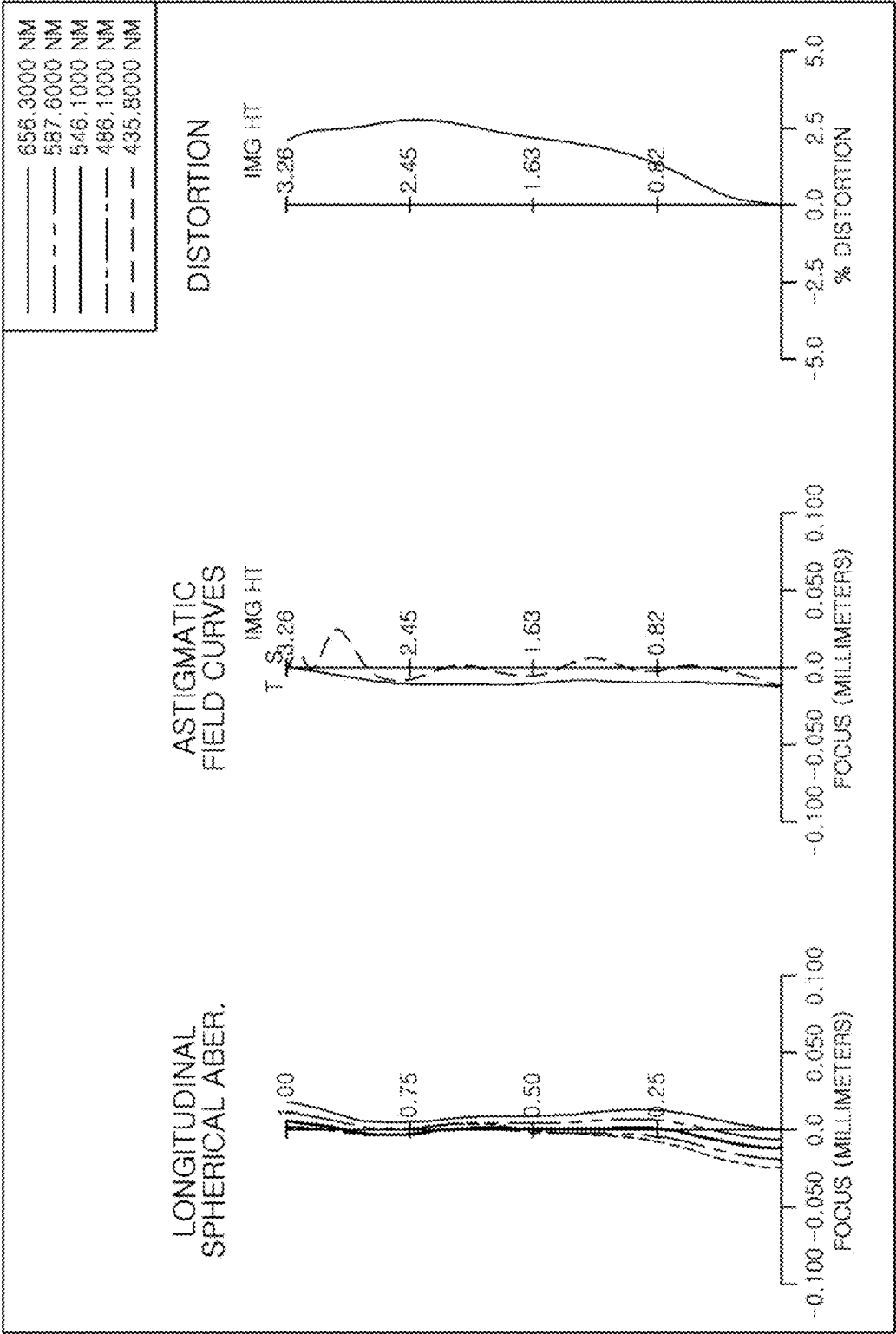


FIG. 2

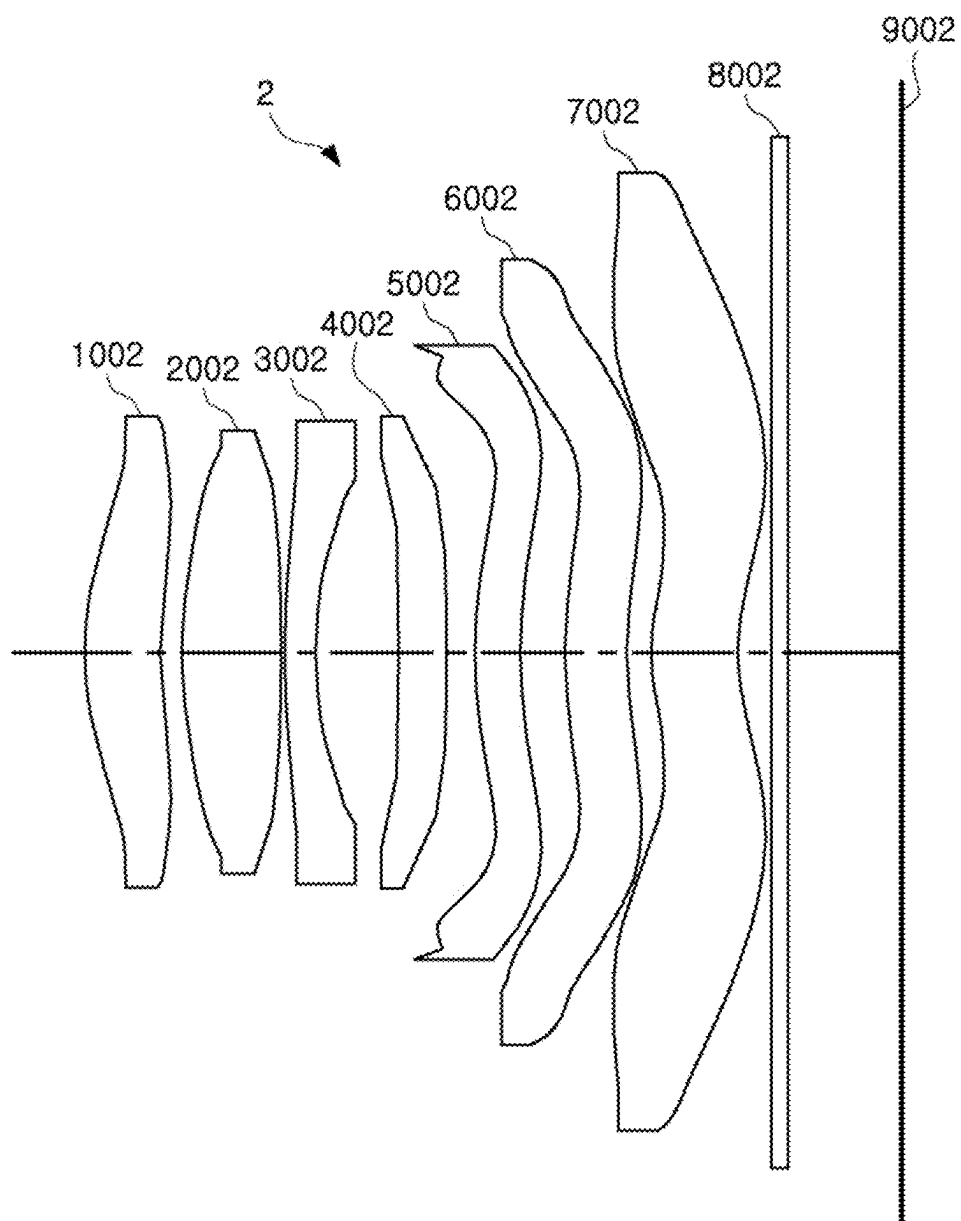


FIG. 3

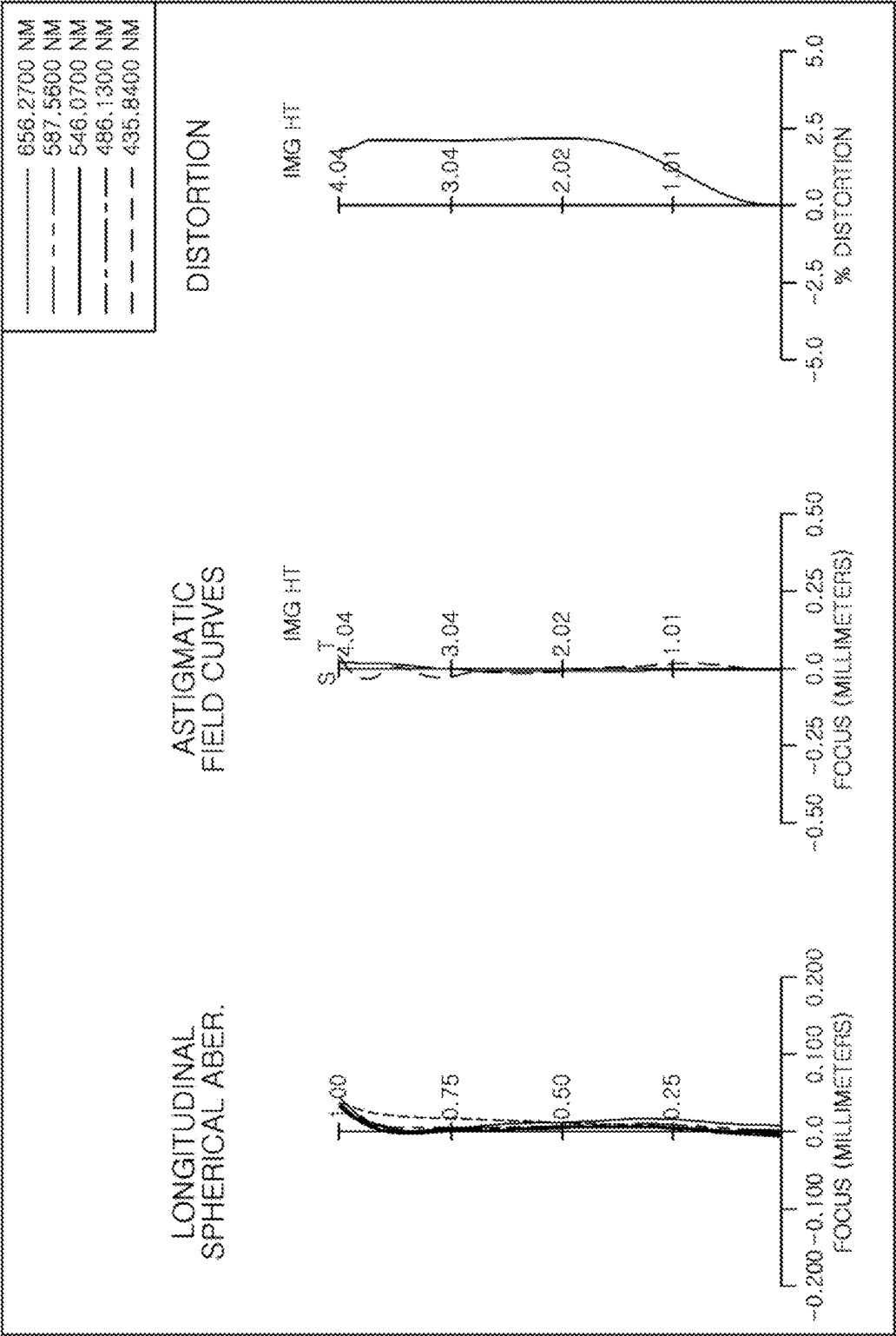


FIG. 4

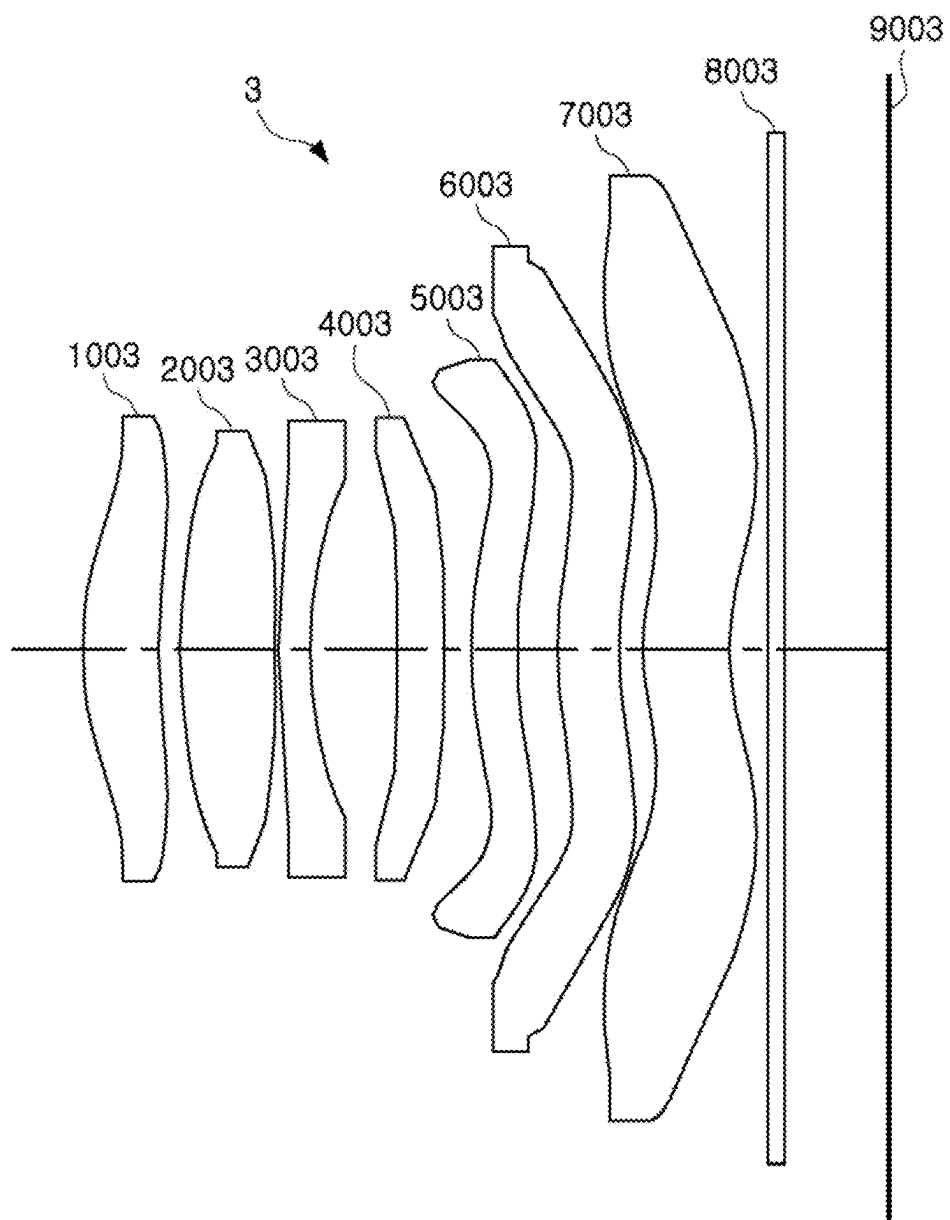


FIG. 5

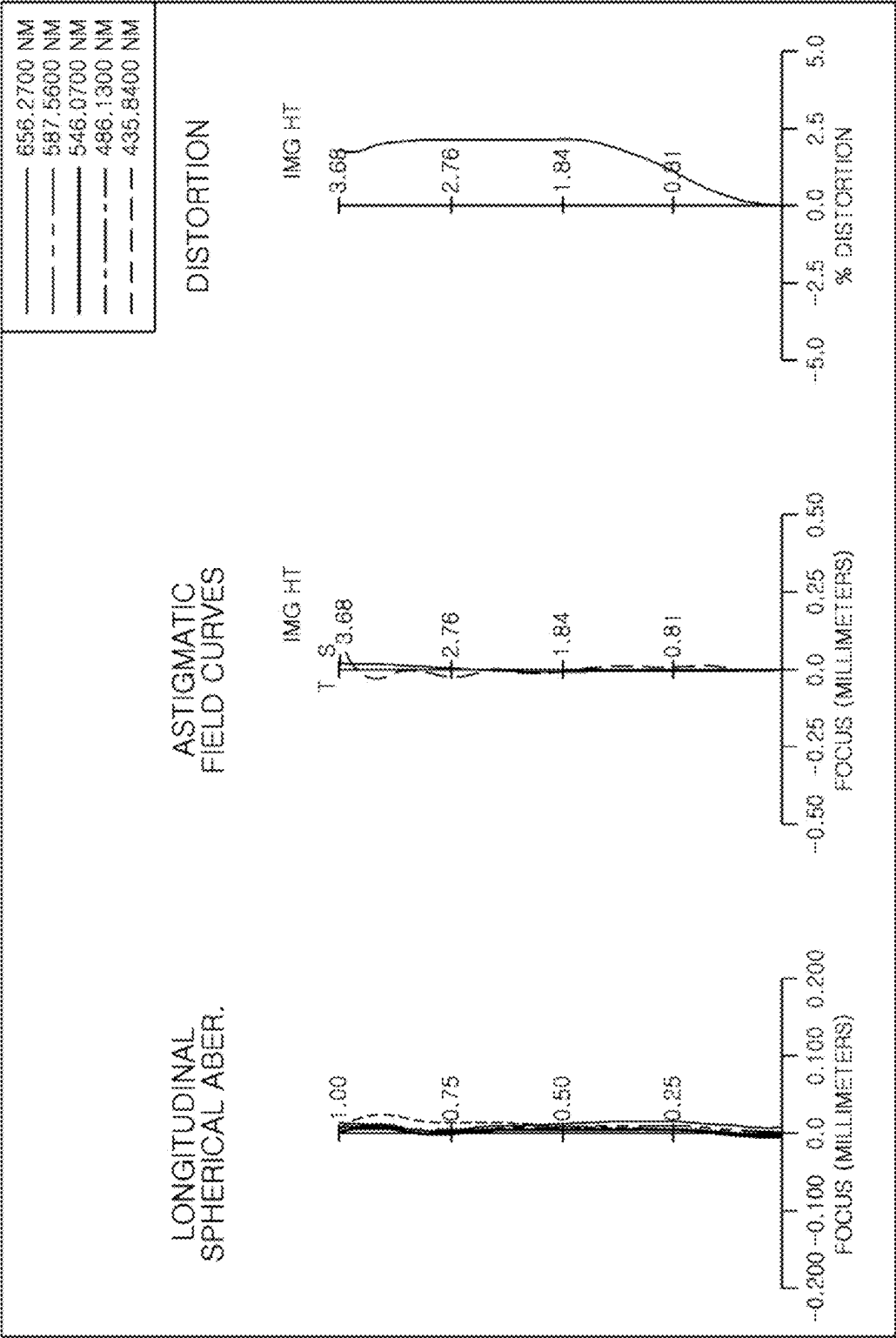


FIG. 6

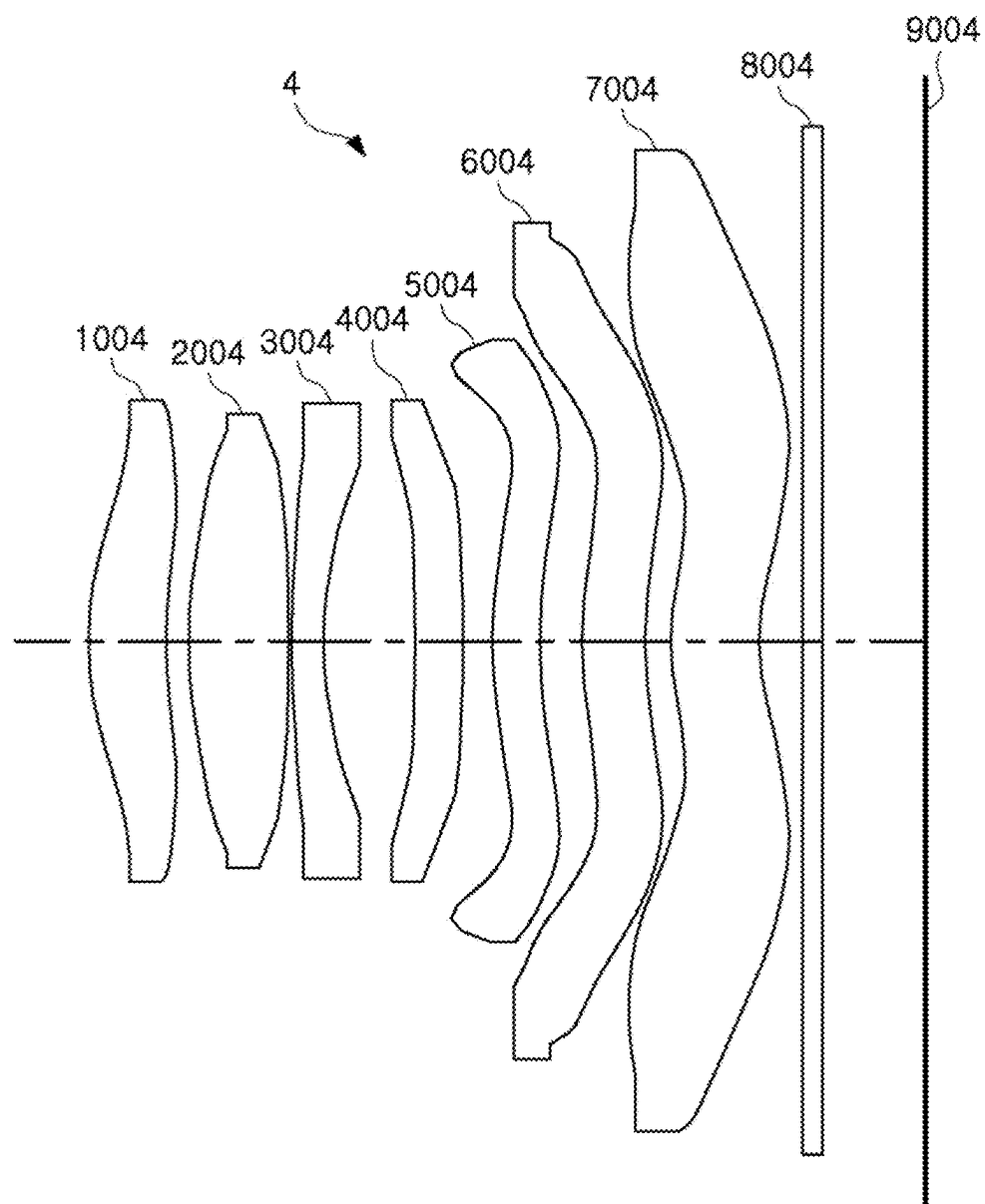


FIG. 7

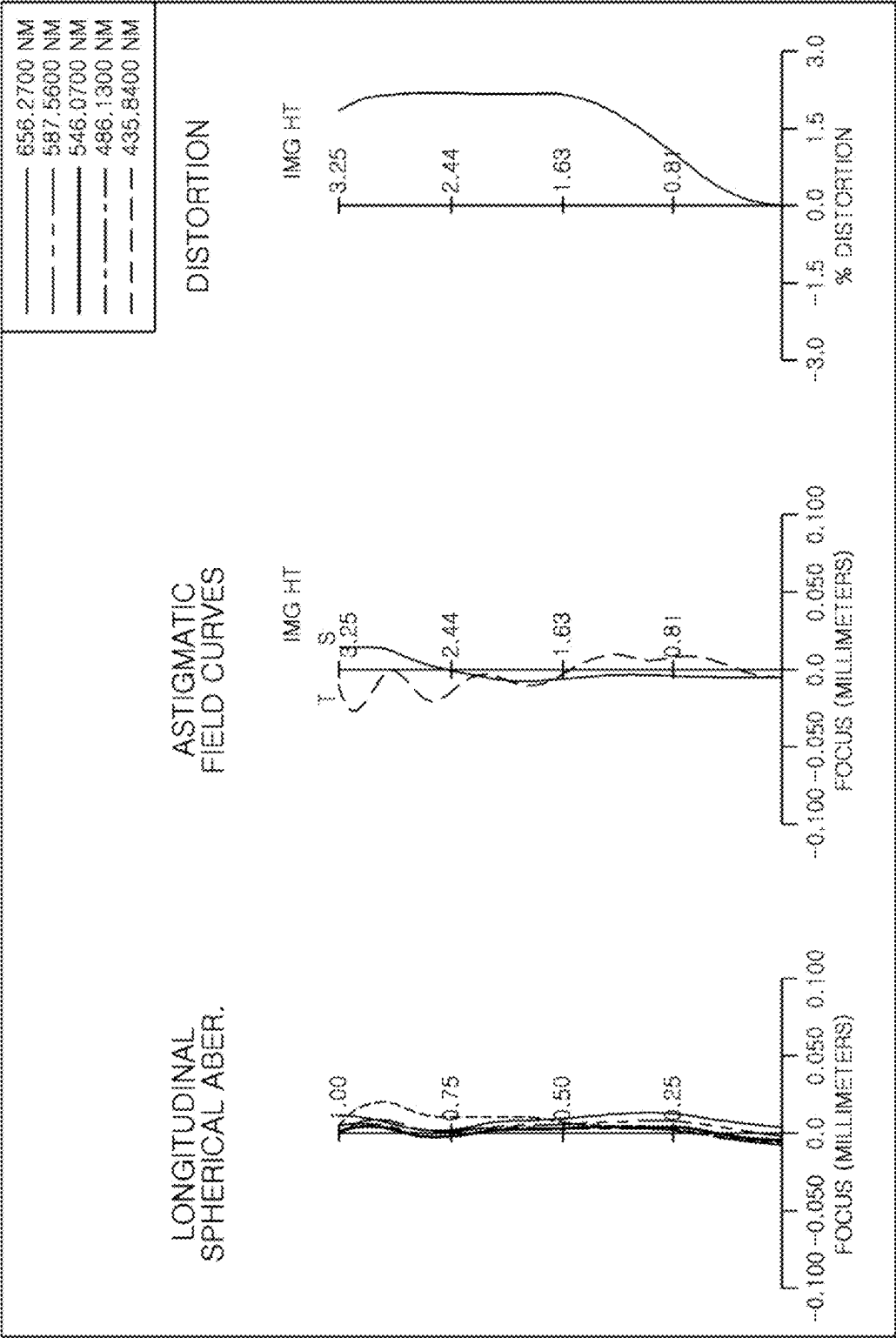


FIG. 8

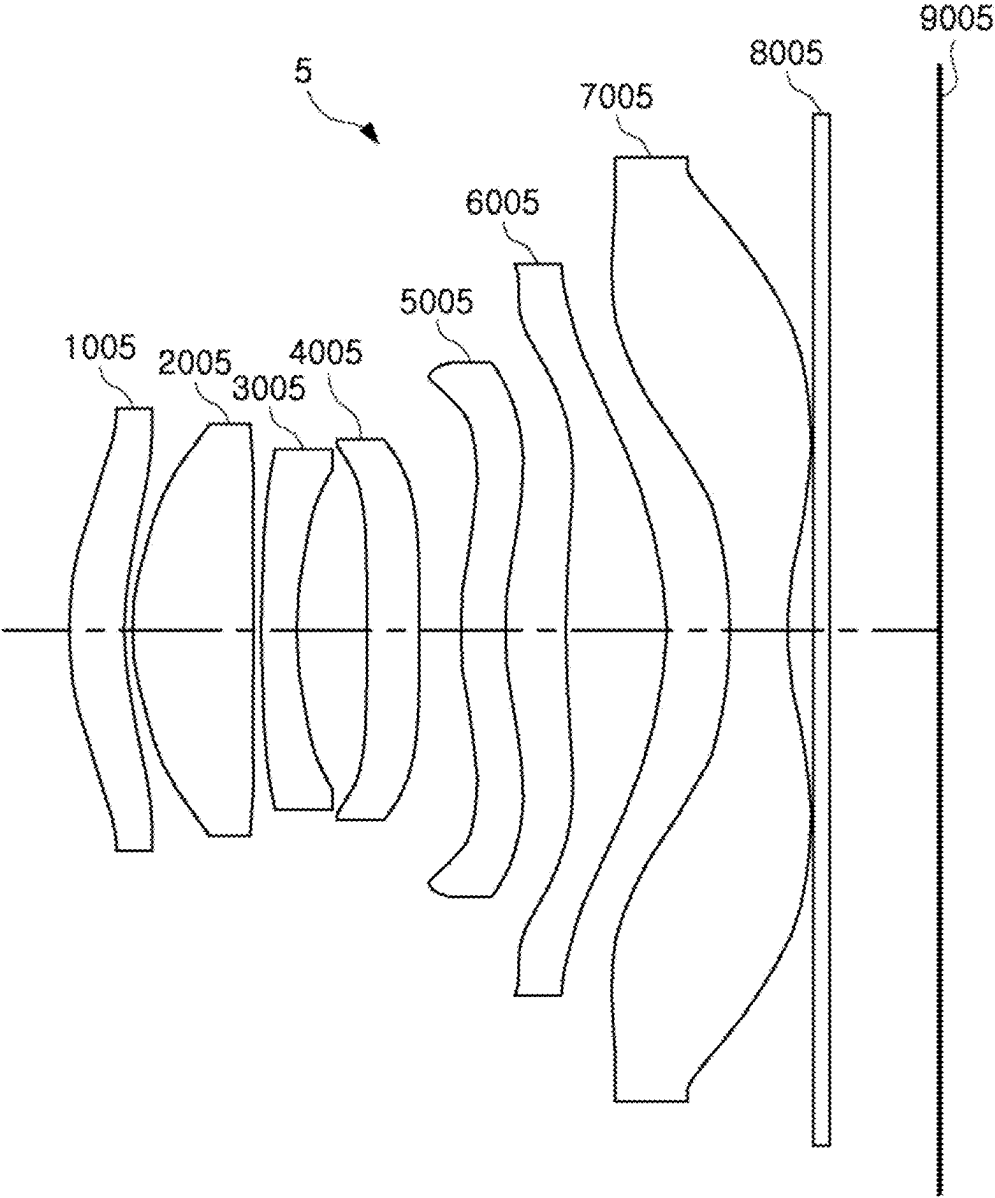


FIG. 9

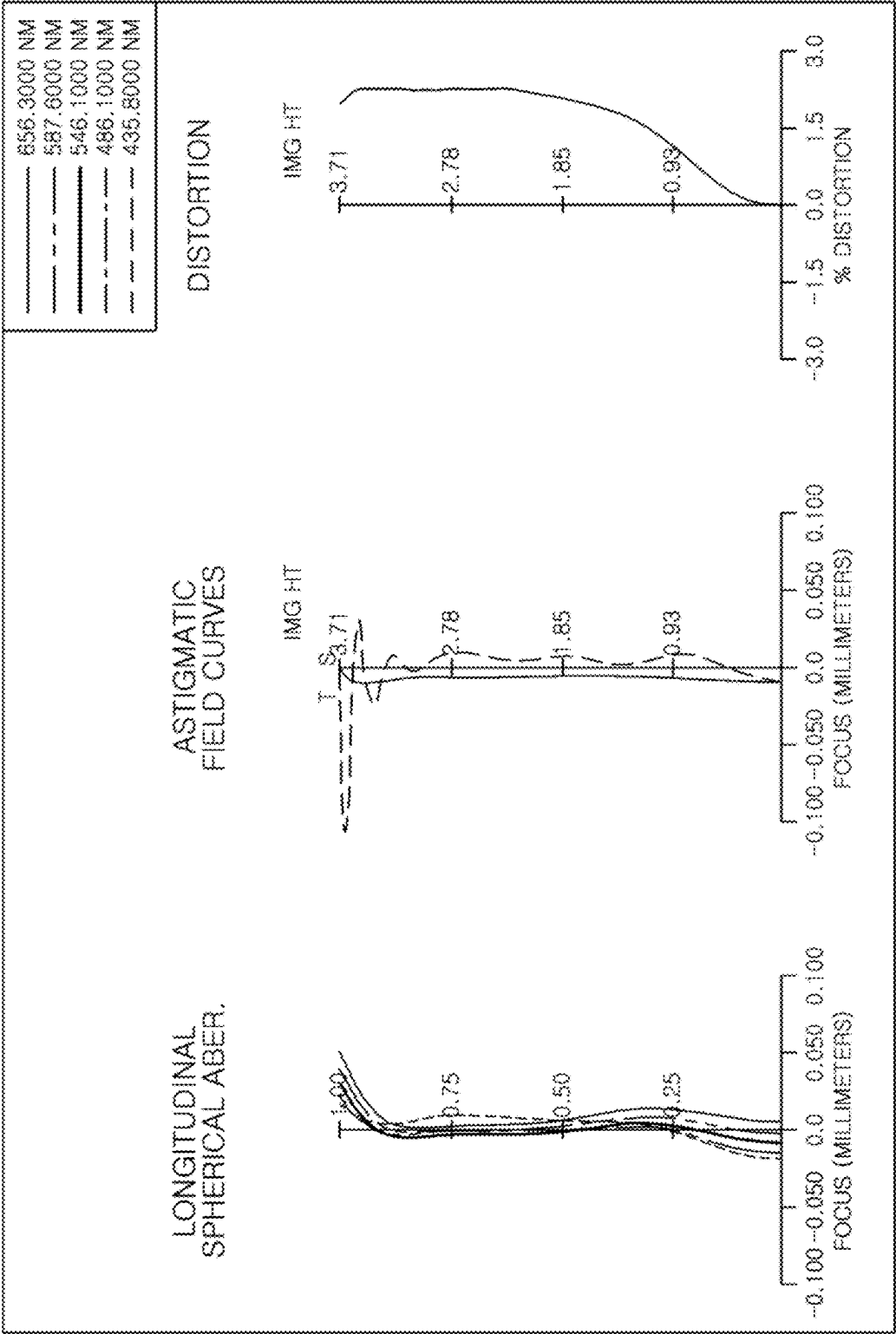


FIG. 10

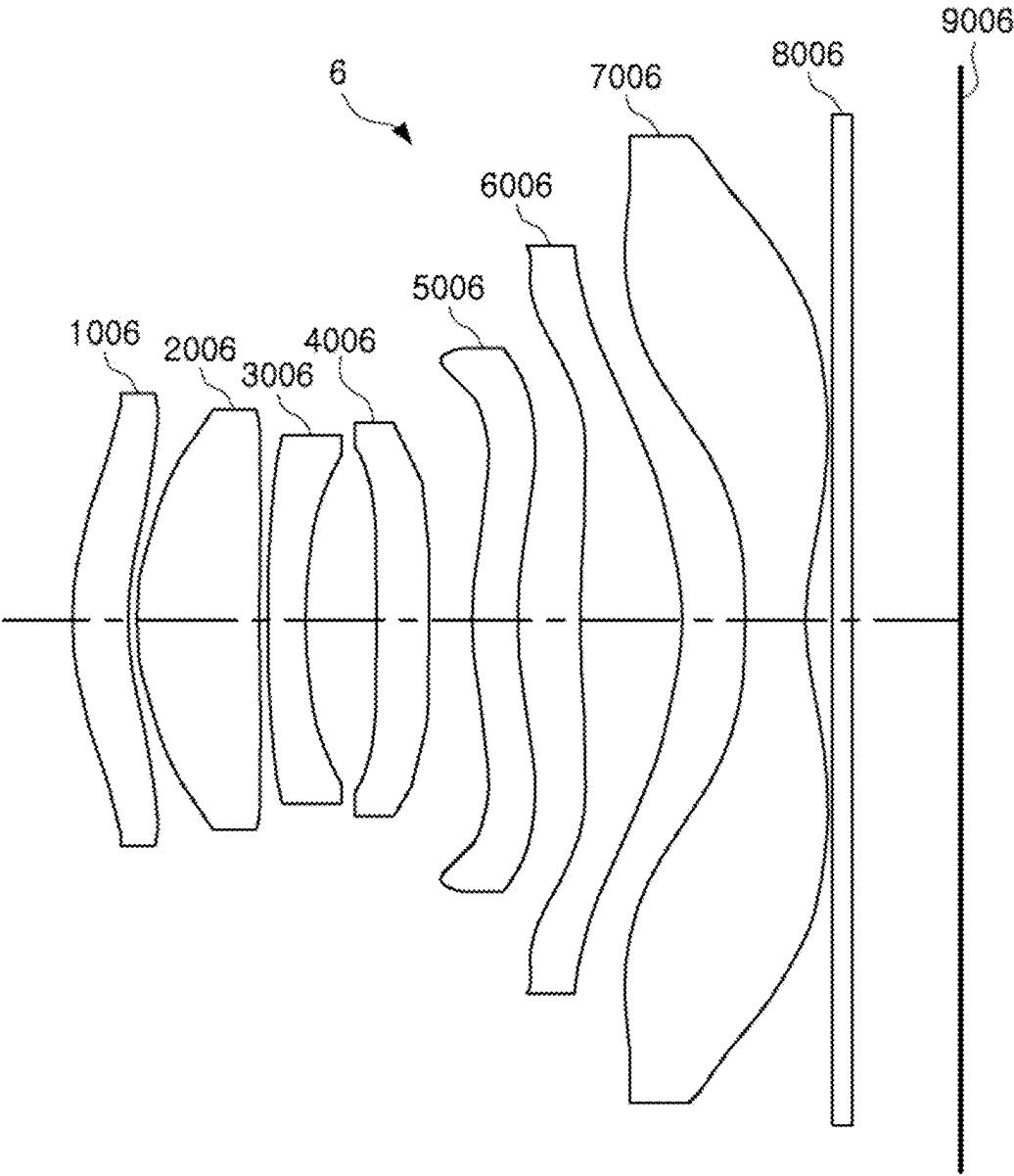


FIG. 11

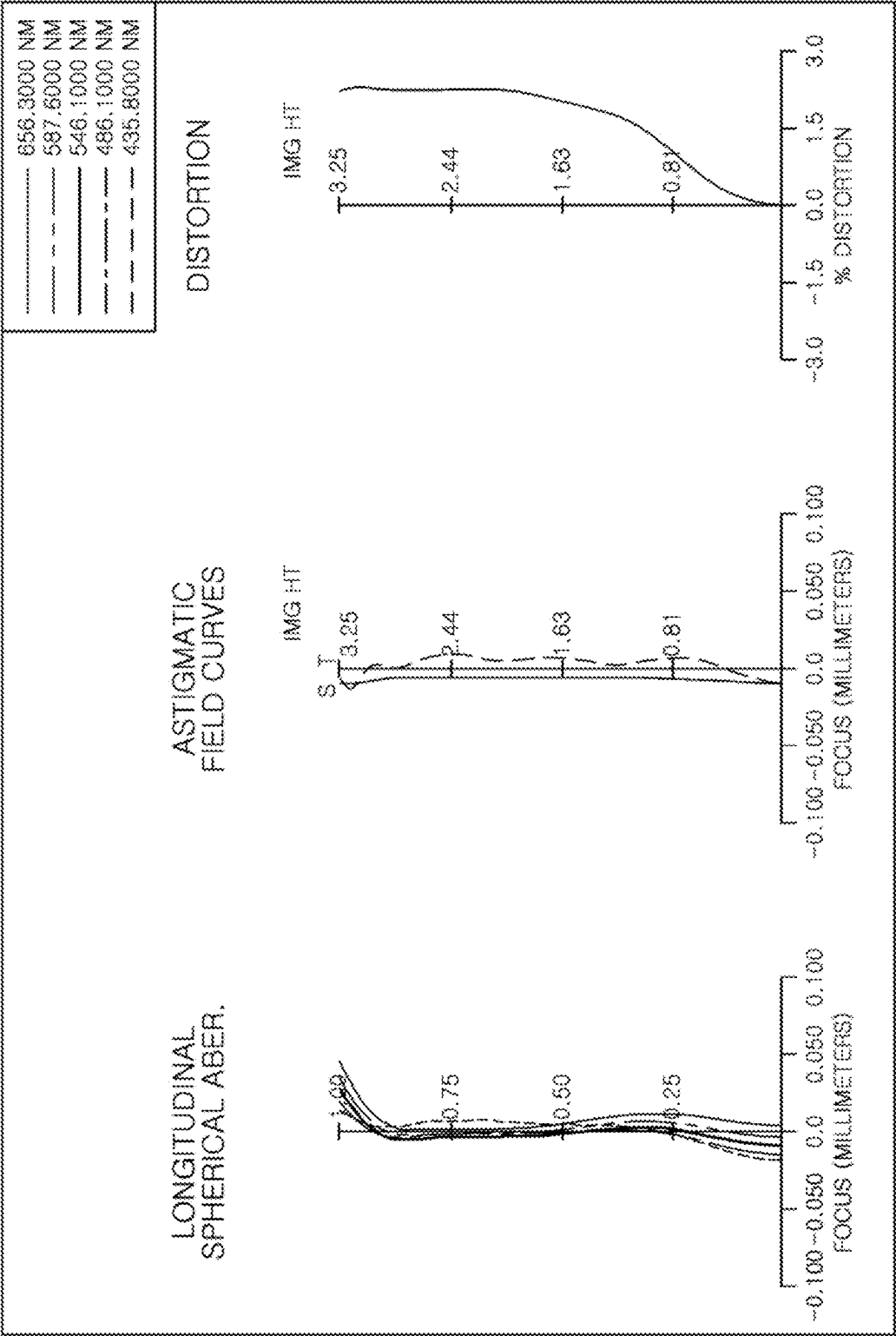


FIG. 12

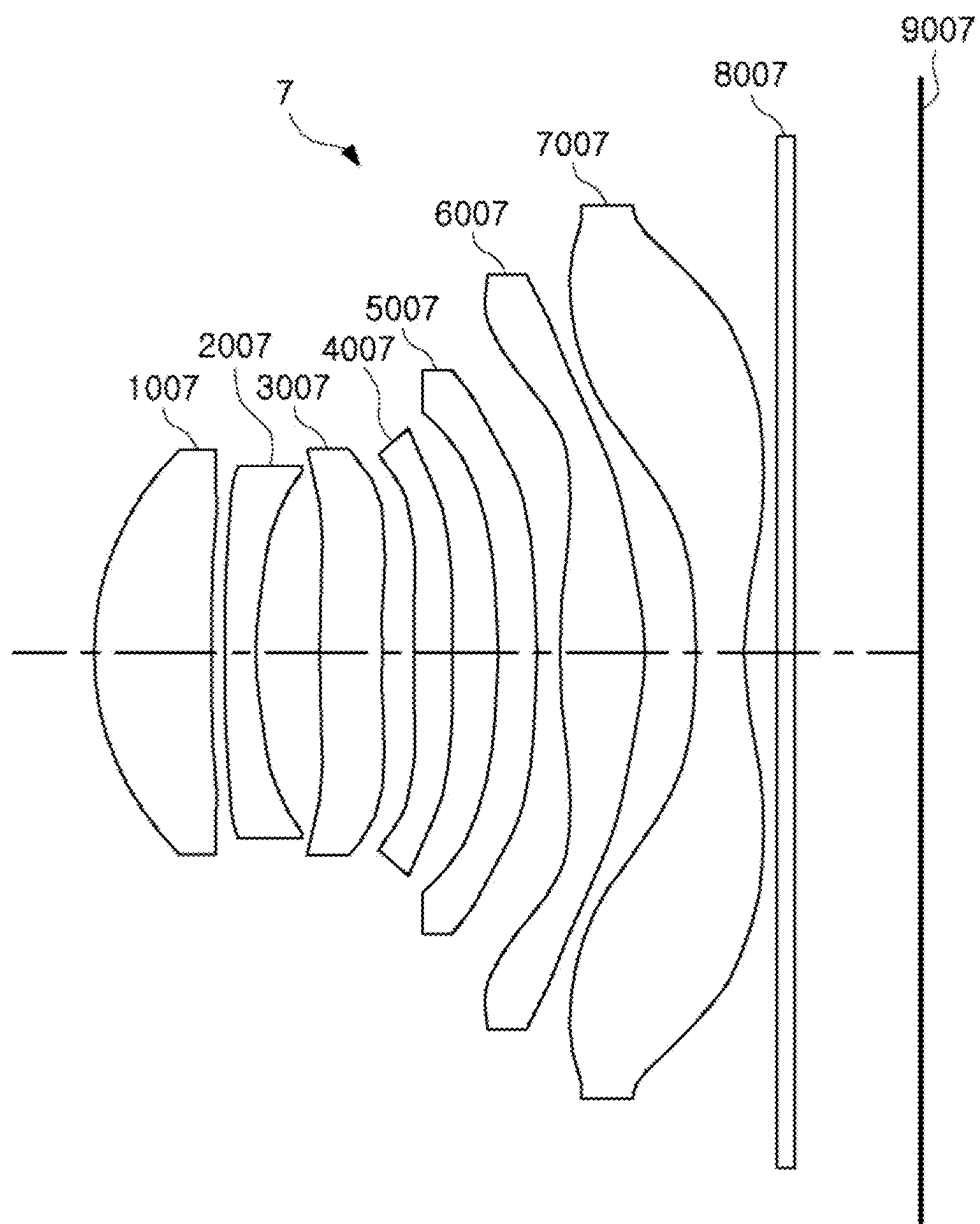


FIG. 13

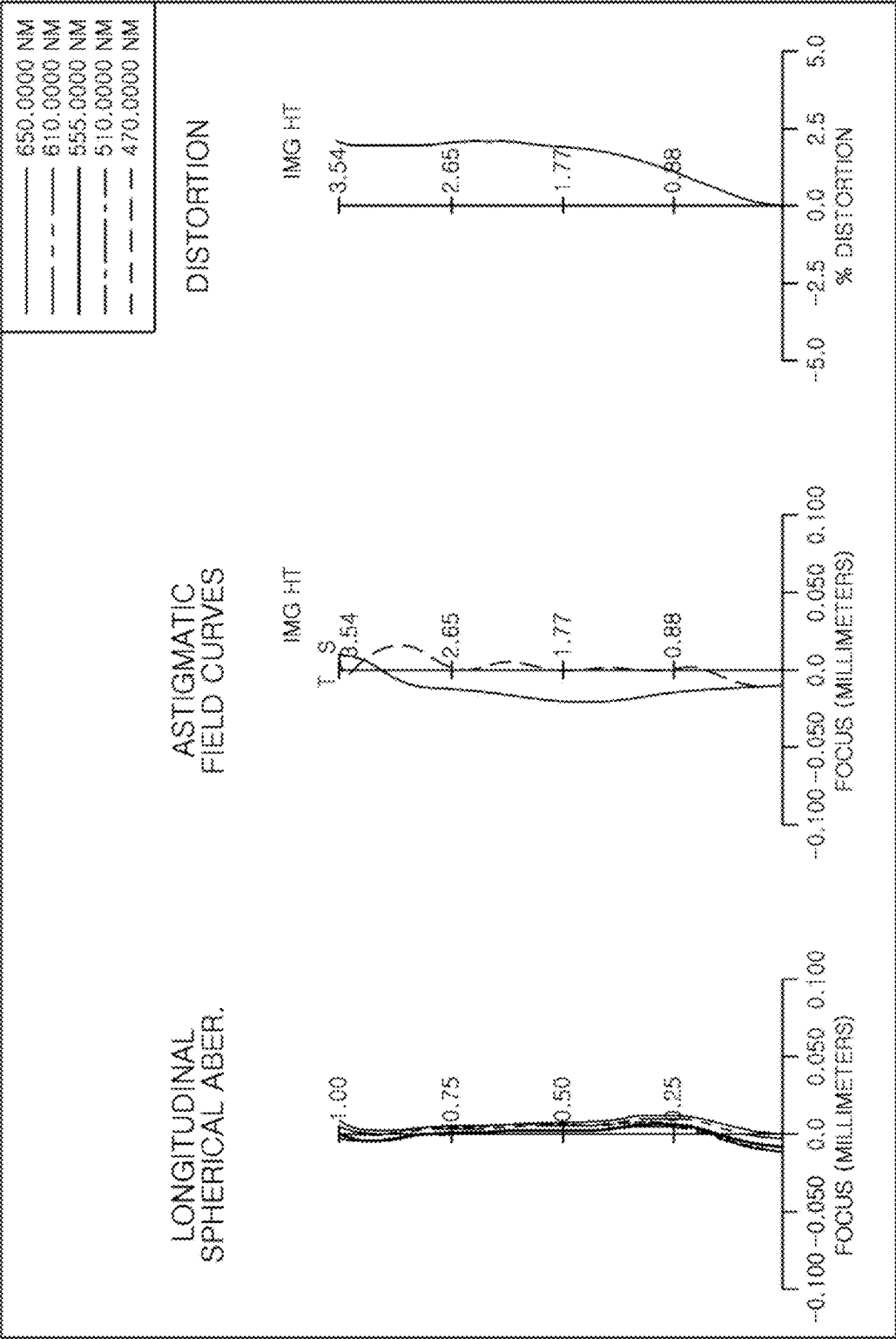


FIG. 14

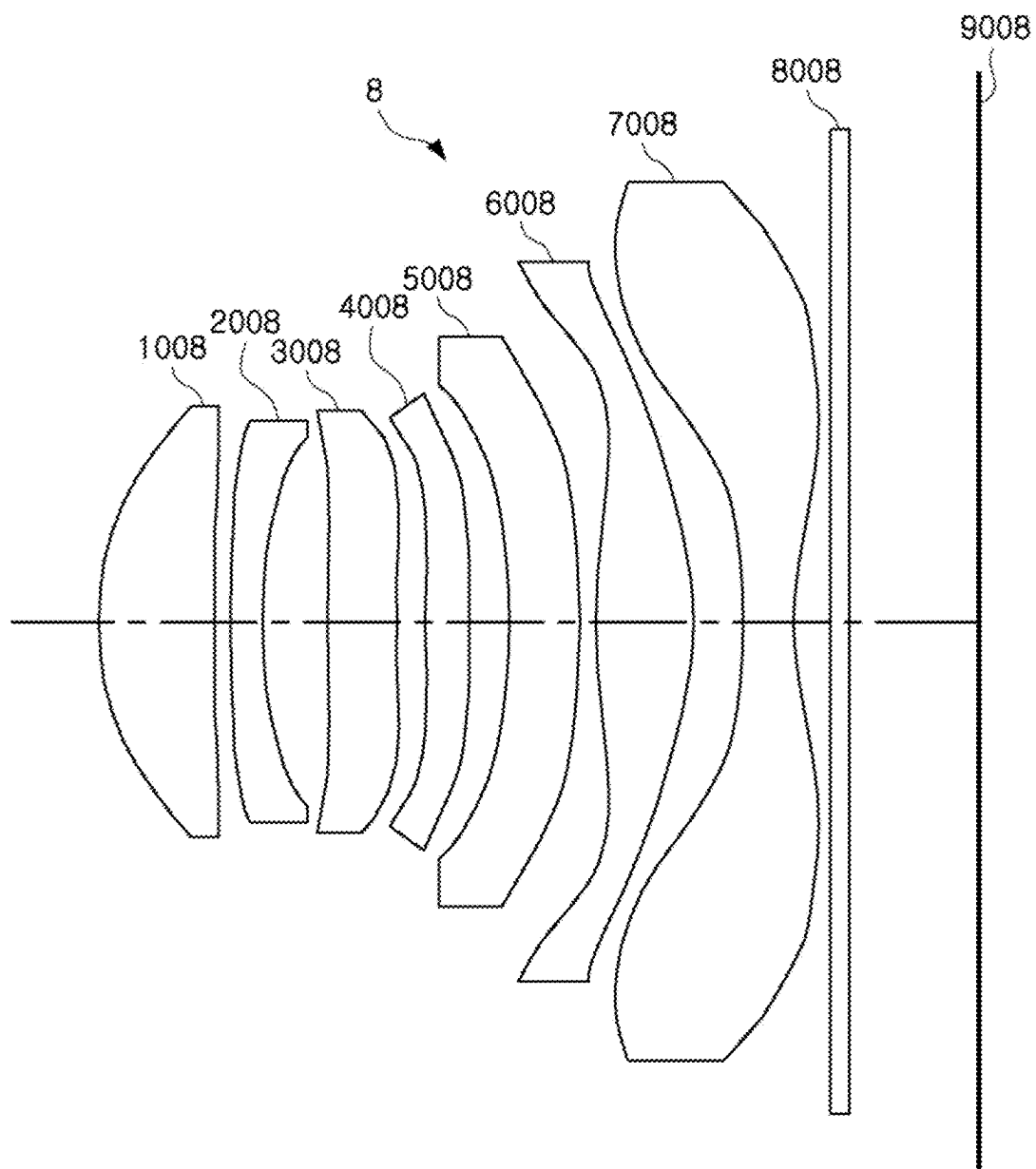


FIG. 15

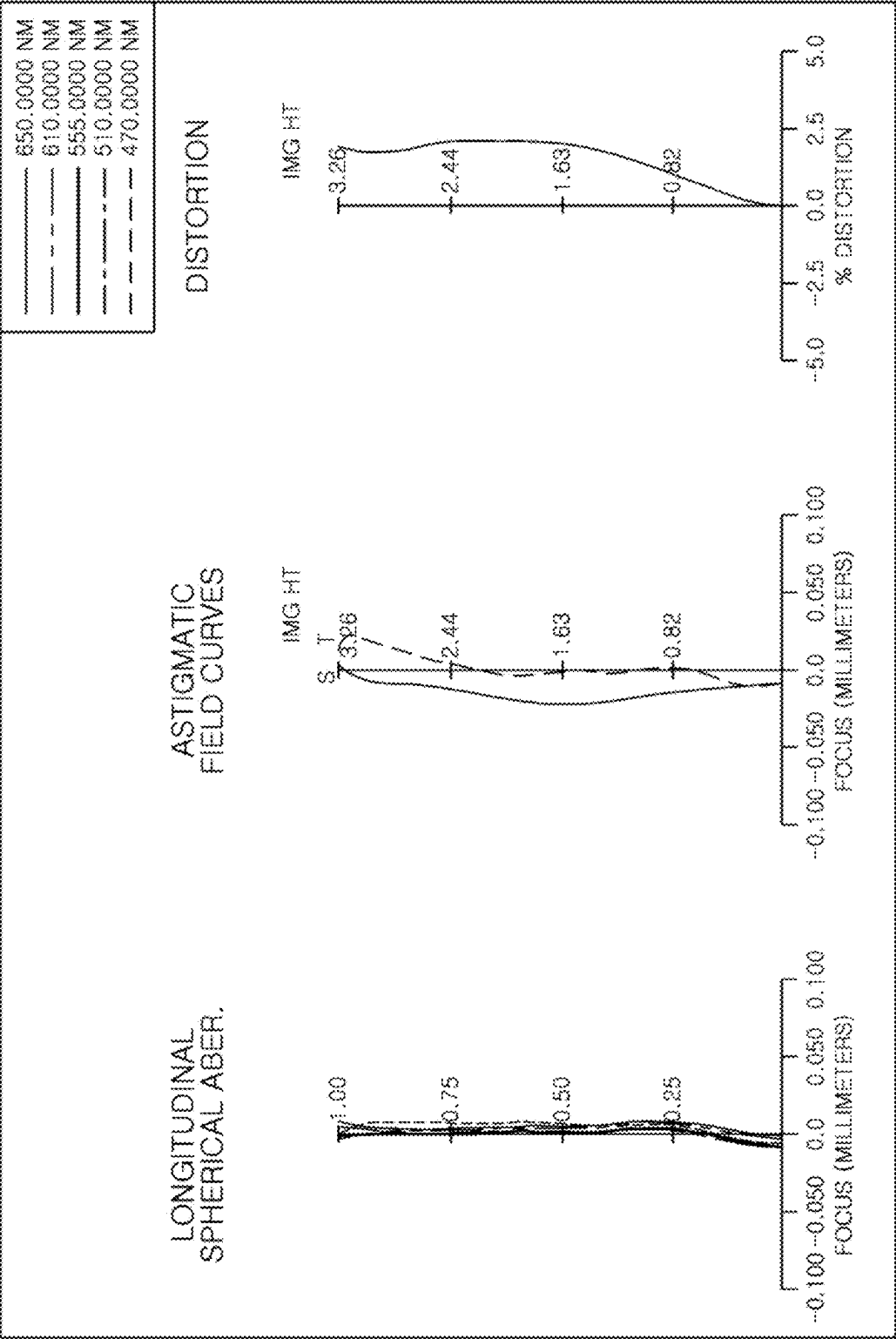


FIG. 16

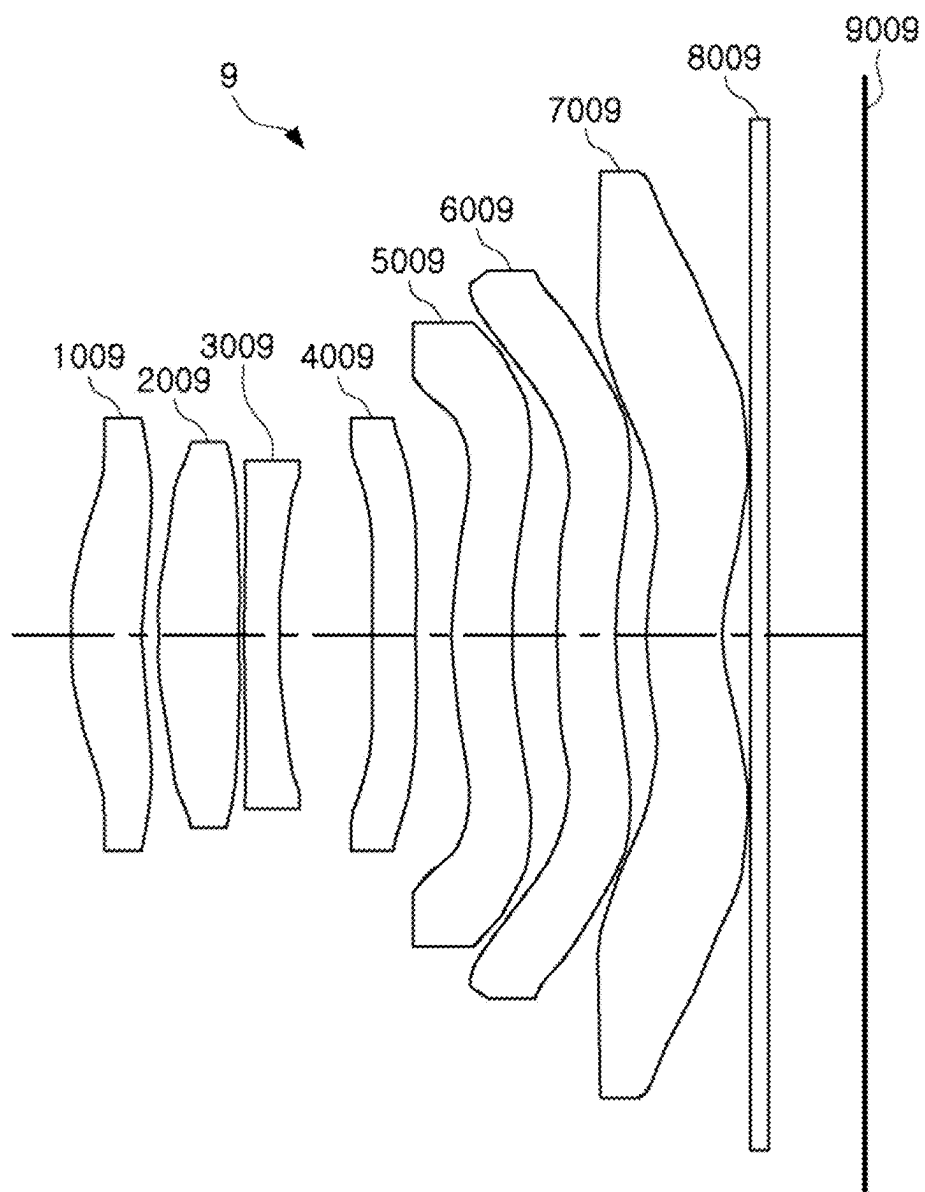


FIG. 17

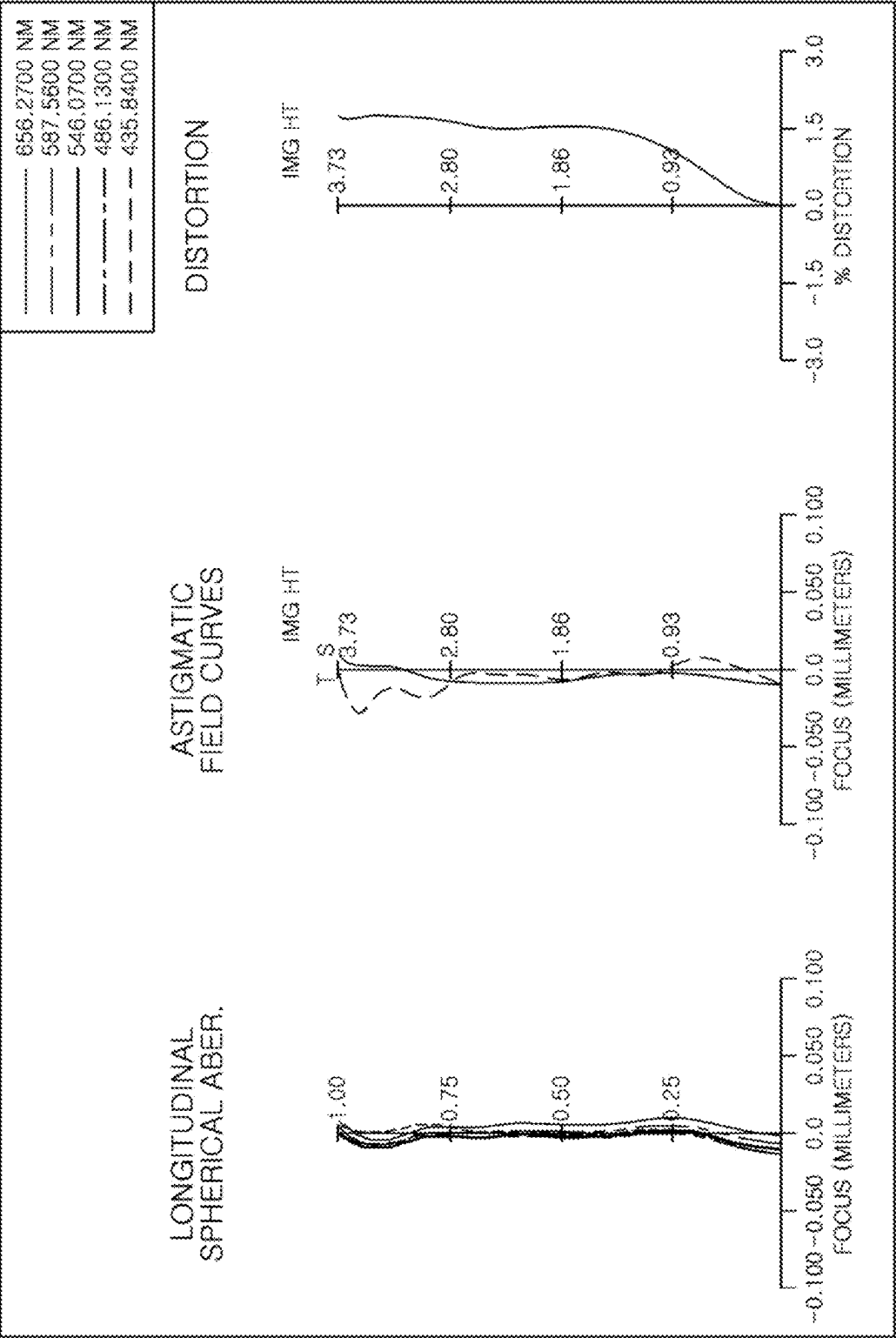


FIG. 18

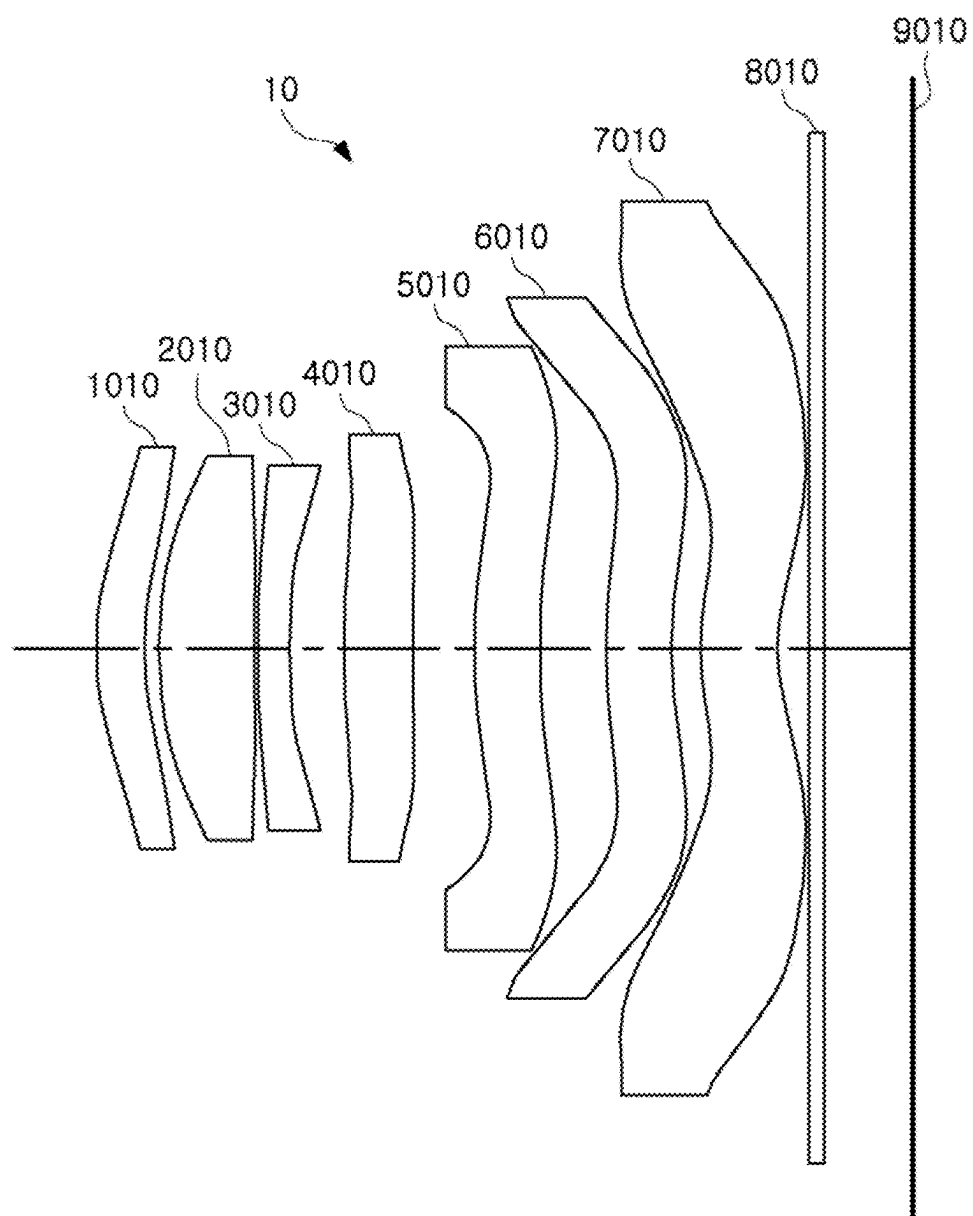


FIG. 19

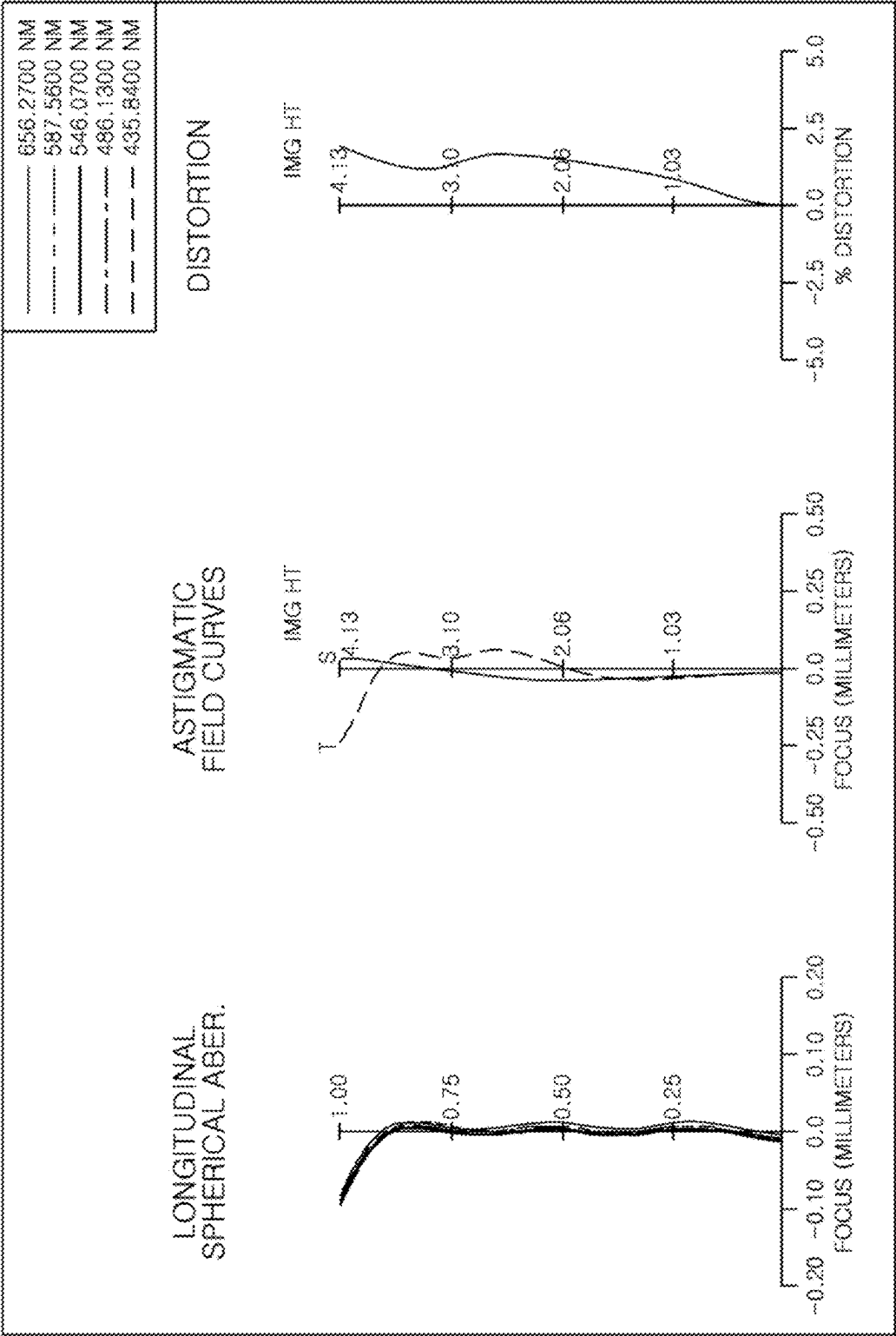


FIG. 20

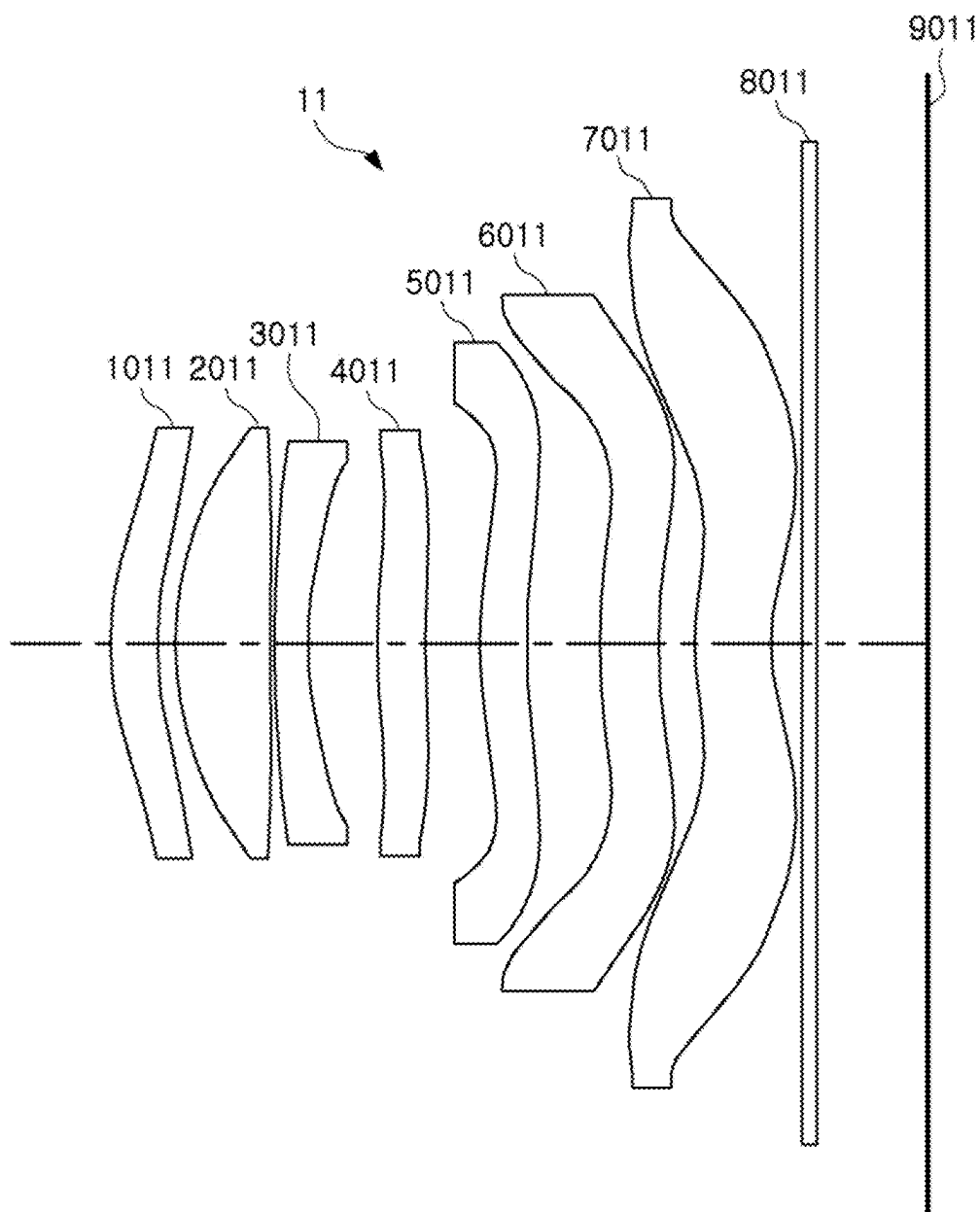


FIG. 21

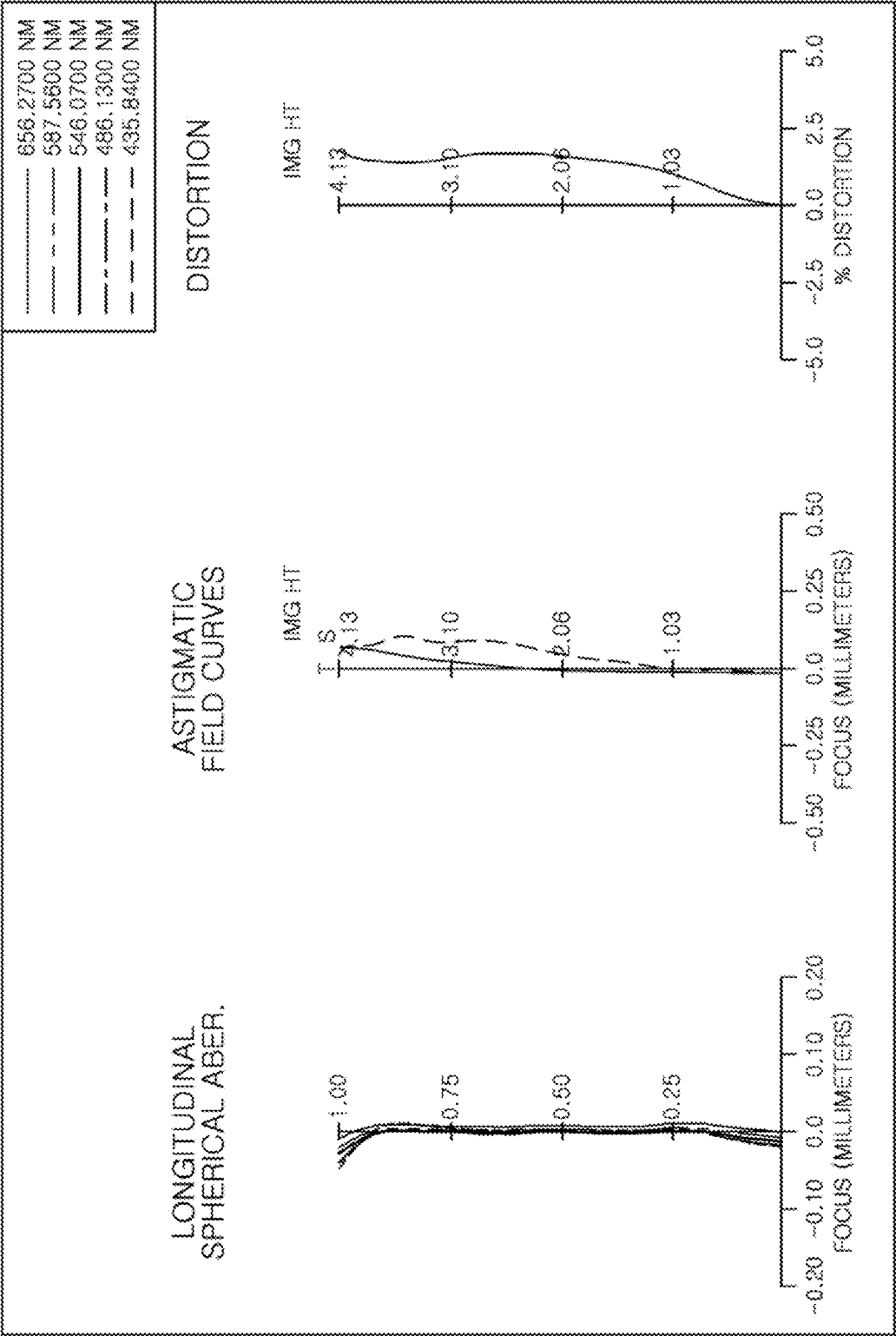


FIG. 22

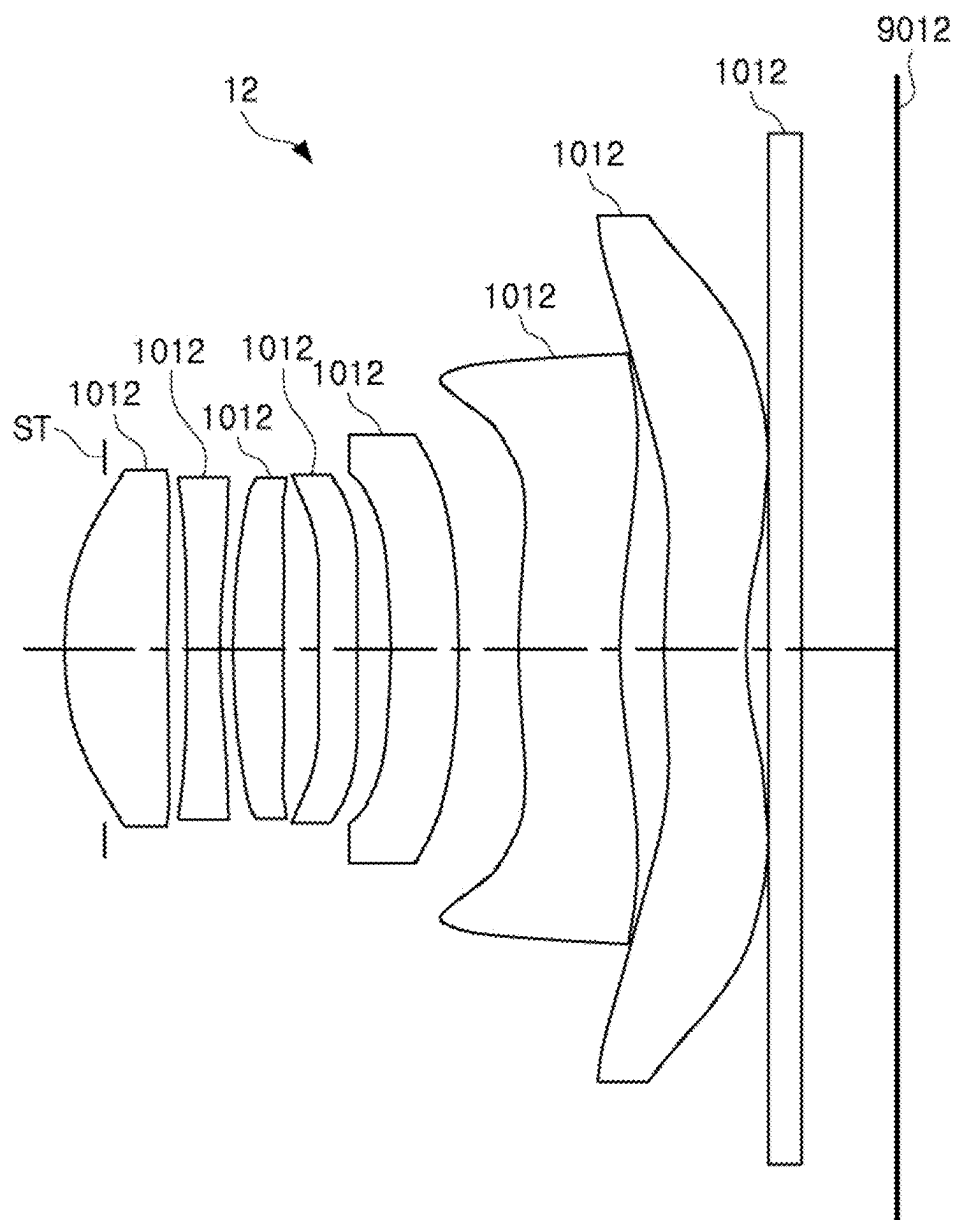


FIG. 23

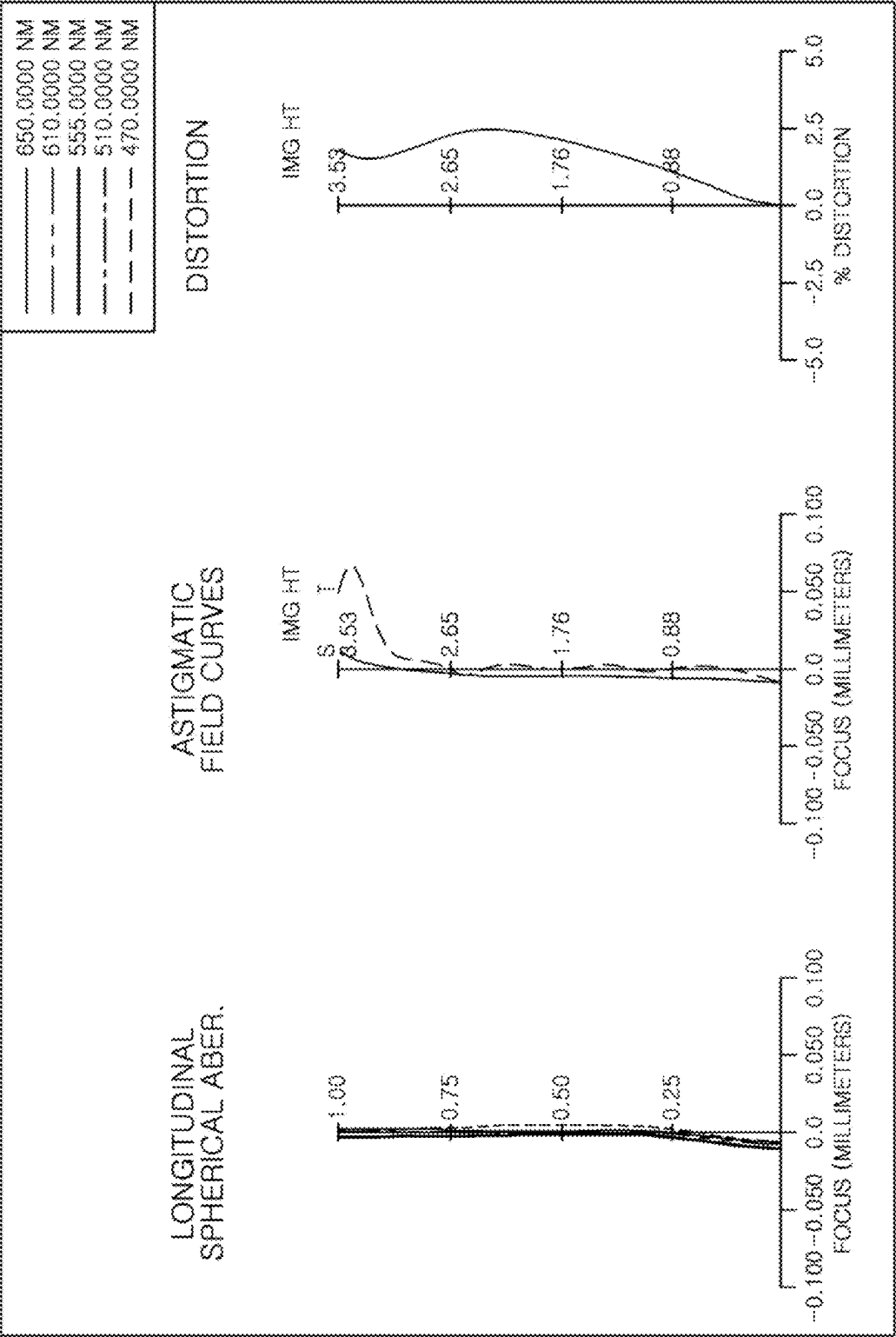


FIG. 24

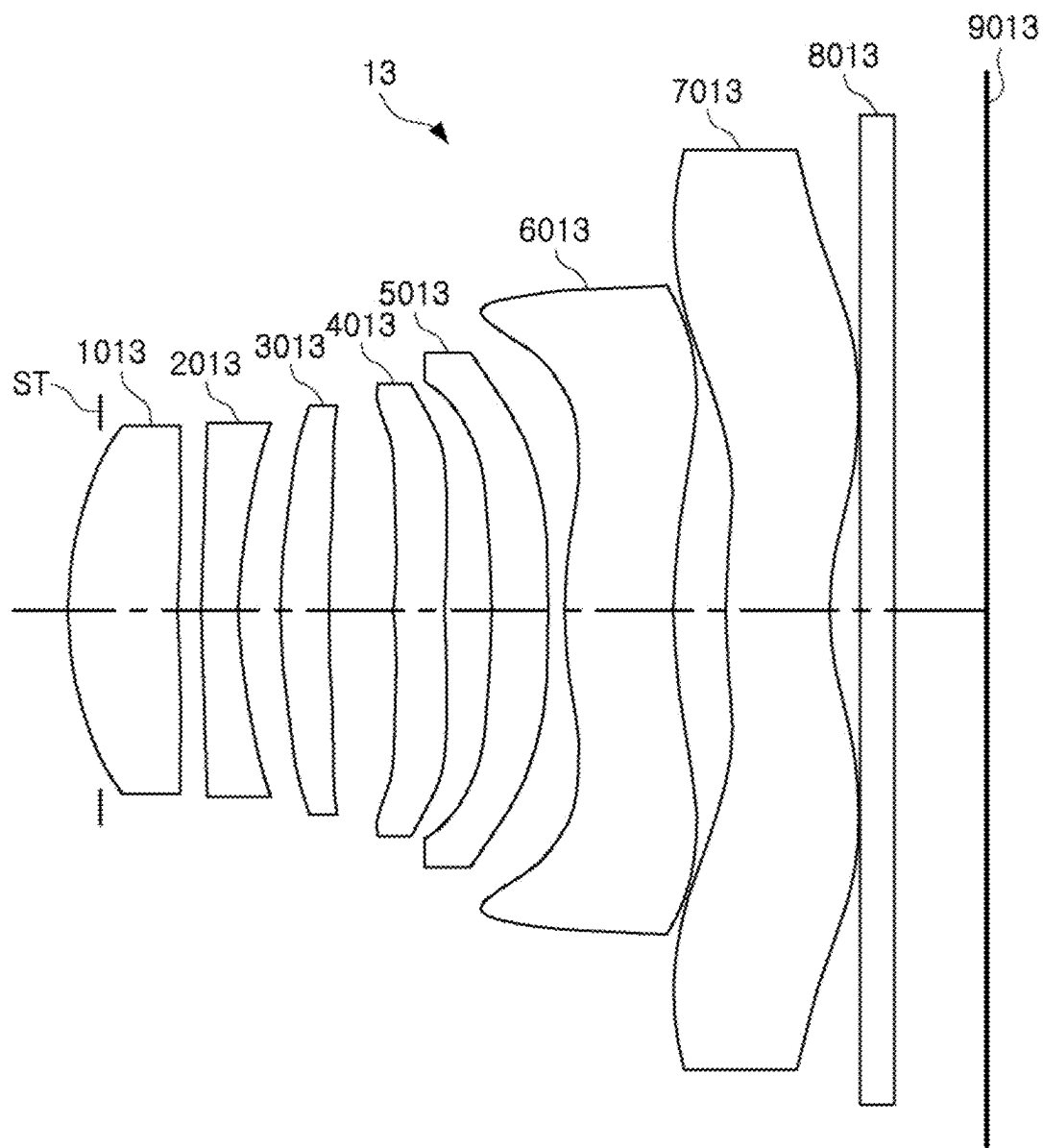


FIG. 25

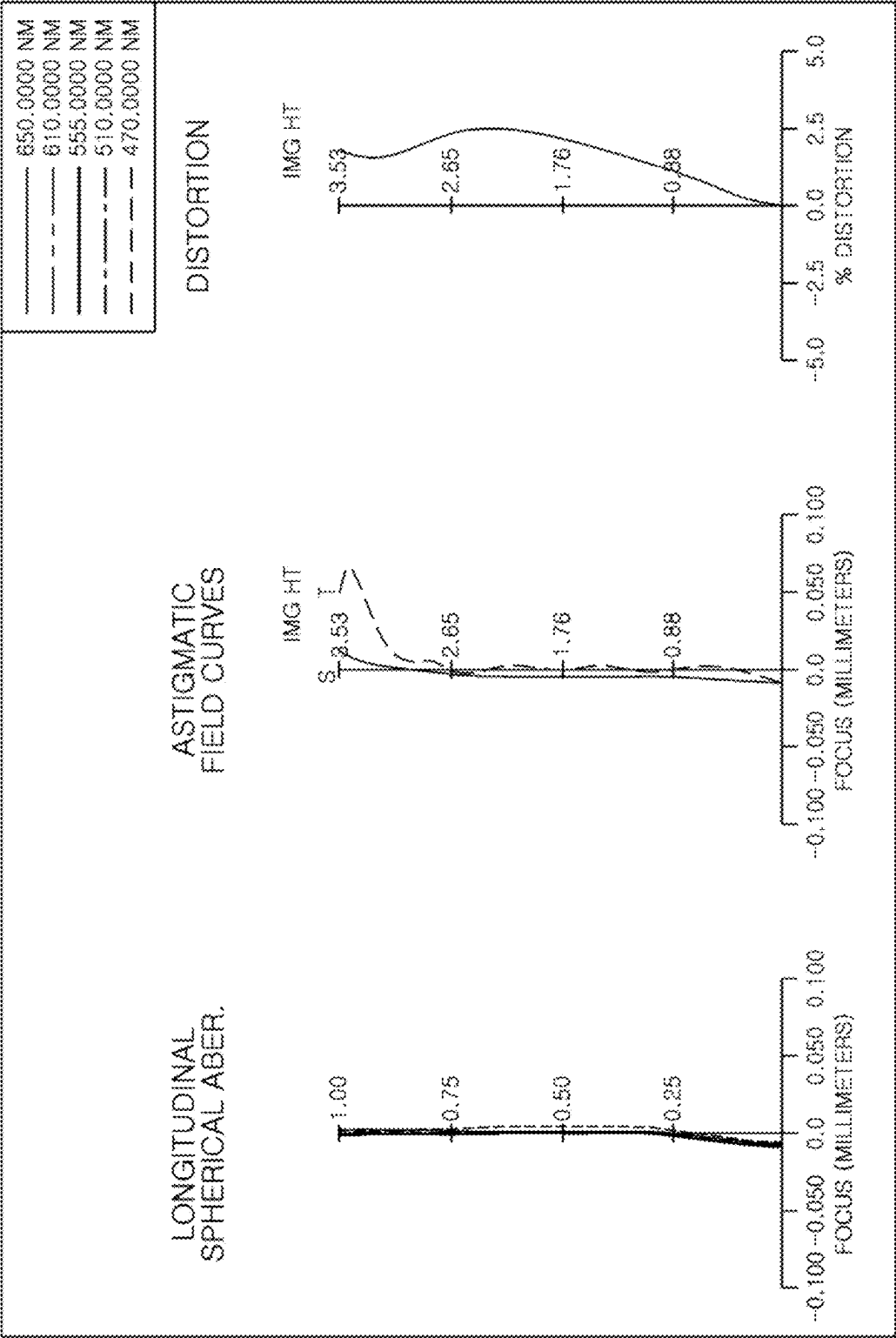


FIG. 26

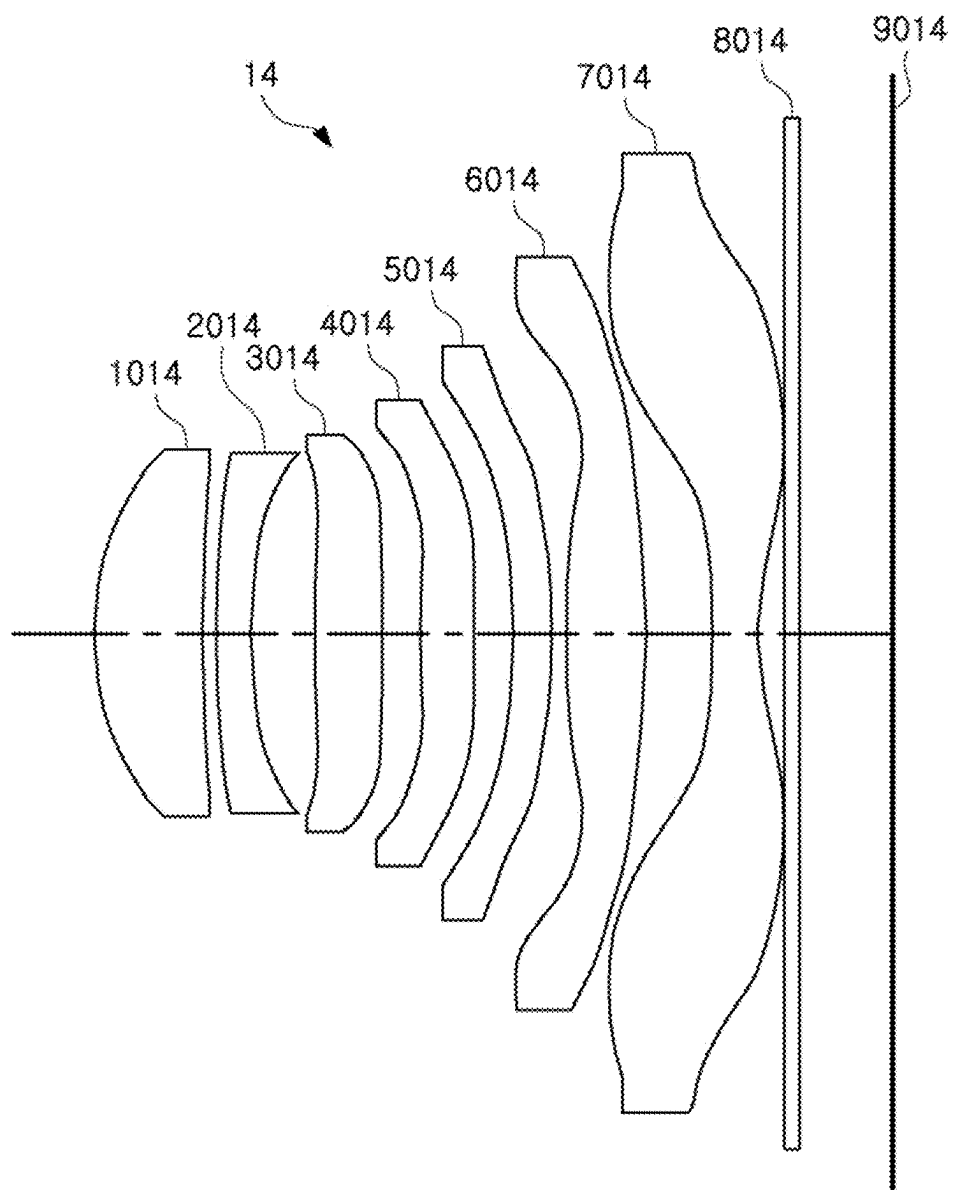


FIG. 27

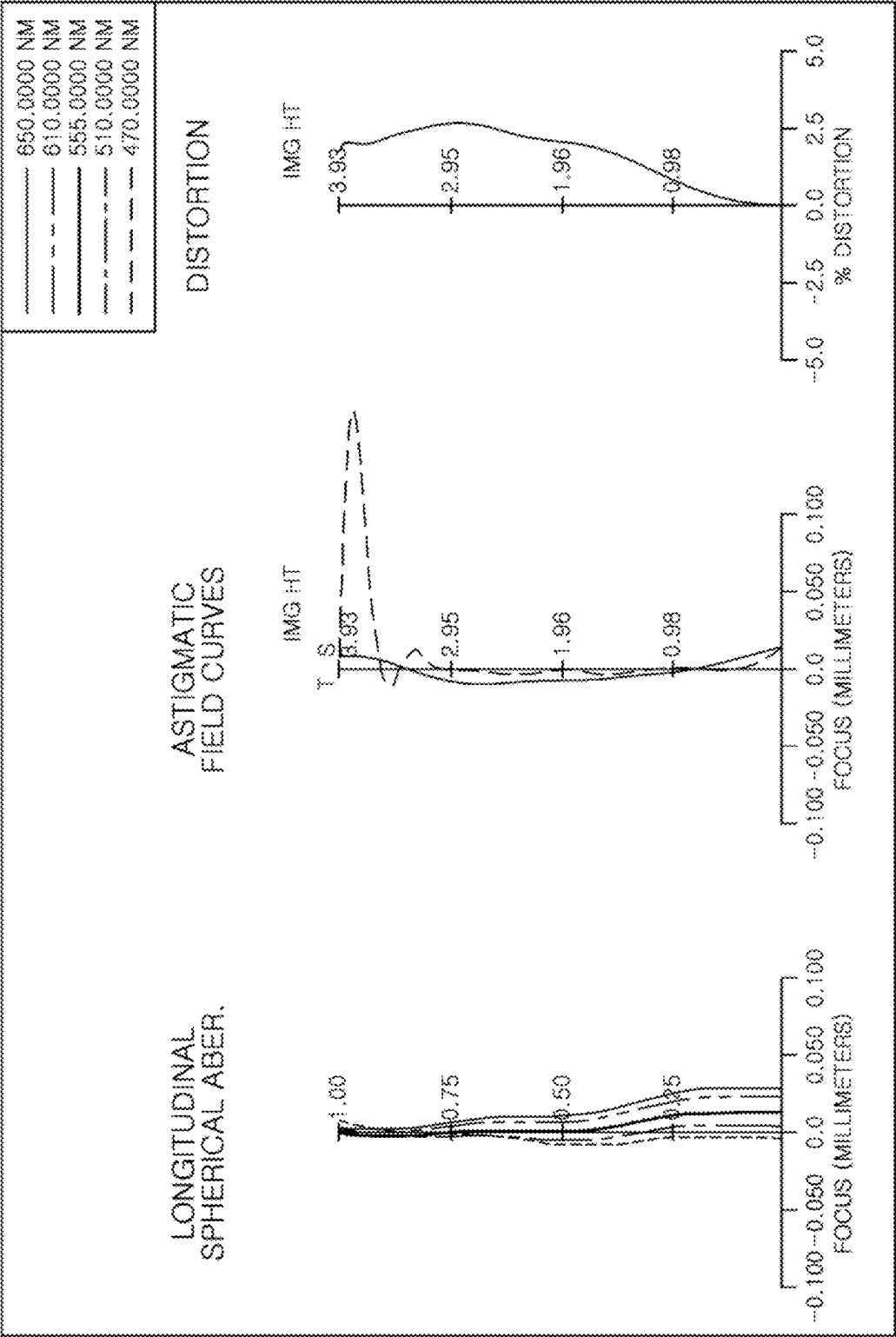


FIG. 28

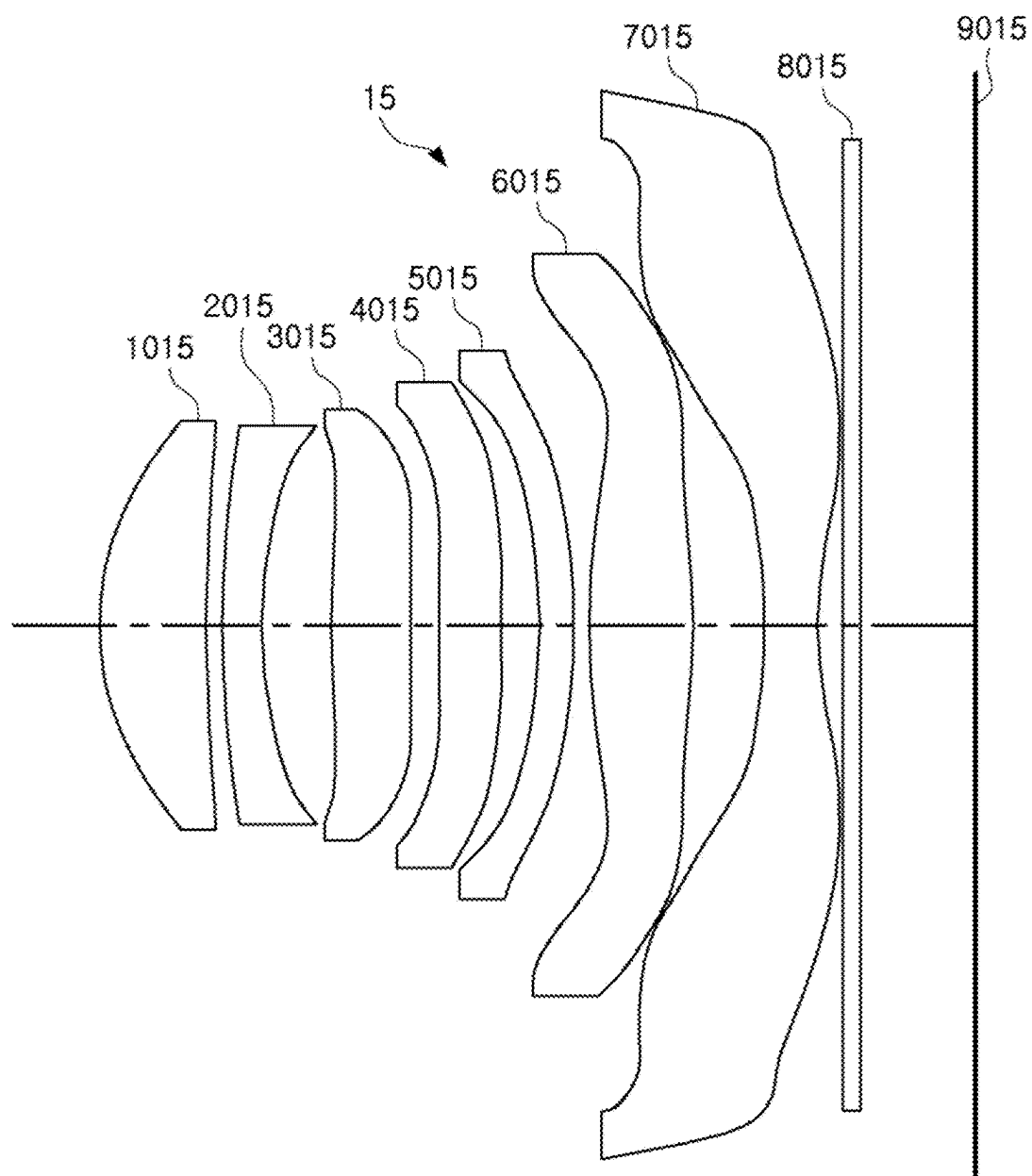


FIG. 29

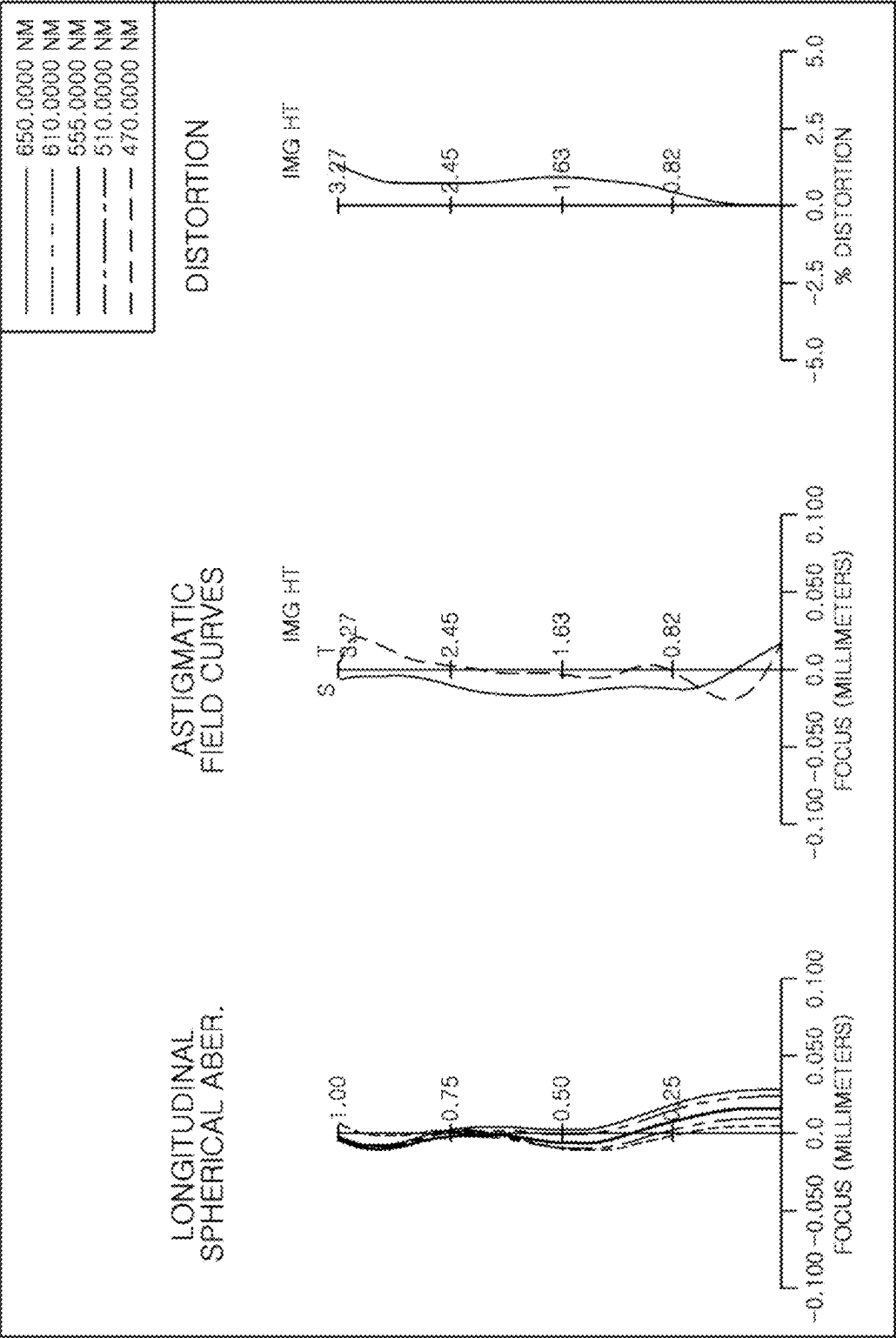


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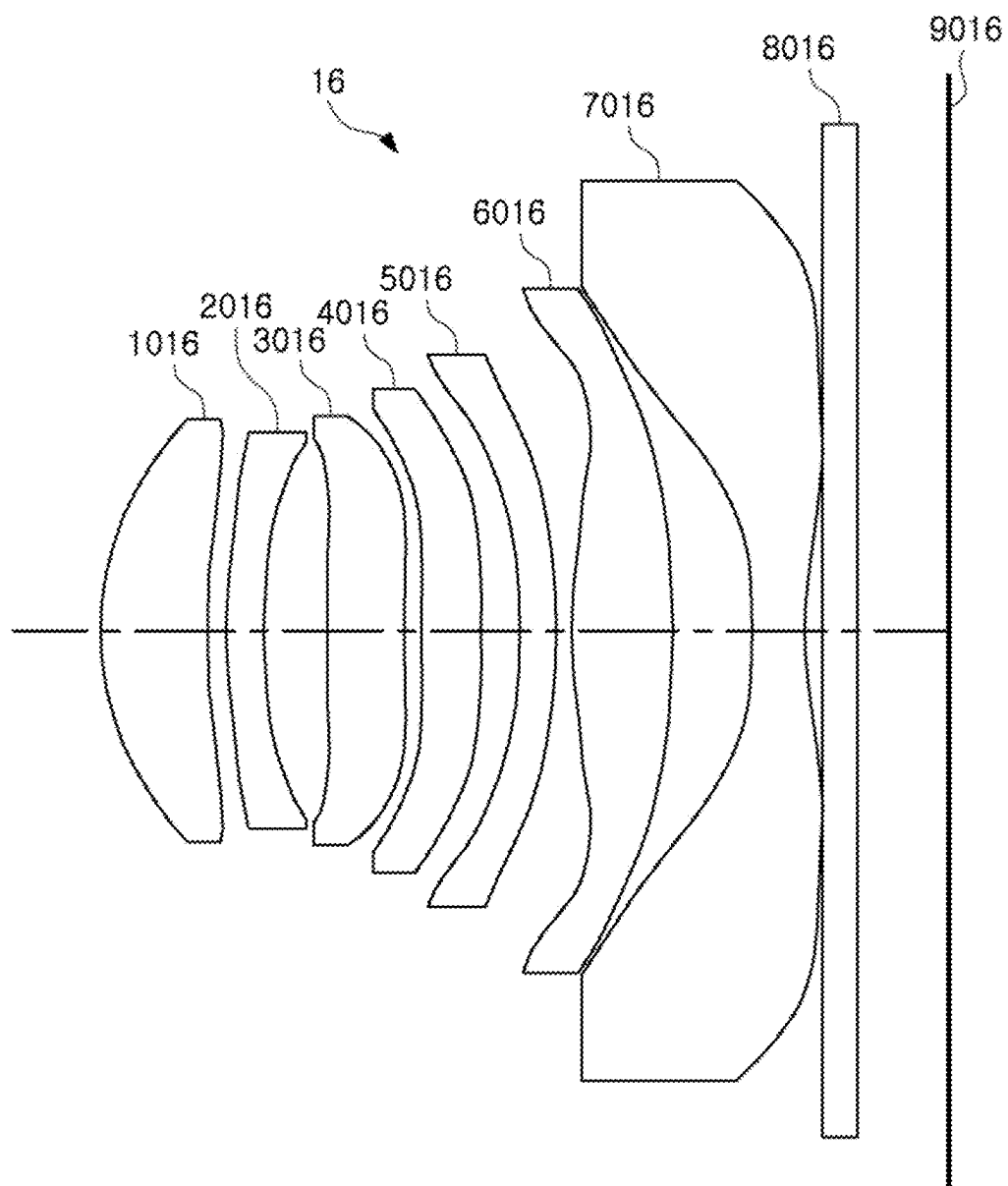


FIG. 31

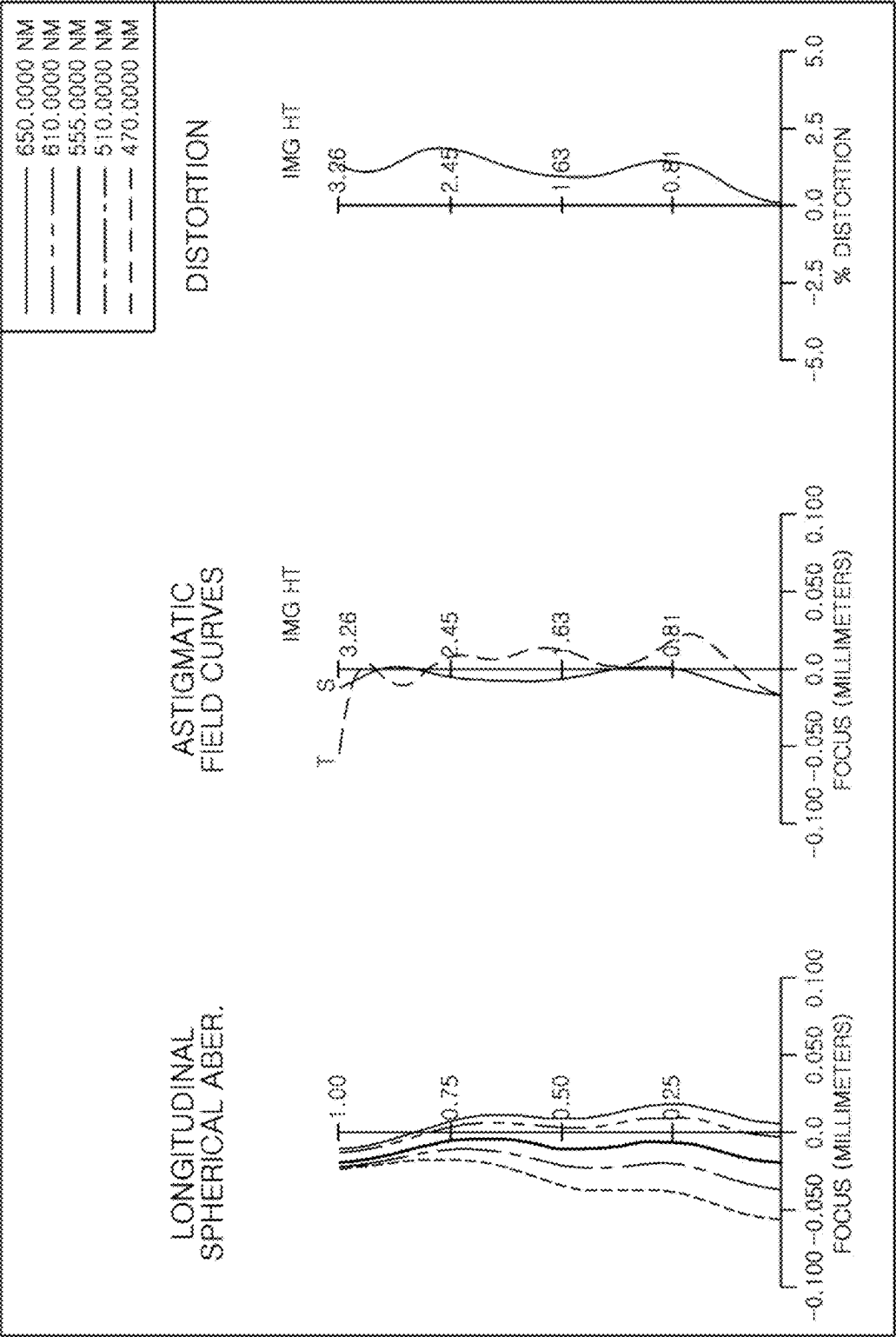


FIG. 32

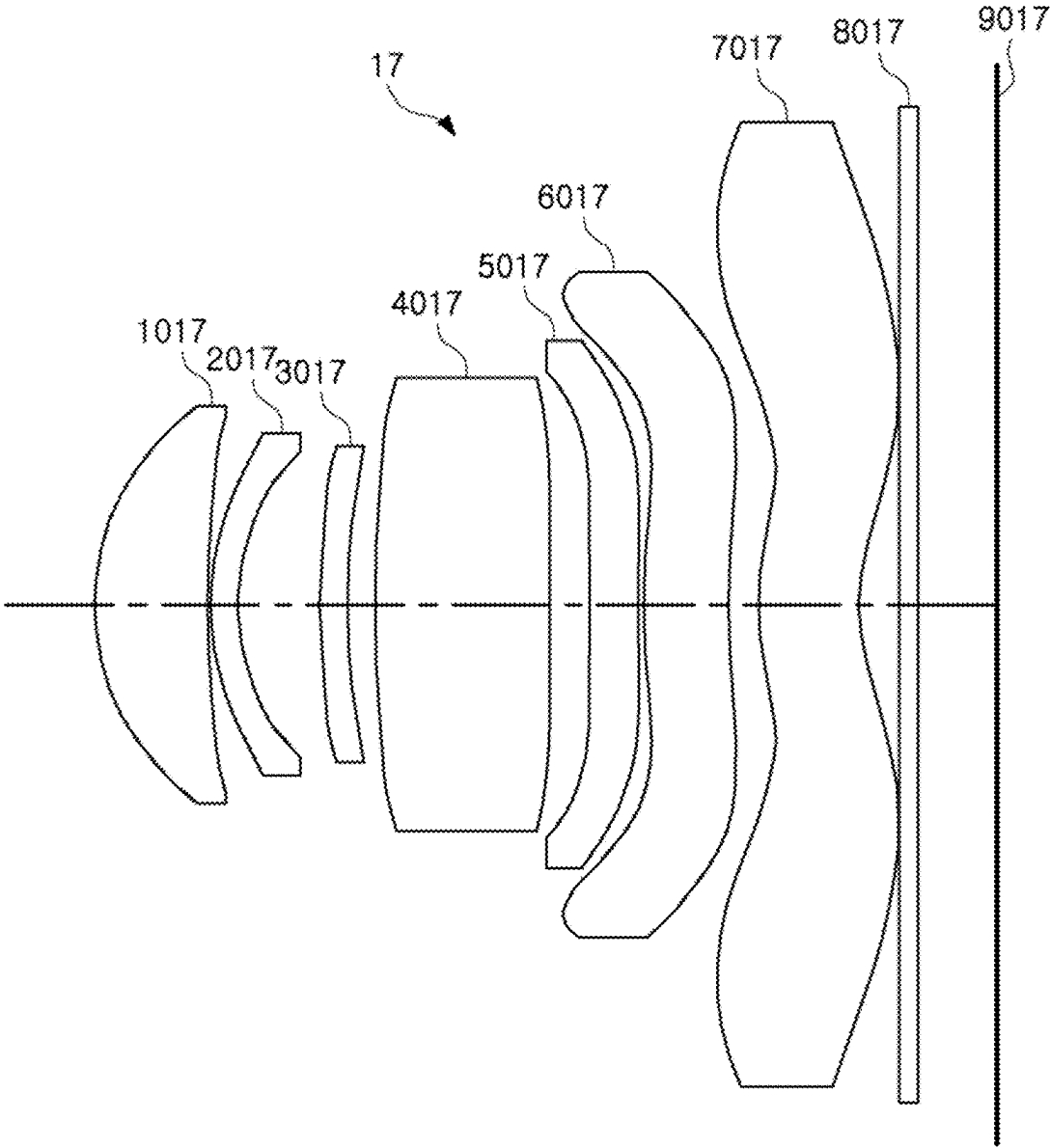


FIG. 33

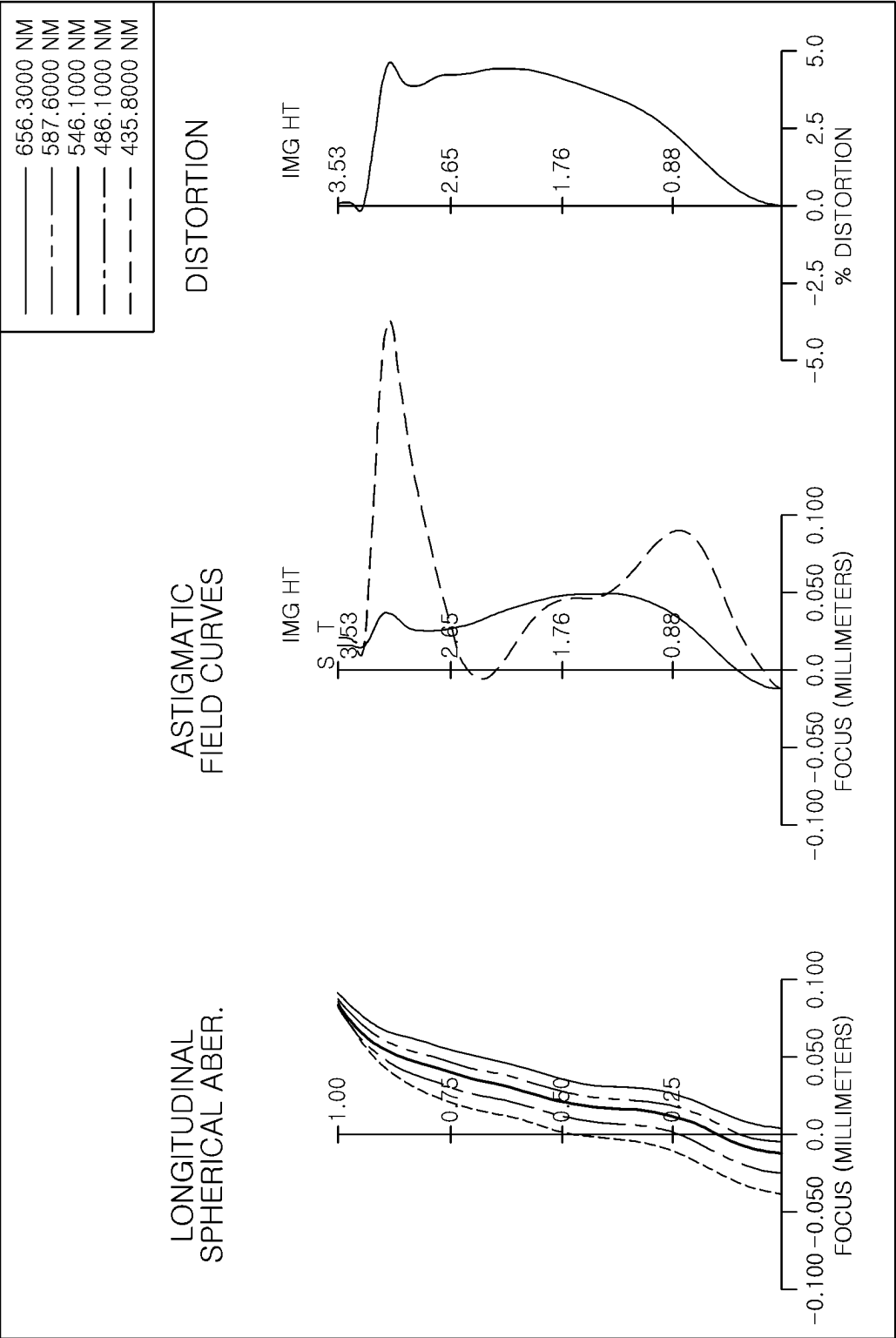


FIG. 34

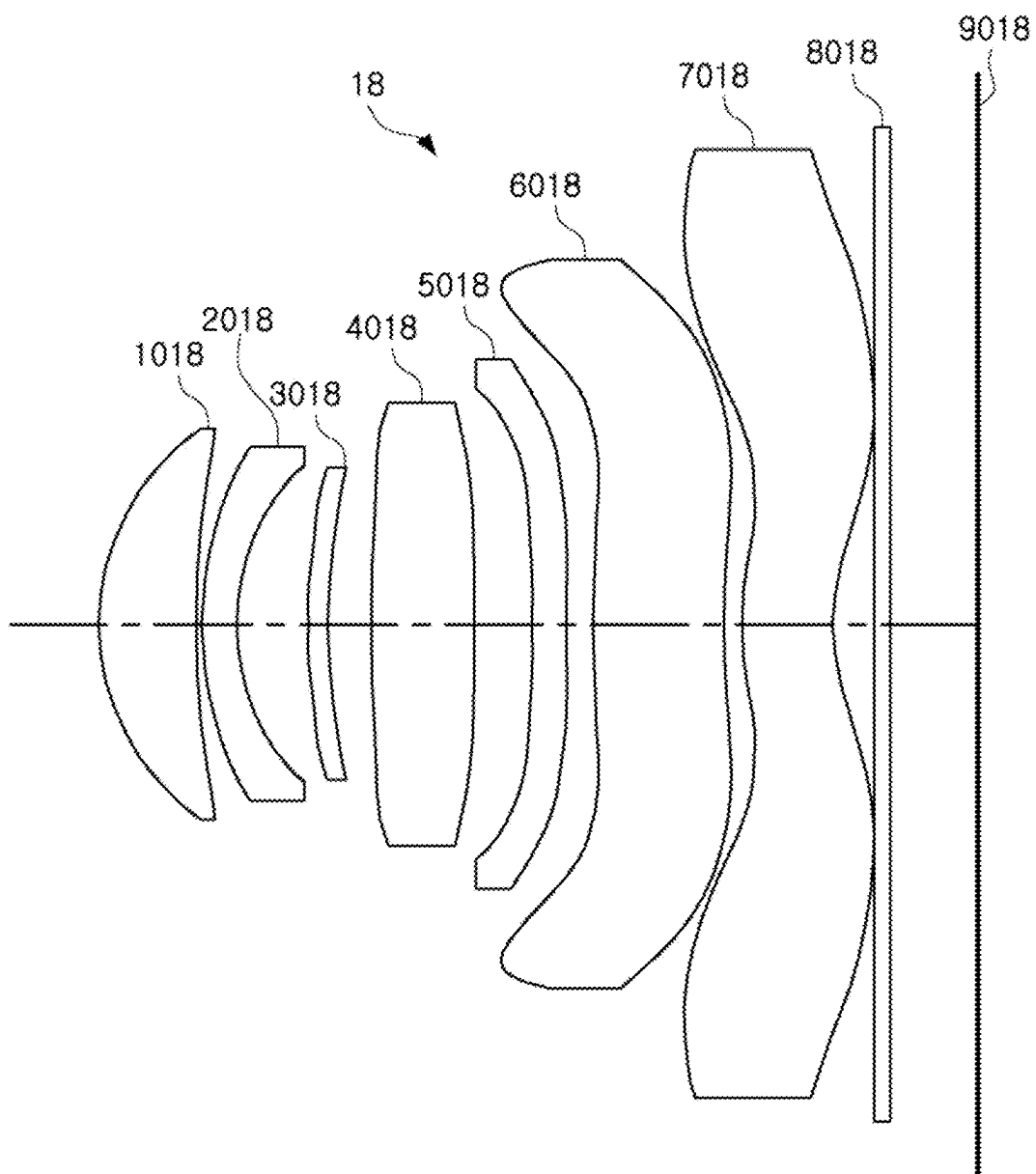


FIG. 35

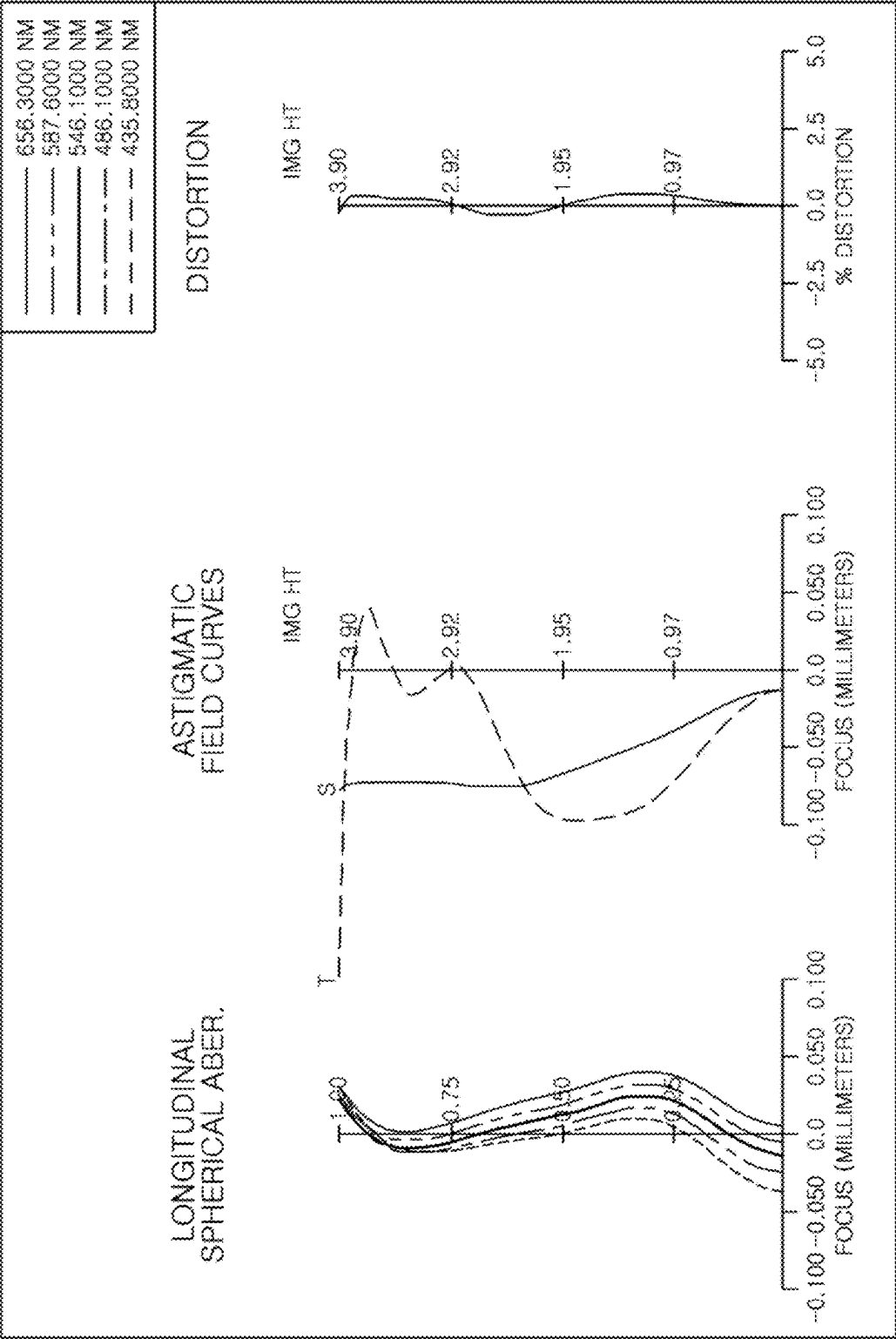


FIG. 36

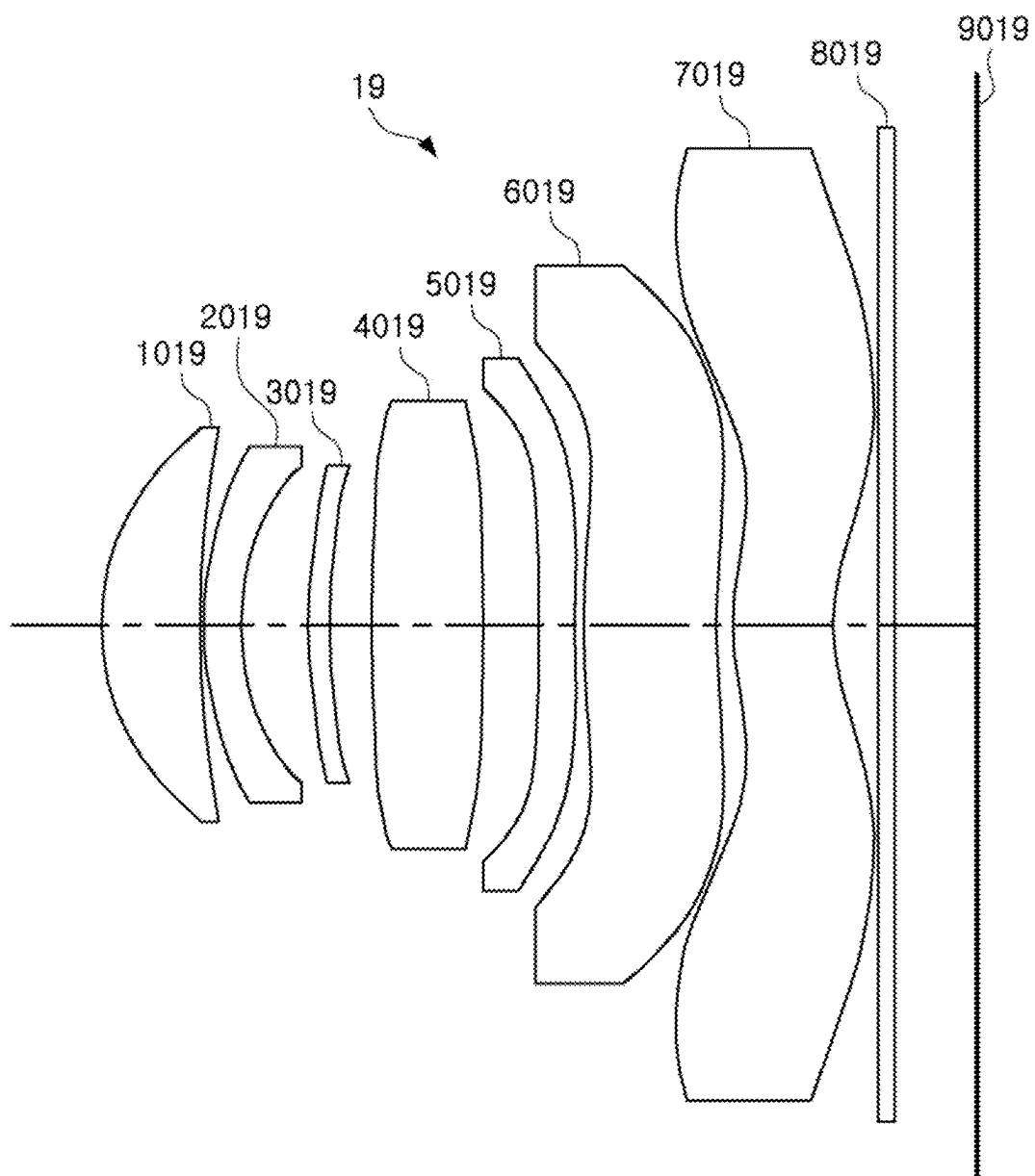


FIG. 37

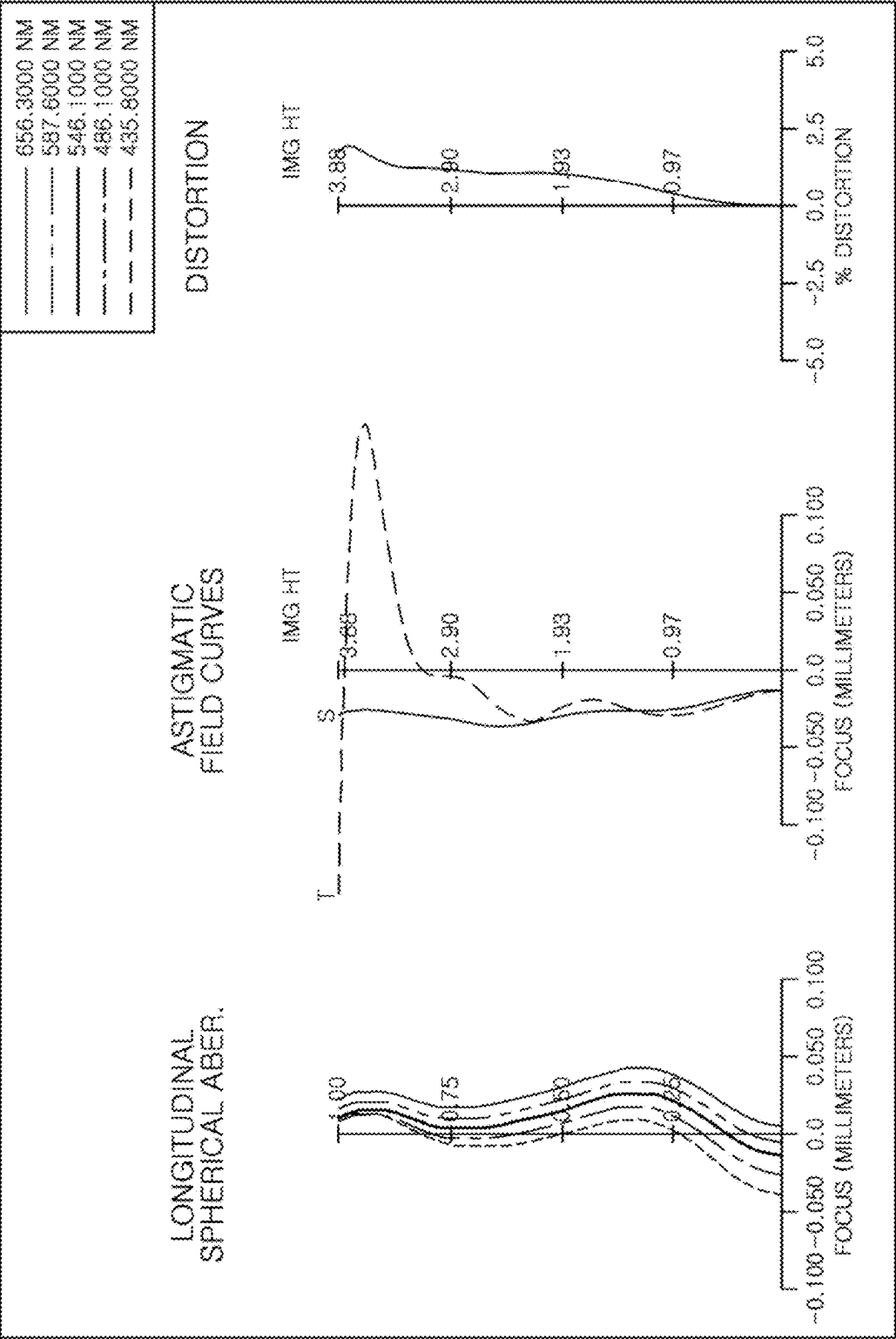


FIG. 38

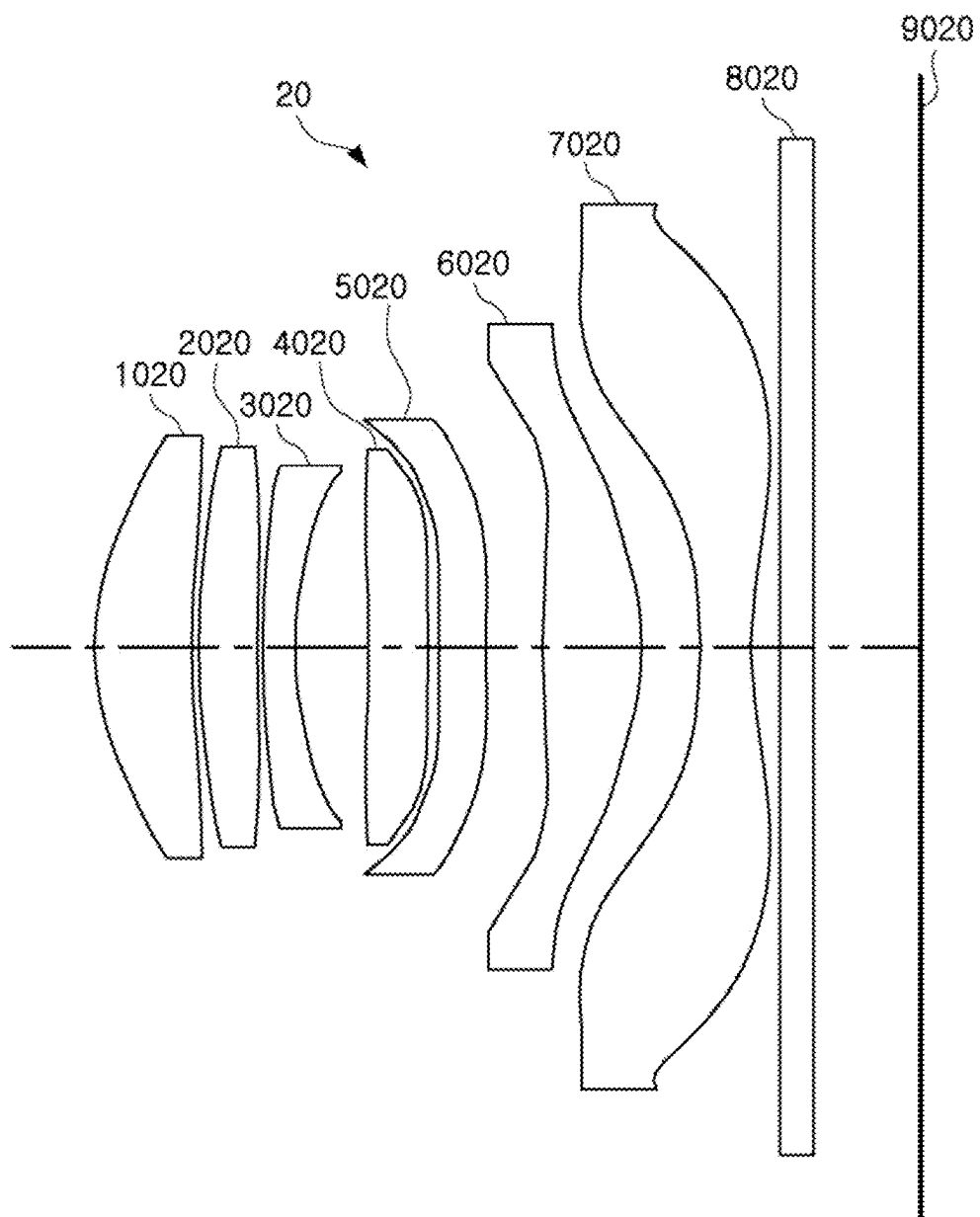


FIG. 39

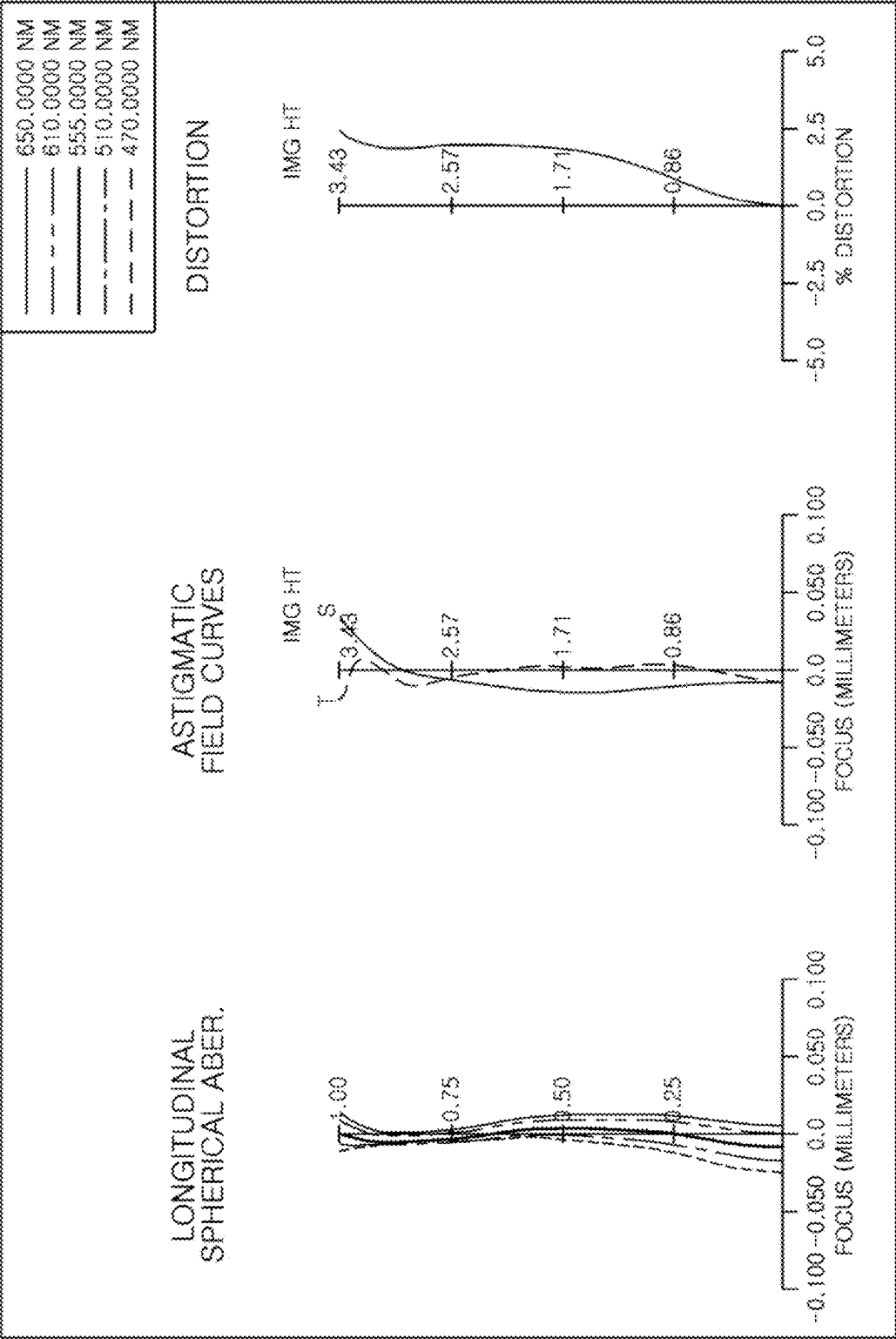


FIG. 40

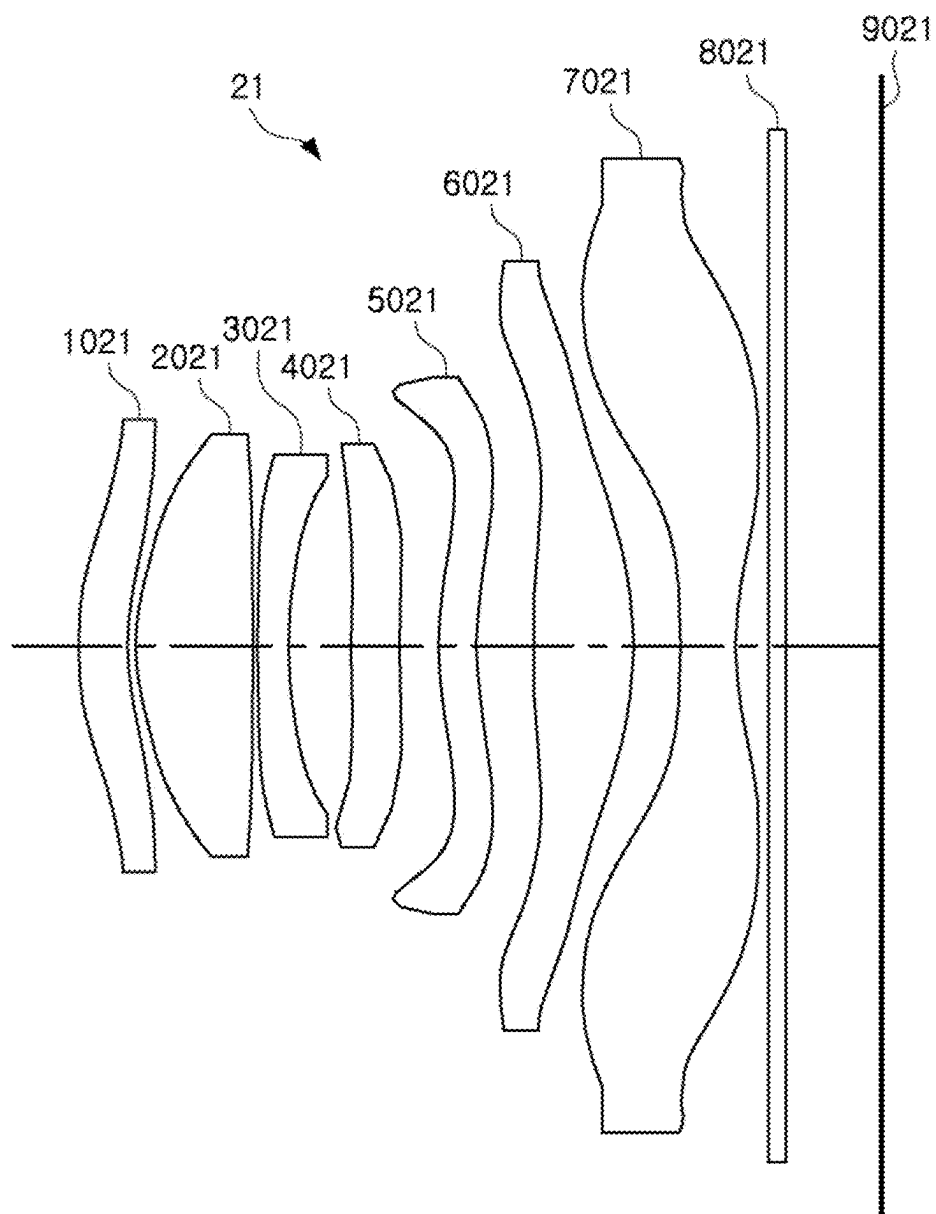


FIG. 41

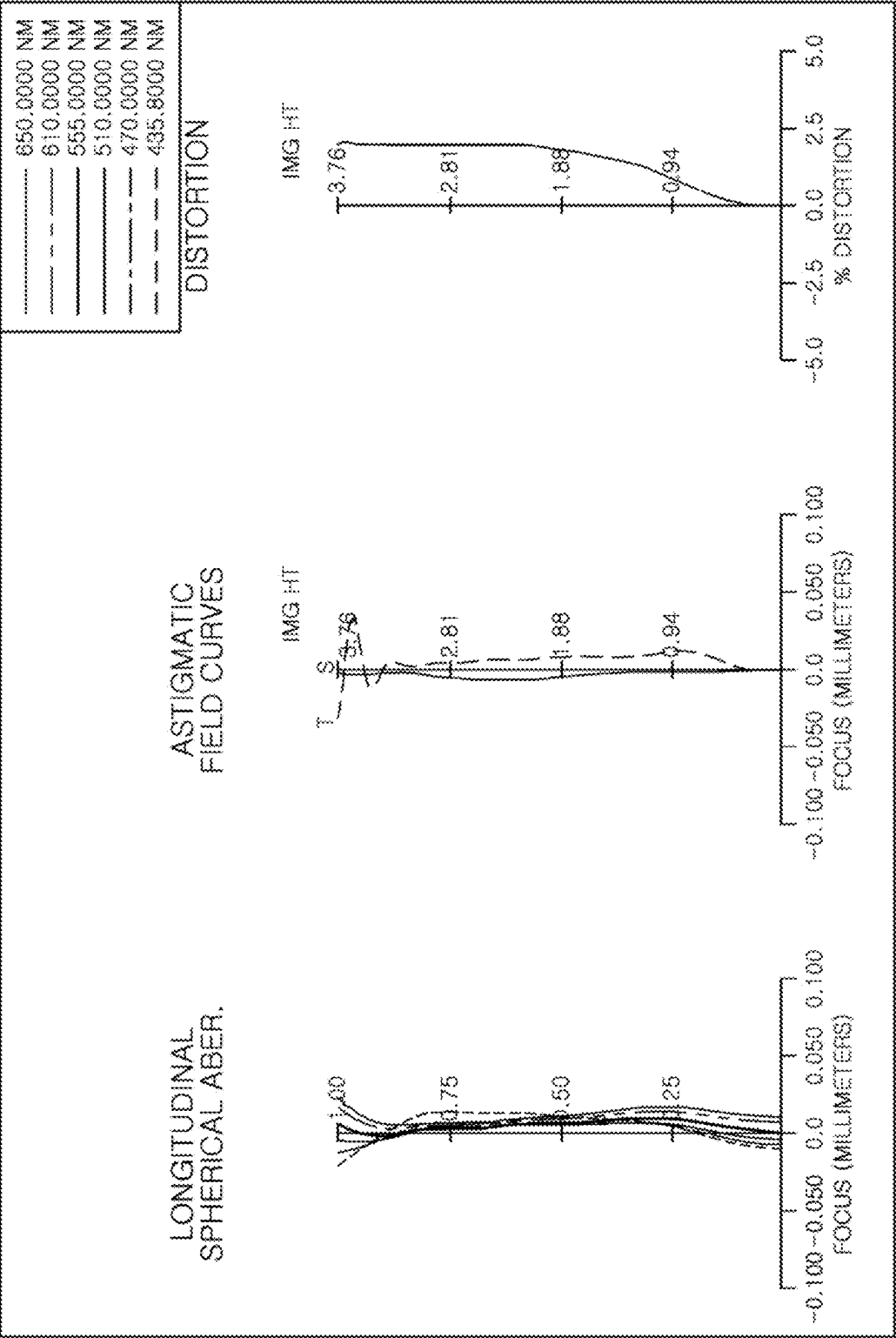


FIG. 42

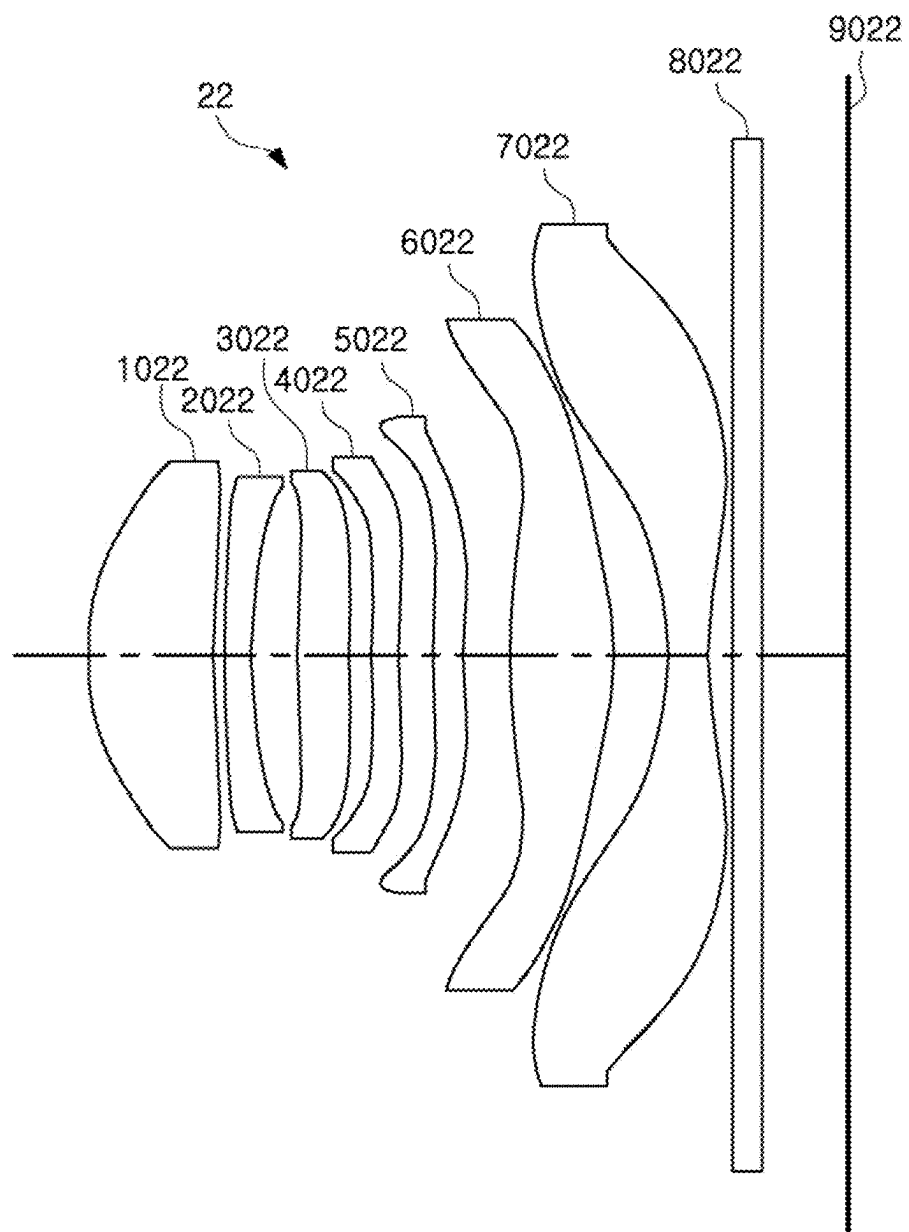


FIG. 43

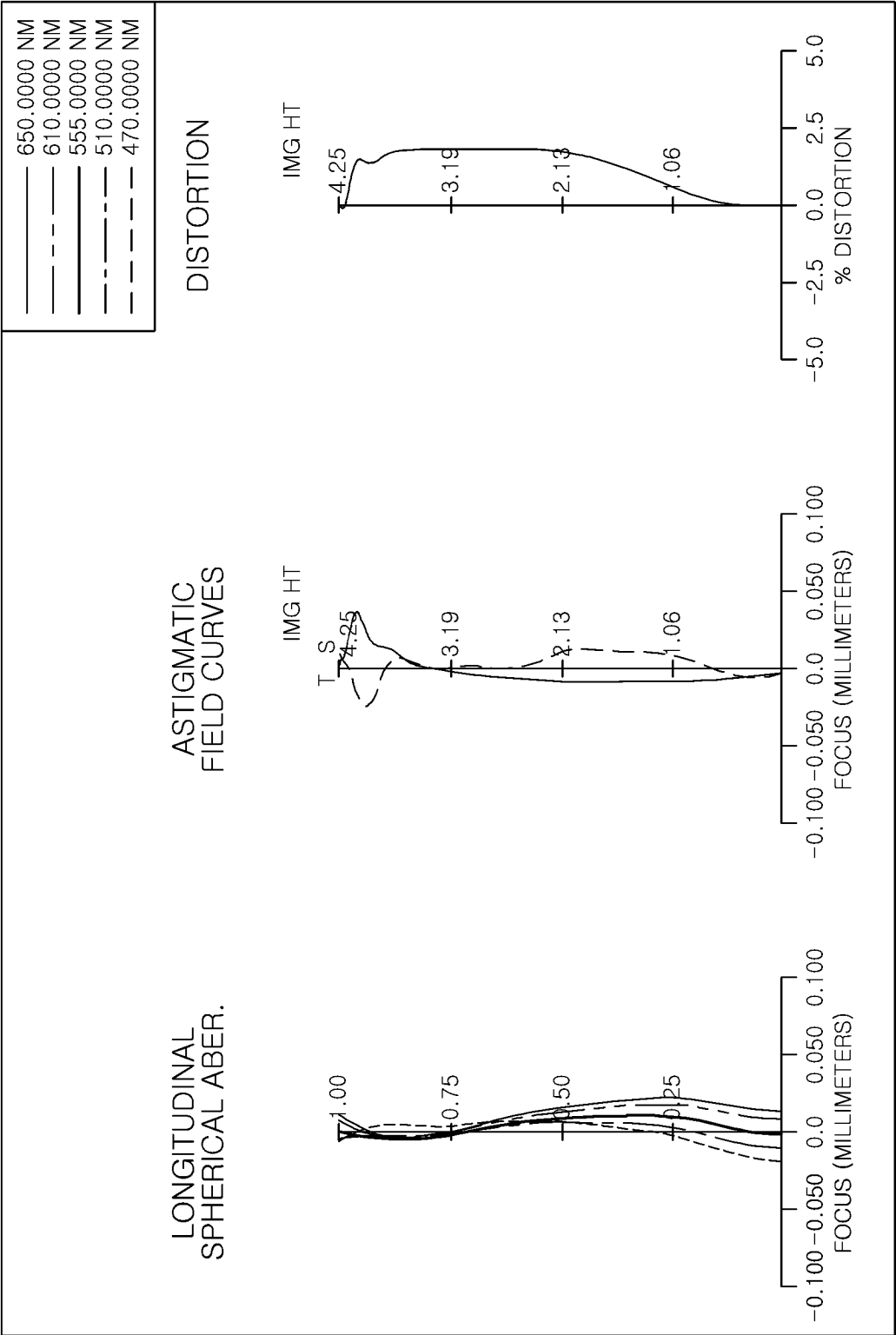


FIG. 44

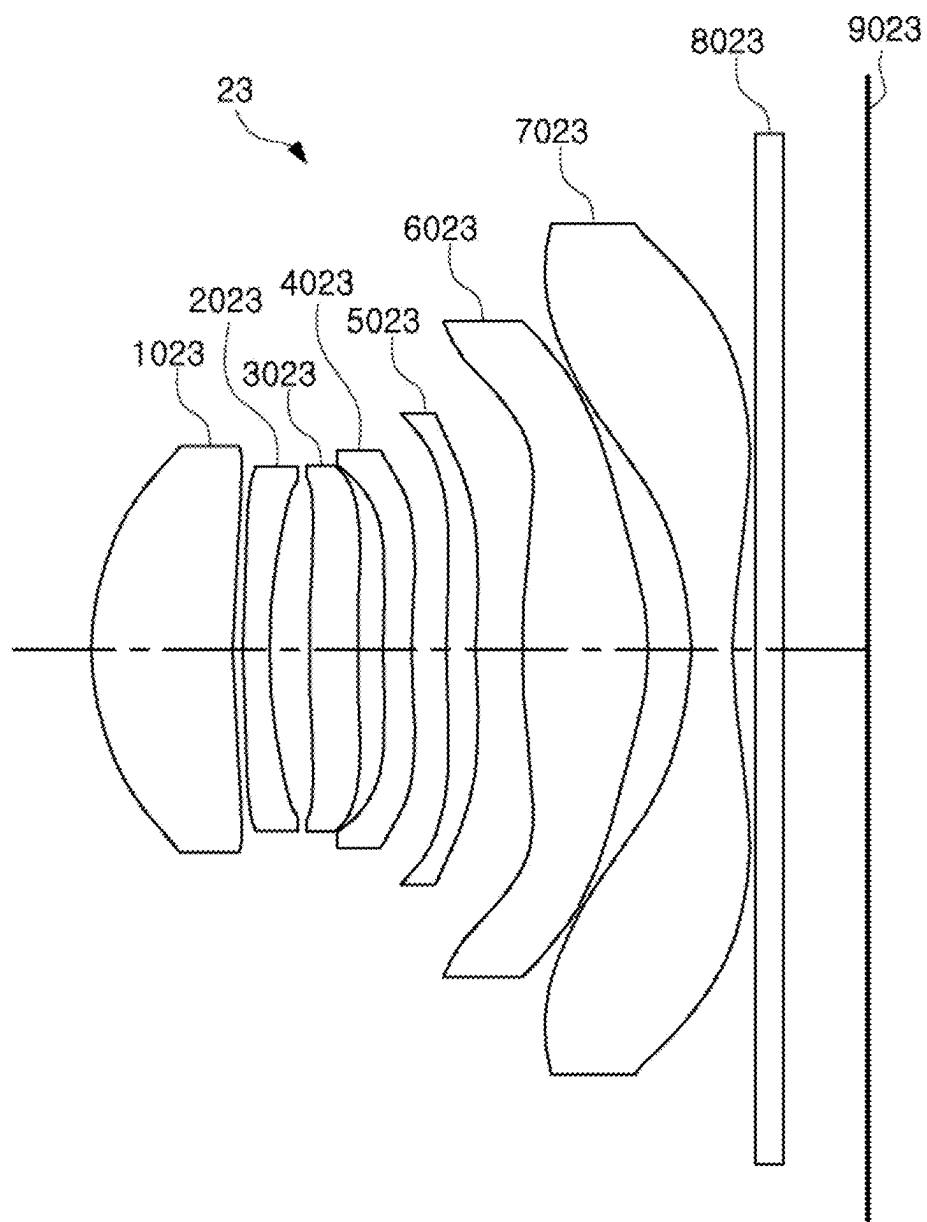


FIG. 45

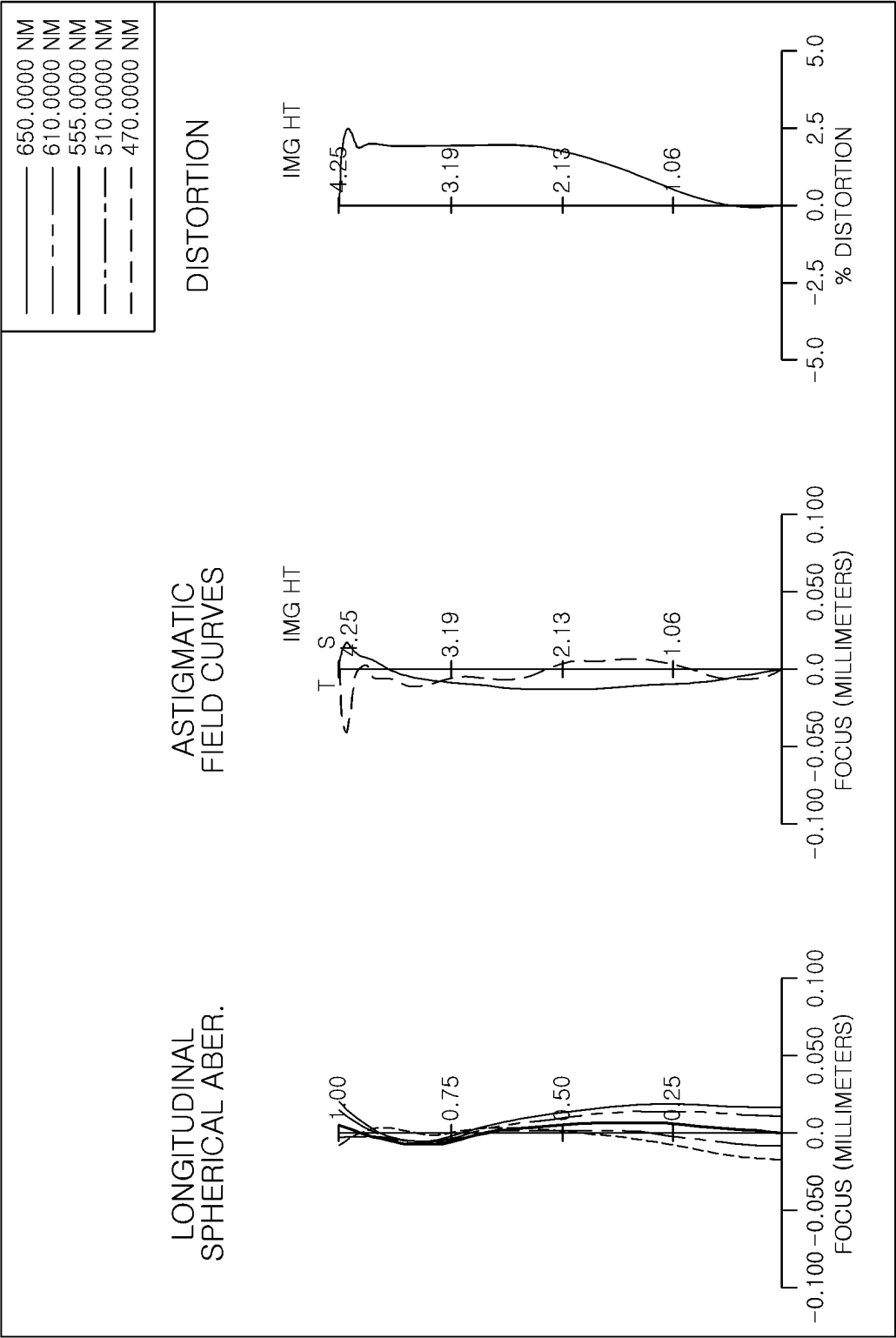


FIG. 46

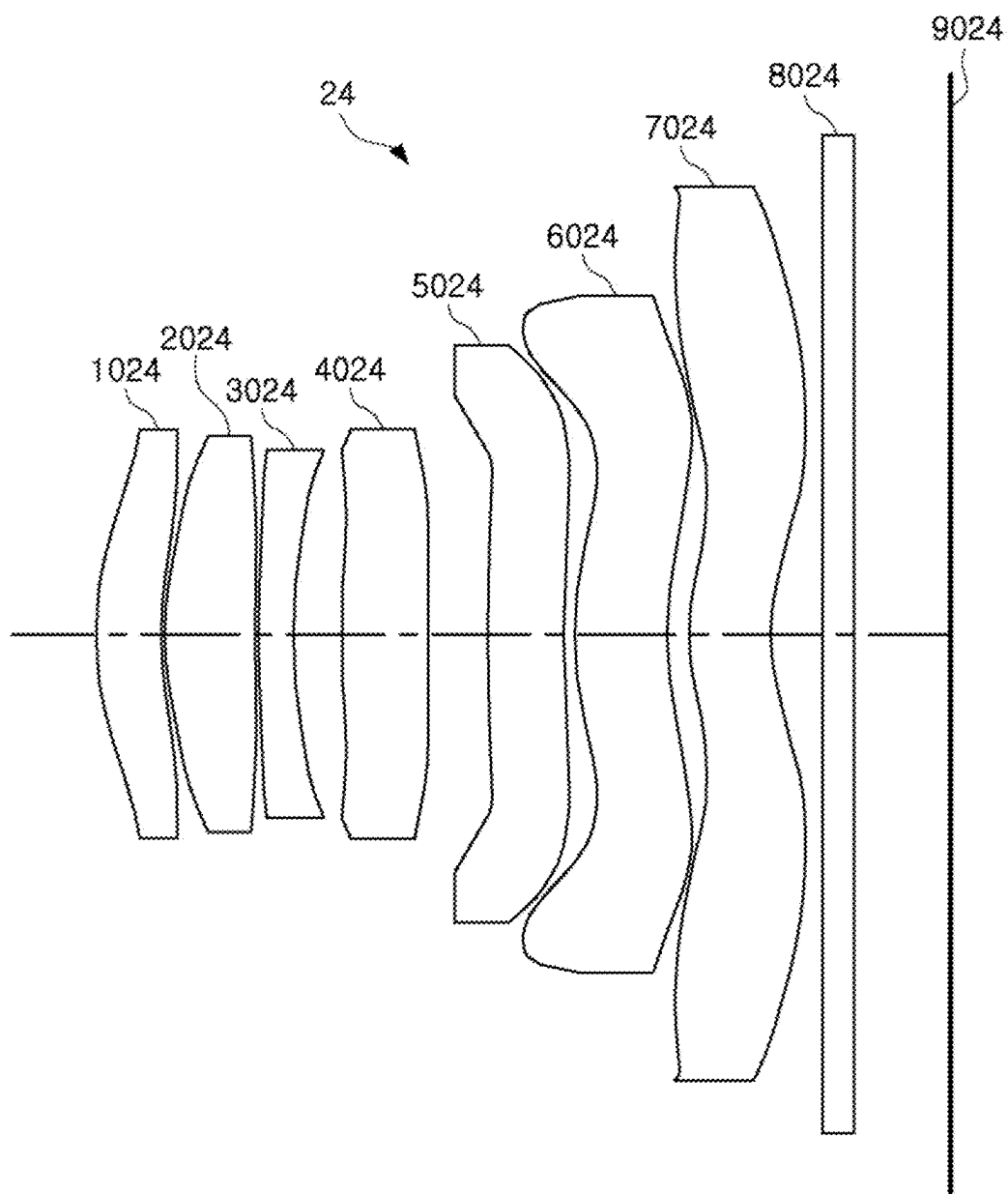


FIG. 47

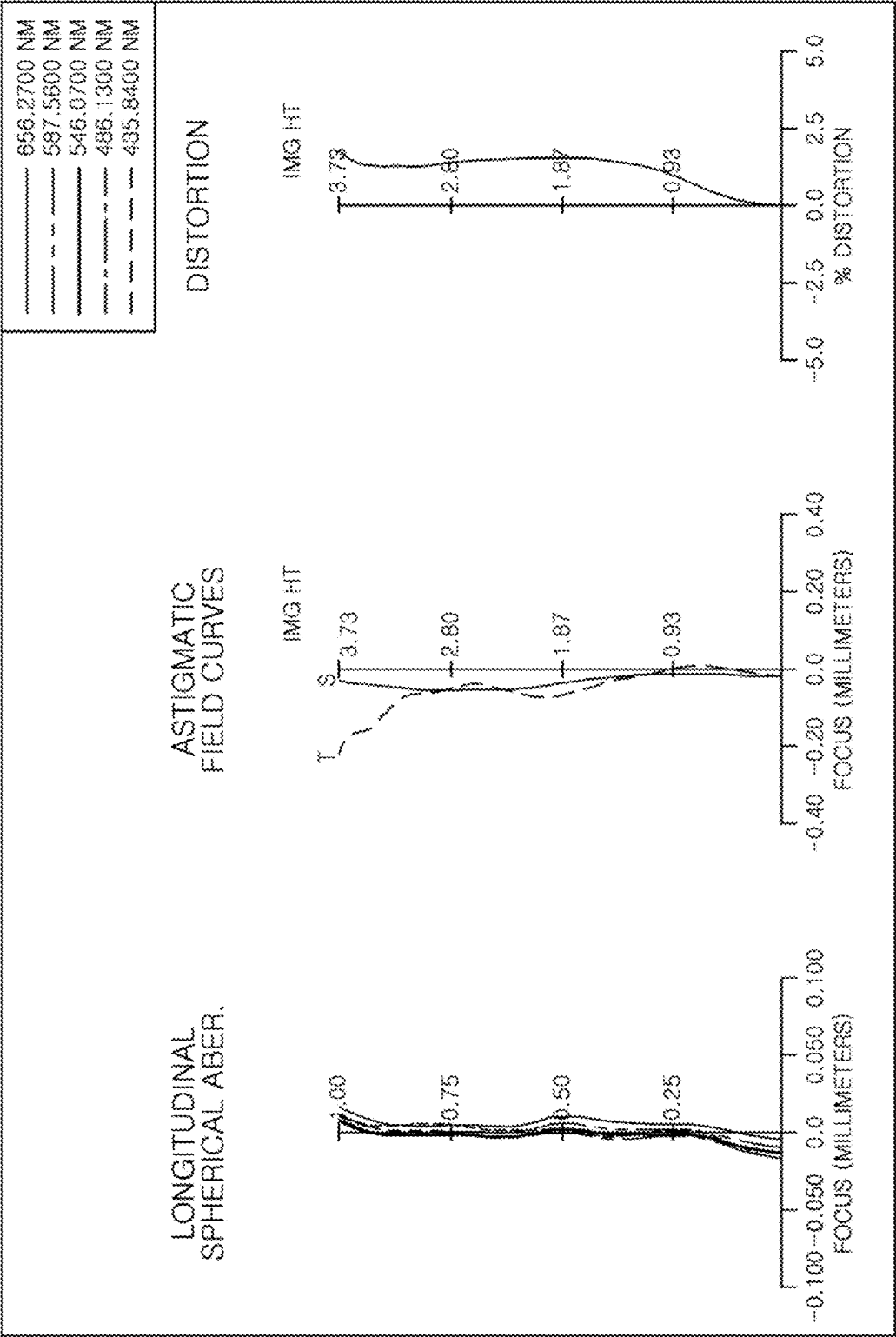


FIG. 48

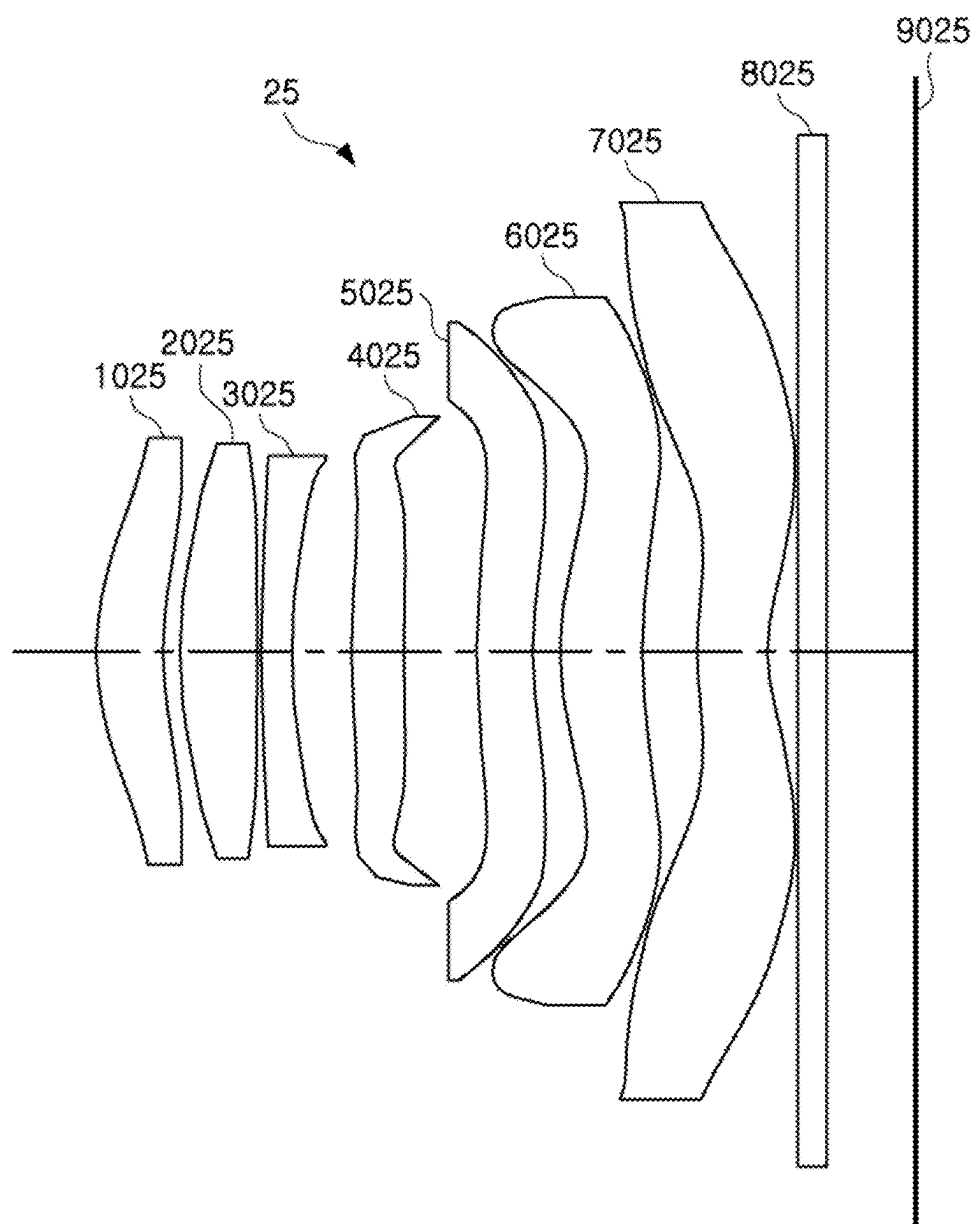


FIG. 49

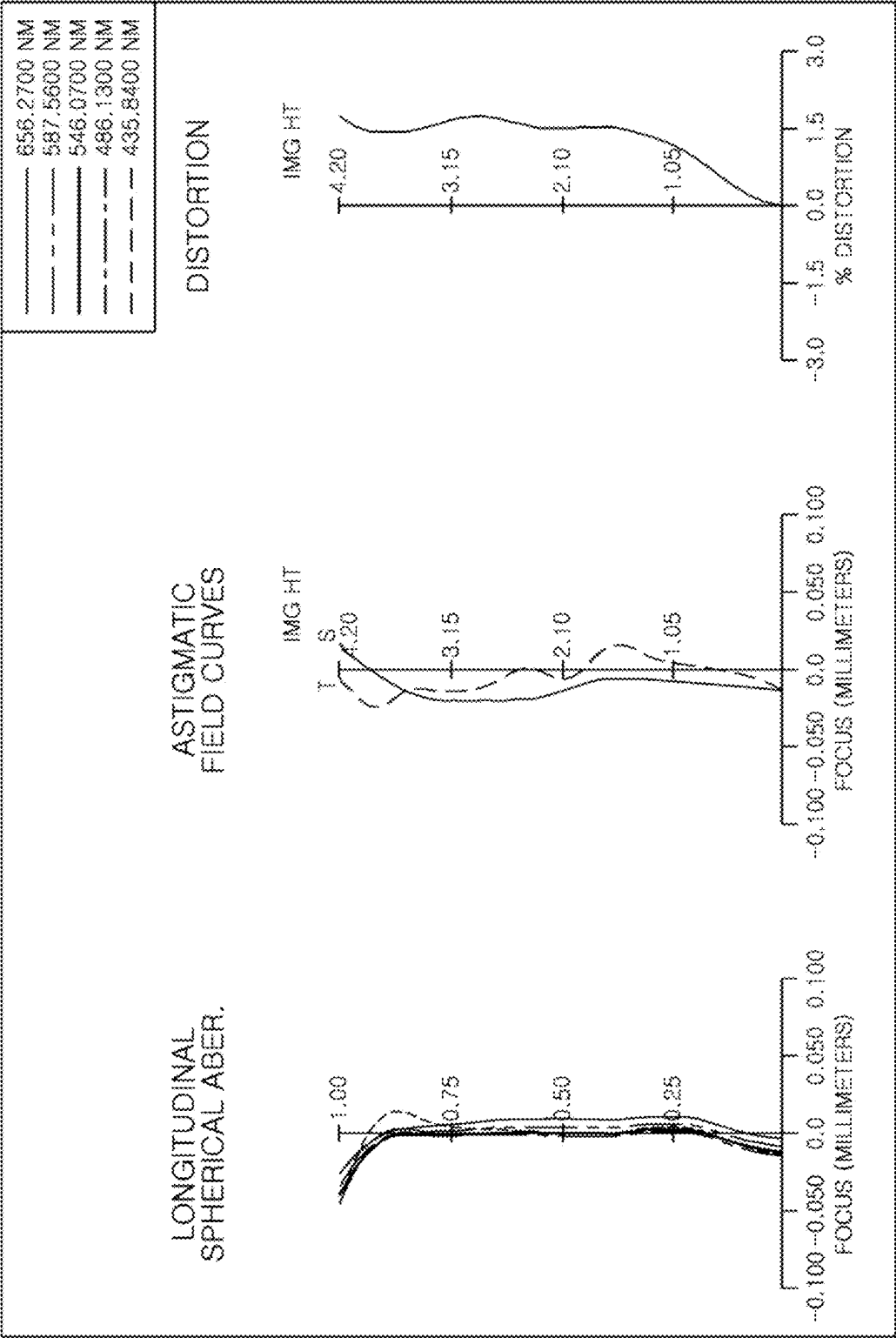


FIG. 50

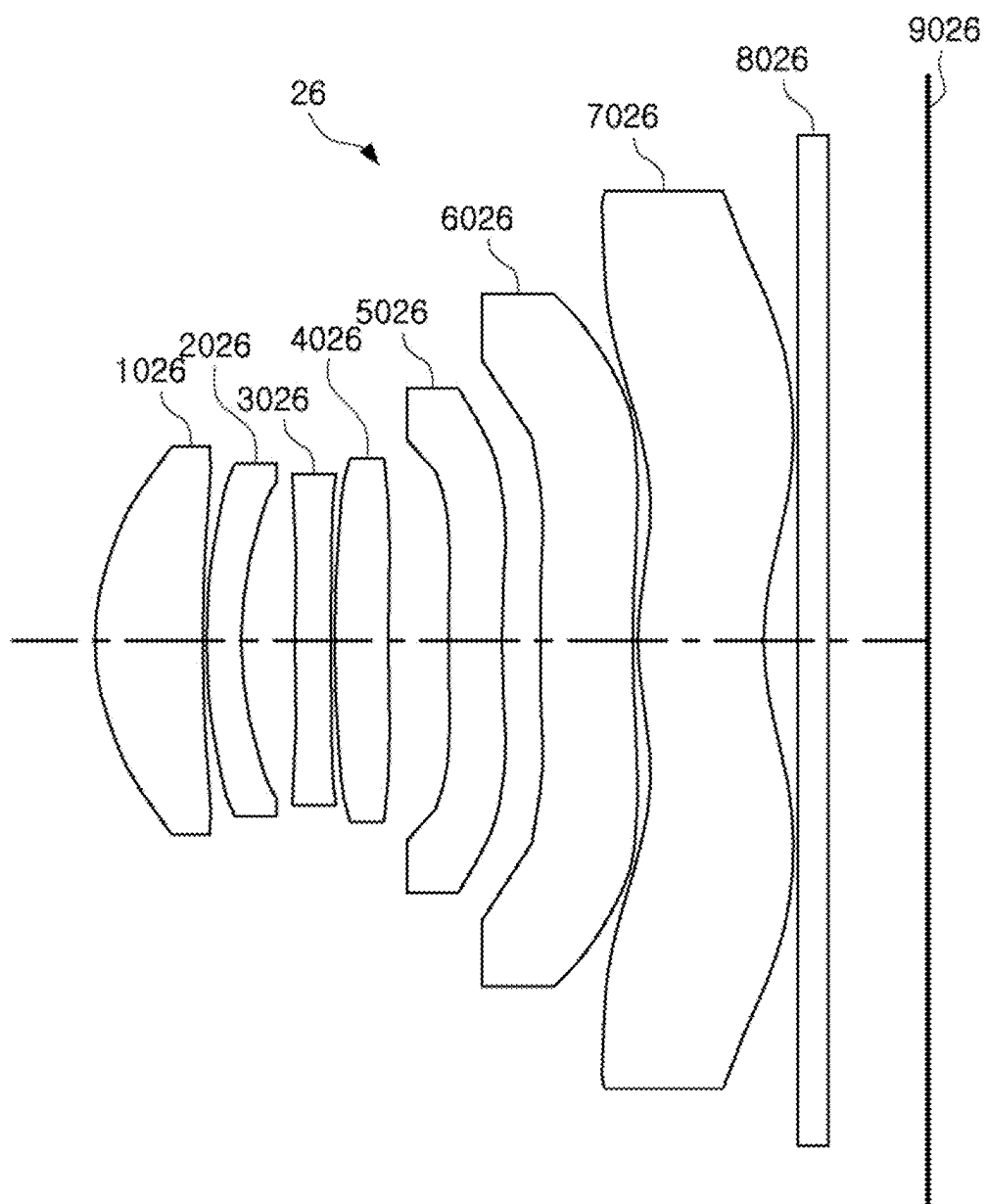


FIG. 51

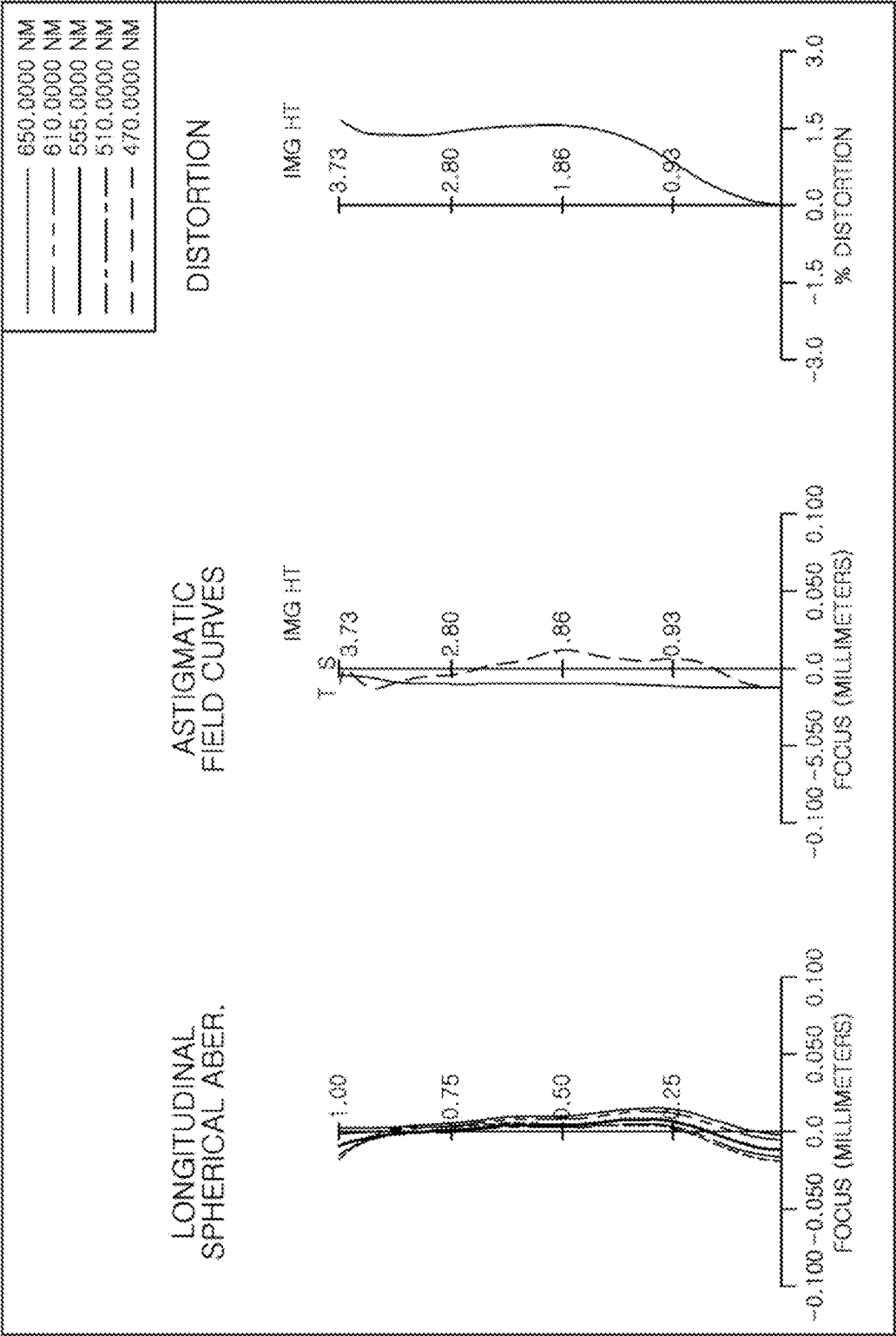


FIG. 52

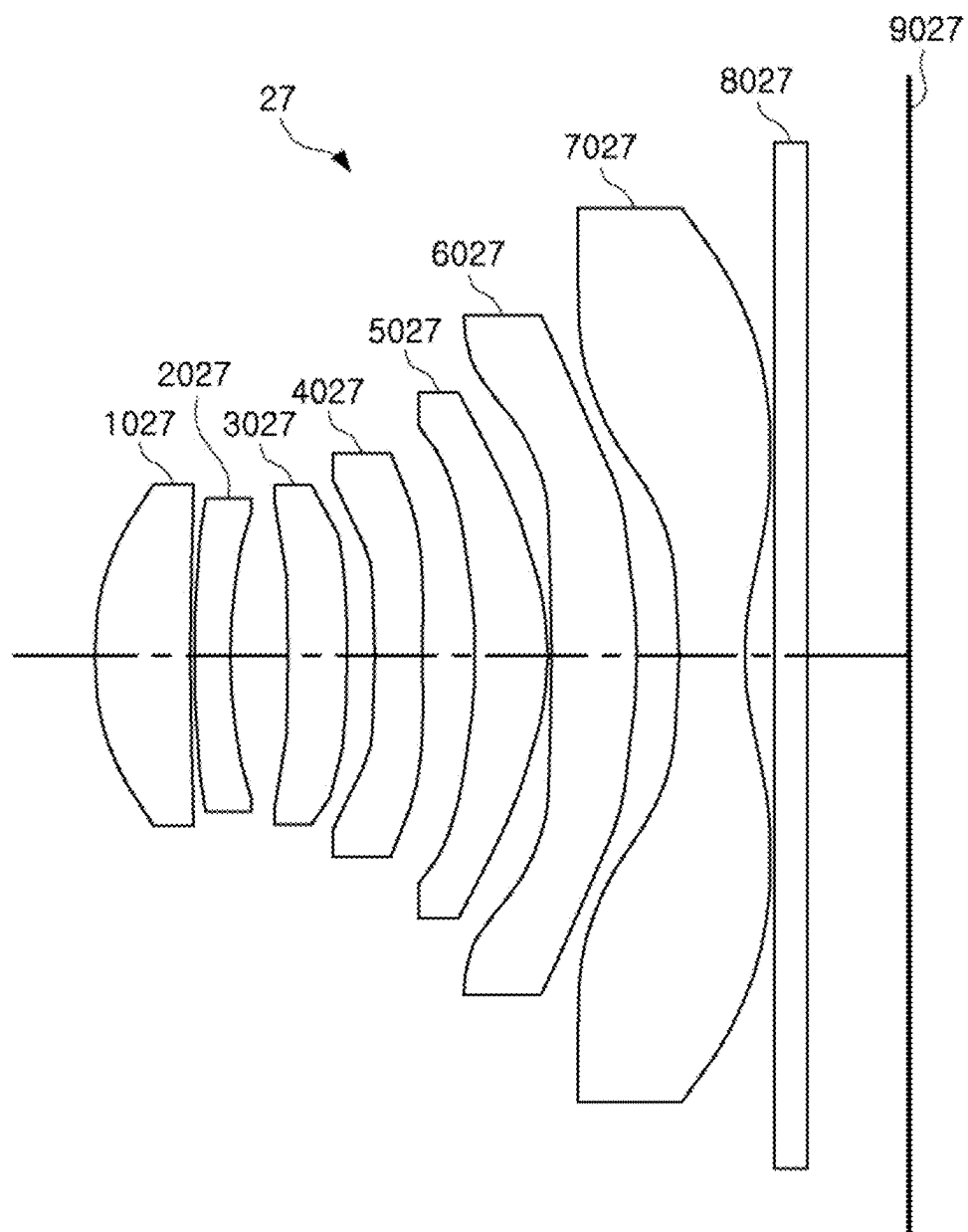


FIG. 53

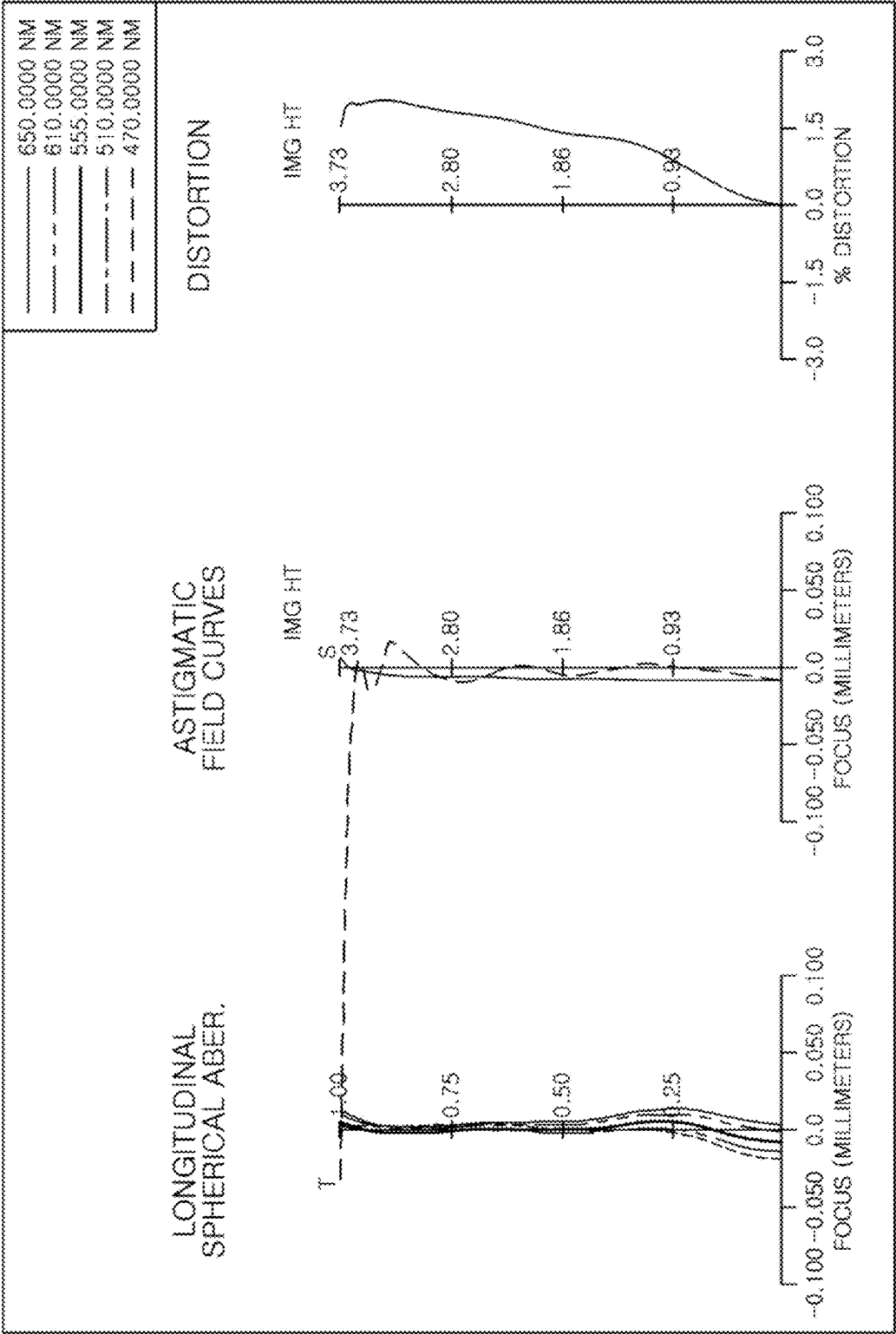


FIG. 54

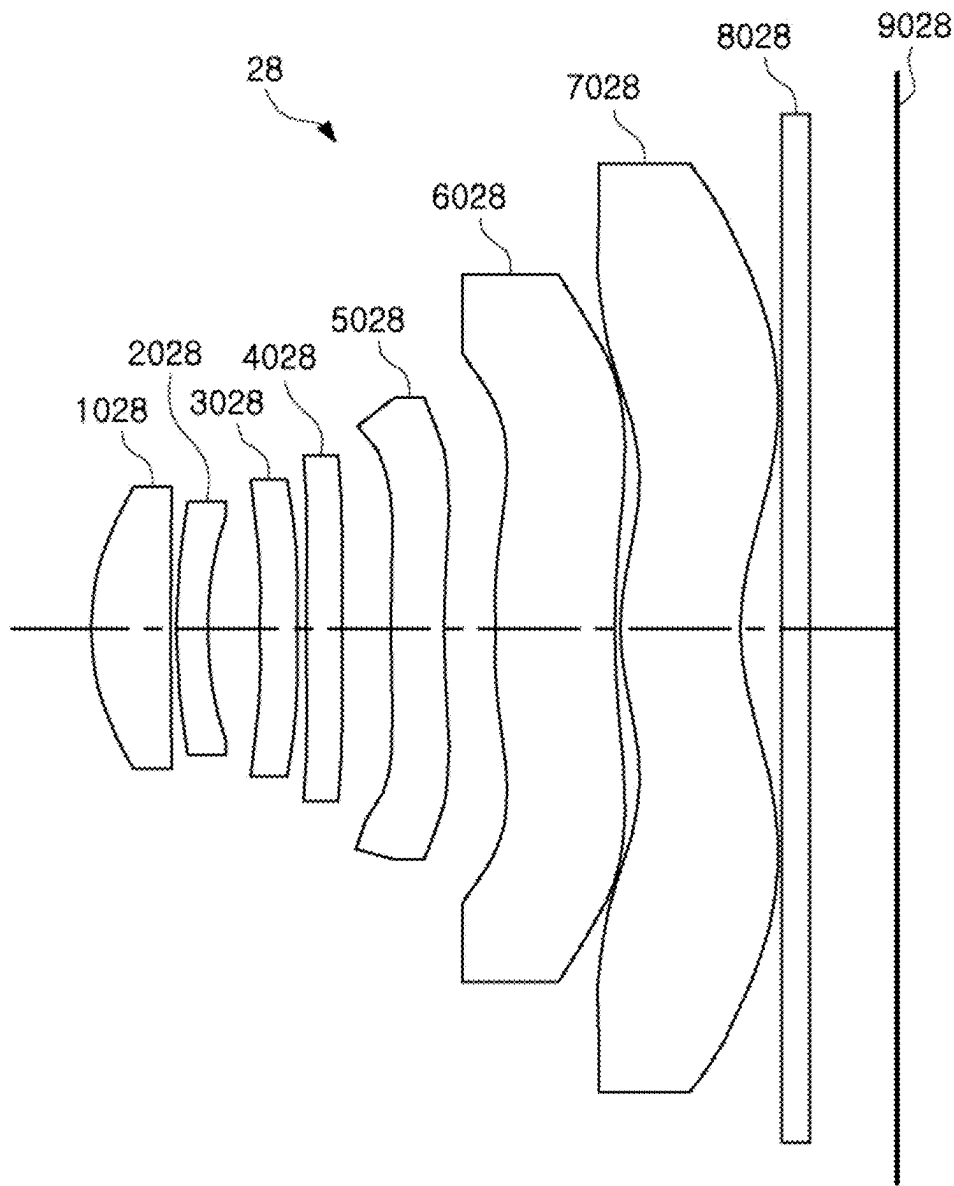


FIG. 55

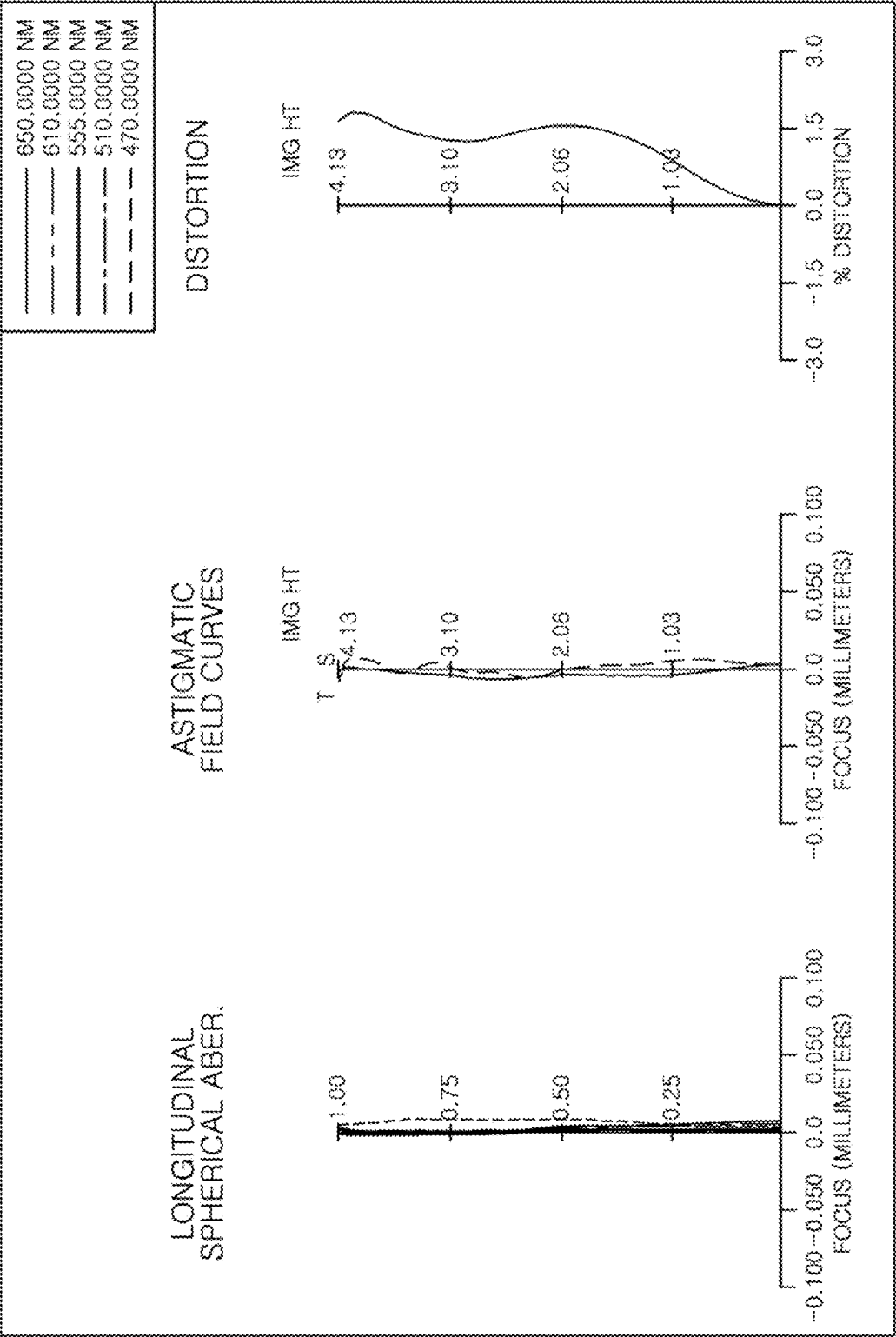


FIG. 56

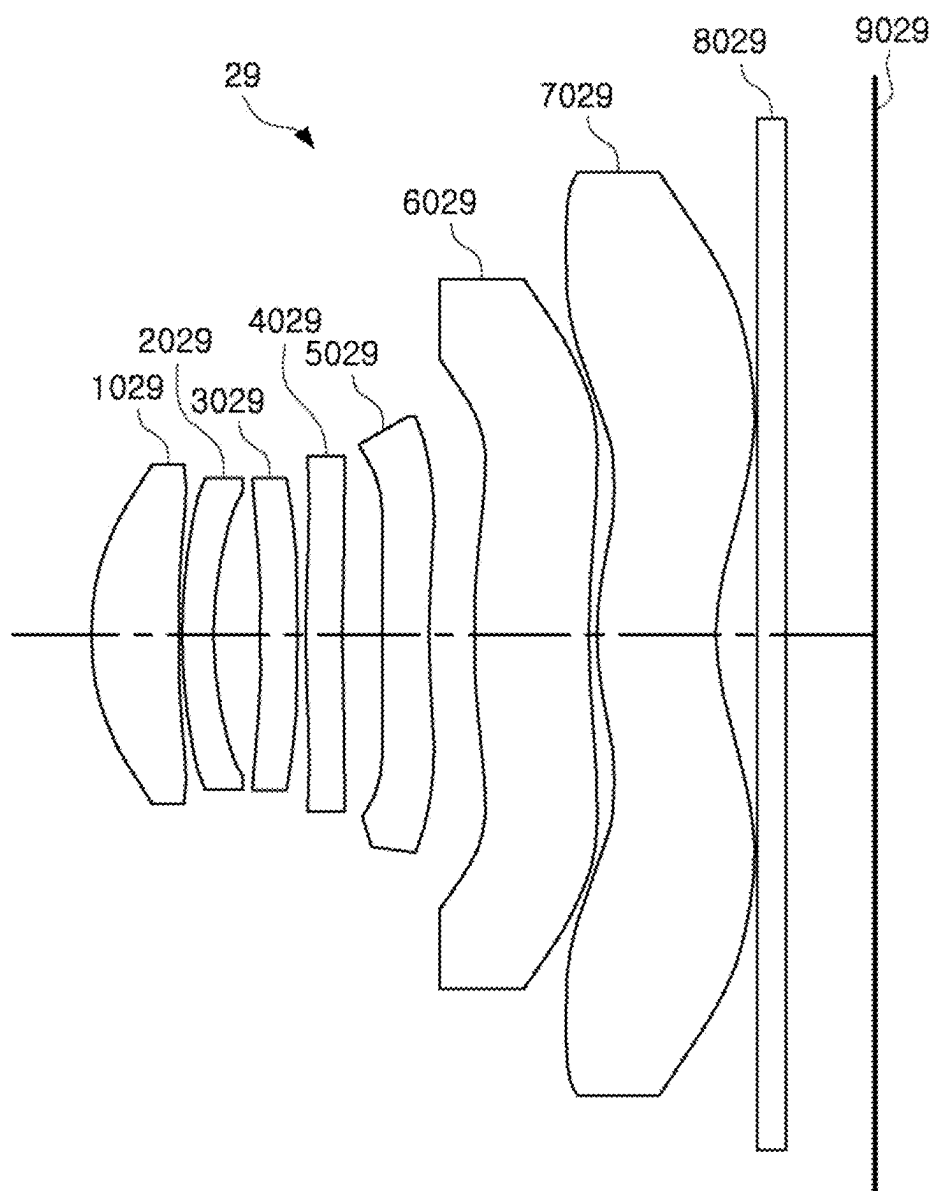


FIG. 57

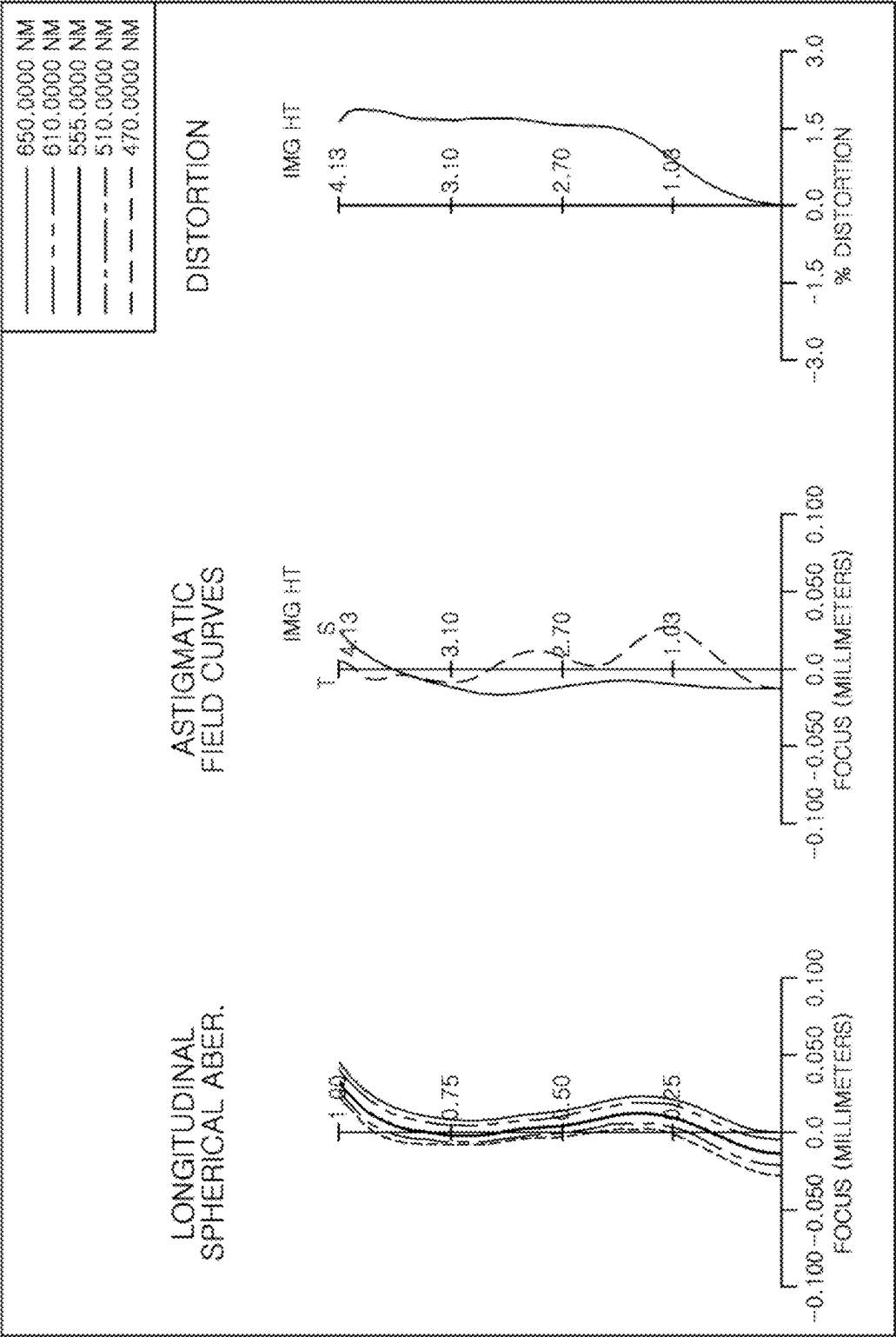


FIG. 58

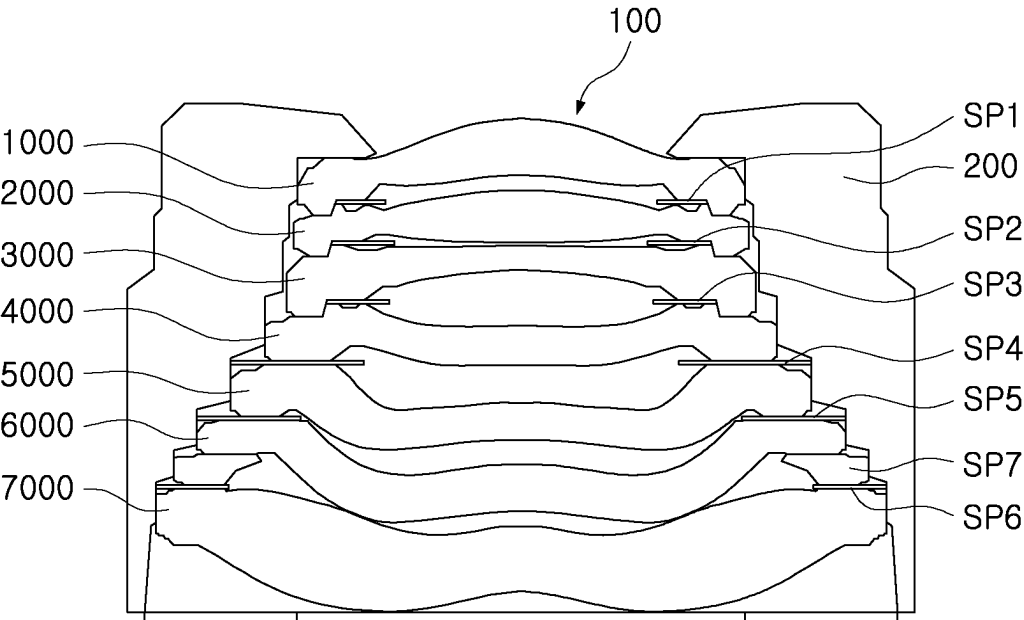


FIG. 59

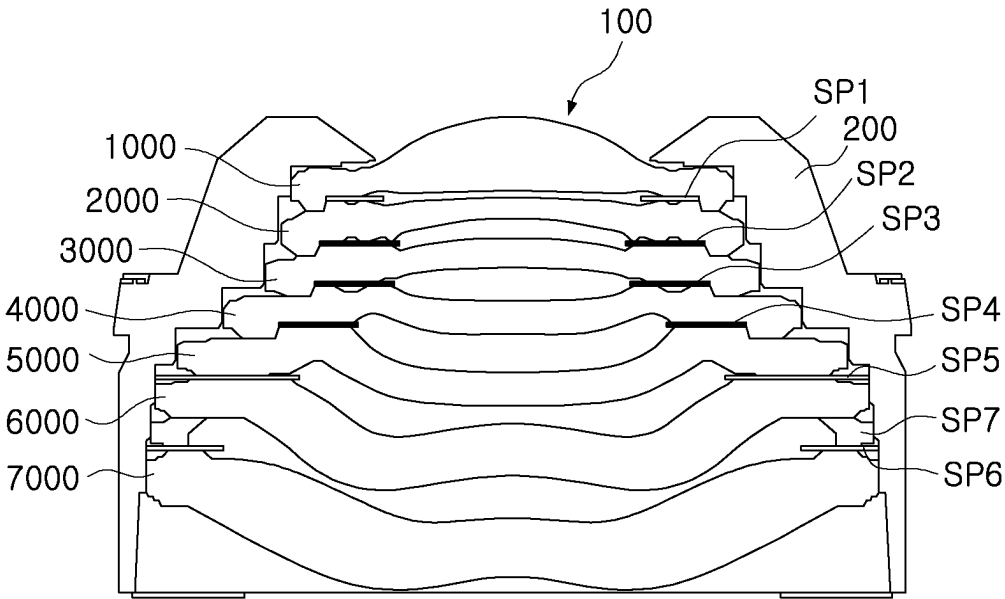


FIG. 60

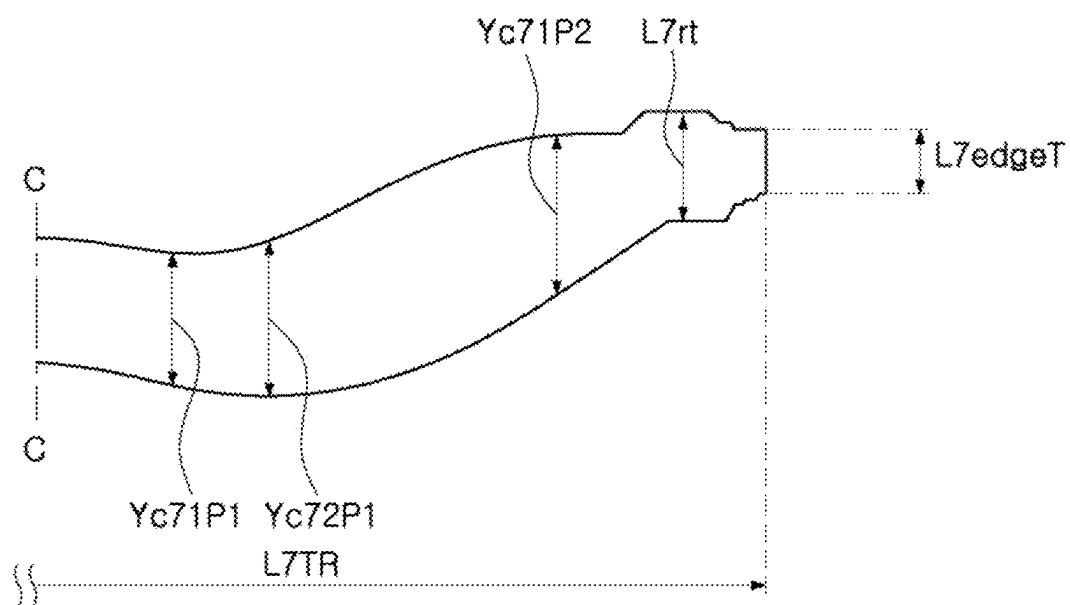


FIG. 61

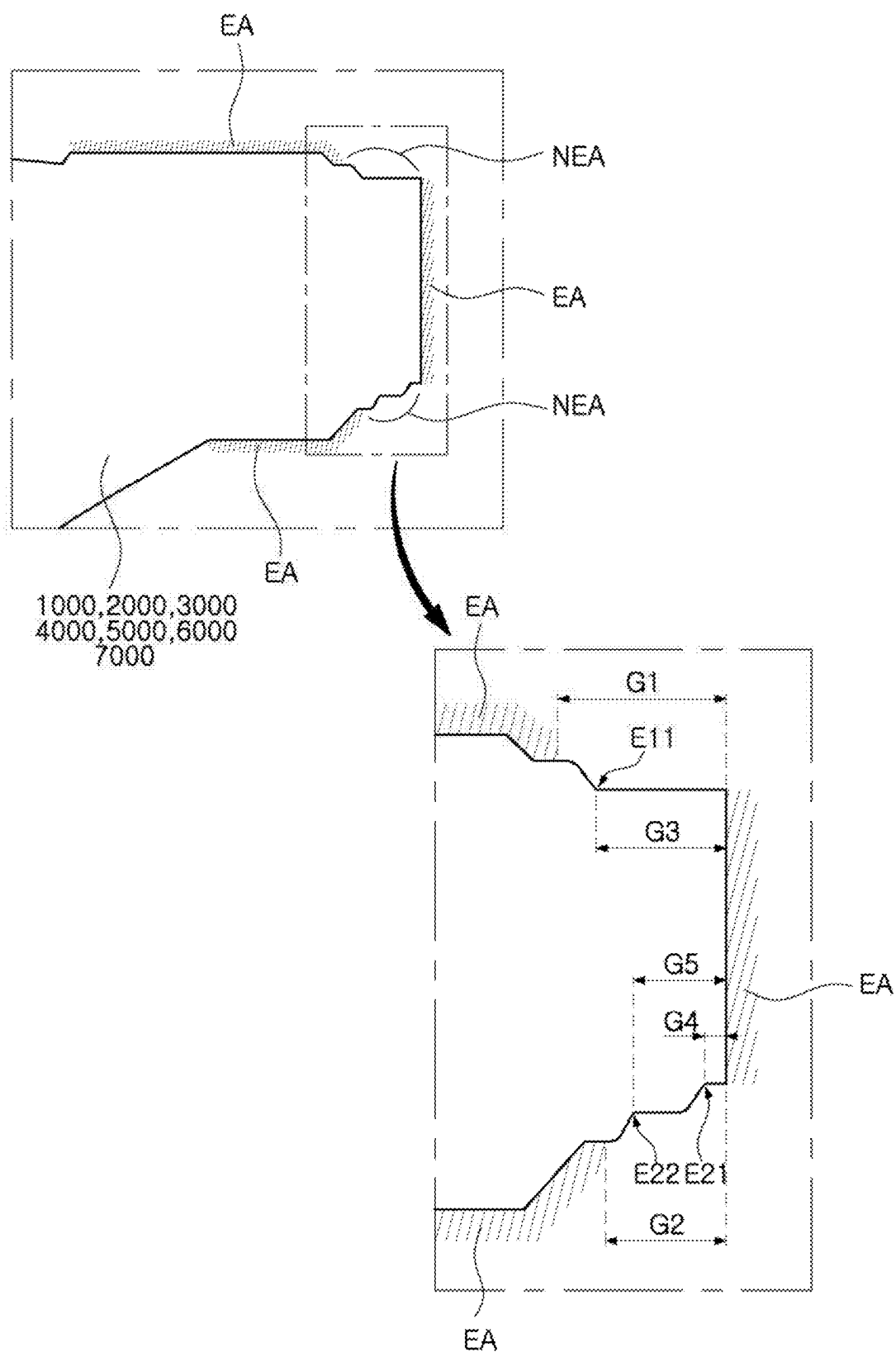


FIG. 62

OPTICAL IMAGING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/670,871 filed on May 22, 2024, which is a continuation of U.S. patent application Ser. No. 18/330,645 filed on Jun. 7, 2023, now U.S. Pat. No. 12,032,226, which is a continuation of U.S. patent application Ser. No. 17/088,665 filed on Nov. 4, 2020, now U.S. Pat. No. 11,714,263, which is a continuation of U.S. patent application Ser. No. 16/424,842 filed on May 29, 2019, now U.S. Pat. No. 11,016,271, which claims the benefit under 35 USC 119(a) of Korean Patent Application Nos. 10-2018-0061397 filed on May 29, 2018, and 10-2018-0106184 filed on Sep. 5, 2018, in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

[0002] This application relates to an optical imaging system including seven lenses.

2. Description of Related Art

[0003] A mobile terminal is commonly provided with a camera for video communications or capturing images. However, it is difficult to achieve high performance in a camera for a mobile terminal due to space limitations inside the mobile terminal.

[0004] Accordingly, a demand for an optical imaging system capable of improving the performance of the camera without increasing a size of the camera has increased as a number of mobile terminals provided with a camera has increased.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In one general aspect, an optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $0.5 < L12345TR_{avg}/L7TR < 0.9$, where $L12345TR_{avg}$ is an average value of overall outer diameters of the first to fifth lenses, $L7TR$ is an overall outer diameter of the seventh lens, and $L12345TR_{avg}$ and $L7TR$ are expressed in a same unit of measurement.

[0007] An object-side surface of the first lens may be convex.

[0008] An image-side surface of the first lens may be concave.

[0009] An image-side surface of the seventh lens may be concave.

[0010] A distance along the optical axis from an object-side surface of the first lens to the imaging plane may be 6.0 mm or less.

[0011] At least one inflection point may be formed on either one or both of an object-side surface and an image-side surface of the sixth lens.

[0012] At least one inflection point may be formed on either one or both of an object-side surface and an image-side surface of the seventh lens.

[0013] The optical imaging system further may satisfy $0.1 < L1w/L7w < 0.3$, where $L1w$ is a weight of the first lens, $L7w$ is a weight of the seventh lens, and $L1w$ and $L7w$ are expressed in a same unit of measurement.

[0014] The optical imaging system may further include a spacer disposed between the sixth and seventh lenses, wherein the optical imaging system further may satisfy $0.5 < S6d/f < 1.2$, where $S6d$ is an inner diameter of the spacer, f is an overall focal length of the optical imaging system, and $S6d$ and f are expressed in a same unit of measurement.

[0015] The optical imaging system further may satisfy $0.4 < L1TR/L7TR < 0.7$, where $L1TR$ is an overall outer diameter of the first lens, and $L1TR$ and $L7TR$ are expressed in a same unit of measurement.

[0016] The optical imaging system further may satisfy $0.5 < L1234TR_{avg}/L7TR < 0.75$, where $L1234TR_{avg}$ is an average value of overall outer diameters of the first to fourth lenses, and $L1234TR_{avg}$ and $L7TR$ are expressed in a same unit of measurement.

[0017] The optical imaging system further may satisfy $0.5 < L12345TR_{avg}/L7TR < 0.76$.

[0018] The second lens may have a positive refractive power.

[0019] The third lens may have a positive refractive power.

[0020] The fifth lens may have a negative refractive power.

[0021] A paraxial region of an object-side surface of the fifth lens may be concave or convex.

[0022] A paraxial region of an image-side surface of the fifth lens may be concave or convex.

[0023] A paraxial region of an object-side surface of the sixth lens may be concave or convex.

[0024] A paraxial region of an image-side surface of the sixth lens may be concave or convex.

[0025] A paraxial region of an object-side surface of the seventh lens may be concave.

[0026] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a view illustrating a first example of an optical imaging system.

[0028] FIG. 2 illustrates aberration curves of the optical imaging system of FIG. 1.

[0029] FIG. 3 is a view illustrating a second example of an optical imaging system.

[0030] FIG. 4 illustrates aberration curves of the optical imaging system of FIG. 3.

[0031] FIG. 5 is a view illustrating a third example of an optical imaging system.

[0032] FIG. 6 illustrates aberration curves of the optical imaging system of FIG. 5.

[0033] FIG. 7 is a view illustrating a fourth example of an optical imaging system.

[0034] FIG. 8 illustrates aberration curves of the optical imaging system of FIG. 7.

[0035] FIG. 9 is a view illustrating a fifth example of an optical imaging system.

[0036] FIG. 10 illustrates aberration curves of the optical imaging system of FIG. 9.

[0037] FIG. 11 is a view illustrating a sixth example of an optical imaging system.

[0038] FIG. 12 illustrates aberration curves of the optical imaging system of FIG. 11.

[0039] FIG. 13 is a view illustrating a seventh example of an optical imaging system.

[0040] FIG. 14 illustrates aberration curves of the optical imaging system of FIG. 13.

[0041] FIG. 15 is a view illustrating an eighth example of an optical imaging system.

[0042] FIG. 16 illustrates aberration curves of the optical imaging system of FIG. 15.

[0043] FIG. 17 is a view illustrating a ninth example of an optical imaging system.

[0044] FIG. 18 illustrates aberration curves of the optical imaging system of FIG. 17.

[0045] FIG. 19 is a view illustrating a tenth example of an optical imaging system.

[0046] FIG. 20 illustrates aberration curves of the optical imaging system of FIG. 19.

[0047] FIG. 21 is a view illustrating an eleventh example of an optical imaging system.

[0048] FIG. 22 illustrates aberration curves of the optical imaging system of FIG. 21.

[0049] FIG. 23 is a view illustrating a twelfth example of an optical imaging system.

[0050] FIG. 24 illustrates aberration curves of the optical imaging system of FIG. 23.

[0051] FIG. 25 is a view illustrating a thirteenth example of an optical imaging system.

[0052] FIG. 26 illustrates aberration curves of the optical imaging system of FIG. 25.

[0053] FIG. 27 is a view illustrating a fourteenth example of an optical imaging system.

[0054] FIG. 28 illustrates aberration curves of the optical imaging system of FIG. 27.

[0055] FIG. 29 is a view illustrating a fifteenth example of an optical imaging system.

[0056] FIG. 30 illustrates aberration curves of the optical imaging system of FIG. 29.

[0057] FIG. 31 is a view illustrating a sixteenth example of an optical imaging system.

[0058] FIG. 32 illustrates aberration curves of the optical imaging system of FIG. 31.

[0059] FIG. 33 is a view illustrating a seventeenth example of an optical imaging system.

[0060] FIG. 34 illustrates aberration curves of the optical imaging system of FIG. 33.

[0061] FIG. 35 is a view illustrating an eighteenth example of an optical imaging system.

[0062] FIG. 36 illustrates aberration curves of the optical imaging system of FIG. 35.

[0063] FIG. 37 is a view illustrating a nineteenth example of an optical imaging system.

[0064] FIG. 38 illustrates aberration curves of the optical imaging system of FIG. 37.

[0065] FIG. 39 is a view illustrating a twentieth example of an optical imaging system.

[0066] FIG. 40 illustrates aberration curves of the optical imaging system of FIG. 39.

[0067] FIG. 41 is a view illustrating a twenty-first example of an optical imaging system.

[0068] FIG. 42 illustrates aberration curves of the optical imaging system of FIG. 41.

[0069] FIG. 43 is a view illustrating a twenty-second example of an optical imaging system.

[0070] FIG. 44 illustrates aberration curves of the optical imaging system of FIG. 43.

[0071] FIG. 45 is a view illustrating a twenty-third example of an optical imaging system.

[0072] FIG. 46 illustrates aberration curves of the optical imaging system of FIG. 45.

[0073] FIG. 47 is a view illustrating a twenty-fourth example of an optical imaging system.

[0074] FIG. 48 illustrates aberration curves of the optical imaging system of FIG. 47.

[0075] FIG. 49 is a view illustrating a twenty-fifth example of an optical imaging system.

[0076] FIG. 50 illustrates aberration curves of the optical imaging system of FIG. 49.

[0077] FIG. 51 is a view illustrating a twenty-sixth example of an optical imaging system.

[0078] FIG. 52 illustrates aberration curves of the optical imaging system of FIG. 51.

[0079] FIG. 53 is a view illustrating a twenty-seventh example of an optical imaging system.

[0080] FIG. 54 illustrates aberration curves of the optical imaging system of FIG. 53.

[0081] FIG. 55 is a view illustrating a twenty-eighth example of an optical imaging system.

[0082] FIG. 56 illustrates aberration curves of the optical imaging system of FIG. 55.

[0083] FIG. 57 is a view illustrating a twenty-ninth example of an optical imaging system.

[0084] FIG. 58 illustrates aberration curves of the optical imaging system of FIG. 57.

[0085] FIGS. 59 and 60 are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.

[0086] FIG. 61 is cross-sectional view illustrating an example of a seventh lens.

[0087] FIG. 62 is a cross-sectional view illustrating an example of a shape of a rib of a lens.

[0088] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0089] The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but

may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known in the art may be omitted for increased clarity and conciseness.

[0090] The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

[0091] Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

[0092] As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

[0093] Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

[0094] Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element’s relationship to another element as shown in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated by 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

[0095] The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

[0096] Thicknesses, sizes, and shapes of lenses illustrated in the drawings may have been slightly exaggerated for convenience of explanation. In addition, the shapes of spherical surfaces or aspherical surfaces of the lenses described in the detailed description and illustrated in the drawings are merely examples. That is, the shapes of the spherical surfaces or the aspherical surfaces of the lenses are not limited to the examples described herein.

[0097] Numerical values of radii of curvature, thicknesses of lenses, distances between elements including lenses or surfaces, effective aperture radii of lenses, focal lengths, and diameters, thicknesses, and lengths of various elements are expressed in millimeters (mm), and angles are expressed in degrees. Thicknesses of lenses and distances between elements including lenses or surfaces are measured along the optical axis of the optical imaging system.

[0098] The term “effective aperture radius” as used in this application refers to a radius of a portion of a surface of a lens or other element (an object-side surface or an image-side surface of a lens or other element) through which light actually passes. The effective aperture radius is equal to a distance measured perpendicular to an optical axis of the surface between the optical axis of the surface and the outermost point on the surface through which light actually passes. Therefore, the effective aperture radius may be equal to a radius of an optical portion of a surface, or may be smaller than the radius of the optical portion of the surface if light does not pass through a peripheral portion of the optical portion of the surface. The object-side surface and the image-side surface of a lens or other element may have different effective aperture radii.

[0099] In this application, unless stated otherwise, a reference to the shape of a lens surface means the shape of a paraxial region of the lens surface. A paraxial region of a lens surface is a central portion of the lens surface surrounding the optical axis of the lens surface in which light rays incident to the lens surface make a small angle θ to the optical axis and the approximations $\sin \theta \approx \theta$, $\tan \theta \approx \theta$, and $\cos \theta \approx 1$ are valid.

[0100] For example, a statement that the object-side surface of a lens is convex means that at least a paraxial region of the object-side surface of the lens is convex, and a statement that the image-side surface of the lens is concave means that at least a paraxial region of the image-side surface of the lens is concave. Therefore, even though the object side-surface of the lens may be described as being convex, the entire object-side surface of the lens may not be convex, and a peripheral region of the object-side surface of the lens may be concave. Also, even though the image-side surface of the lens may be described as being concave, the entire image-side surface of the lens may not be concave, and a peripheral region of the image-side surface of the lens may be convex.

[0101] An optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system. Thus, the first lens is a lens closest to an object (or a subject) to be imaged by the optical imaging system, while the seventh lens is a lens closest to the imaging plane.

[0102] Each lens of the optical imaging system includes an optical portion and a rib. The optical portion of the lens is a portion of the lens that is configured to refract light, and is generally formed in a central portion of the lens. The rib of the lens is an edge portion of the lens that enables the lens to be mounted in a lens barrel and the optical axis of the lens to be aligned with the optical axis of the optical imaging system. The rib of the lens extends radially outward from the optical portion. The optical portions of the lenses are generally not in contact with each other. For example, the first to seventh lenses are mounted in the lens barrel so that they are spaced apart from one another by predetermined distances along the optical axis of the optical imaging system. The ribs of the lenses may be in selective contact with each other. For example, the ribs of the first to fourth lenses, or the first to fifth lenses, or the second to fourth lenses, may be in contact with each other so that the optical axes of these lenses may be easily aligned with the optical axis of the optical imaging system.

[0103] Next, a configuration of the optical imaging system will be described.

[0104] The optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system.

[0105] The optical imaging system further includes an image sensor and a filter. The image sensor forms an imaging plane, and converts light refracted by the first to seventh lenses into an electric signal. The filter is disposed between the seventh lens and the imaging plane, and blocks infrared rays in the light refracted by the first to seventh lenses from being incident on the imaging plane.

[0106] The optical imaging system further includes a stop and spacers. The stop is disposed in front of the first lens, or at a position of either an object-side surface or an image-side surface of one of the first to seventh lenses, or between two adjacent lenses of the first to seventh lenses, or between the object-side surface and the image-side surface of one of the first to seventh lenses, to adjust the amount of light incident on the imaging plane. Some examples may include two stops, one of which may be disposed in front of the first lens, or at the position of the object-side surface of the first lens, or between the object-side surface and the image-side surface of the first lens. Each of the spacers is disposed at a respective position between two lenses of the first to seventh lenses to maintain a predetermined distance between the two lenses. In addition, the spacers may be made of a light-shielding material to block extraneous light penetrating into the ribs of the lenses. There may be six or seven spacers. For example, a first spacer is disposed between the first lens and the second lens, a second spacer is disposed between the second lens and the third lens, a third spacer is disposed between the third lens and the fourth lens, a fourth spacer is disposed between the fourth lens and the fifth lens, a fifth spacer is disposed between the fifth lens and the sixth lens, and a sixth spacer is disposed between the sixth lens and the seventh lens. In addition, the optical imaging system may further include a seventh spacer disposed between the sixth lens and the sixth spacer.

[0107] Next, the lenses of the optical imaging system will be described.

[0108] The first lens has a refractive power. For example, the first lens has a positive refractive power or a negative refractive power. One surface of the first lens may be convex. For example, an object-side surface of the first lens may be convex. One surface of the first lens may be concave. For example, an image-side surface of the first lens may be concave. The first lens may have an aspherical surface. For example, one surface or both surfaces of the first lens may be aspherical.

[0109] The second lens has a refractive power. For example, the second lens has a positive refractive power or a negative refractive power. At least one surface of the second lens may be convex. For example, an object-side surface of the second lens may be convex, or both the object-side surface and an image-side surface of the second lens may be convex. One surface of the second lens may be concave. For example, the image-side surface of the second lens may be concave. The second lens may have an aspherical surface. For example, one surface or both surfaces of the second lens may be aspherical.

[0110] The third lens has a refractive power. For example, the third lens has a positive refractive power or a negative refractive power. One surface of the third lens may be convex. For example, an object-side surface or an image-side surface of the third lens may be convex. One surface of the third lens may be concave. For example, the object-side surface or the image-side surface of the third lens may be concave. The third lens may have an aspherical surface. For example, one surface or both surfaces of the third lens may be aspherical.

[0111] The fourth lens has a refractive power. For example, the fourth lens has a positive refractive power or a negative refractive power. At least one surface of the fourth lens may be convex. For example, an object-side surface or an image-side surface of the fourth lens may be convex, or both the object-side surface and the image-side surface of the fourth lens may be convex. One surface of the fourth lens may be concave. For example, the object-side surface or the image-side surface of the fourth lens may be concave. The fourth lens may have an aspherical surface. For example, one surface or both surfaces of the fourth lens may be aspherical.

[0112] The fifth lens has a refractive power. For example, the fifth lens has a positive refractive power or a negative refractive power. One surface of the fifth lens may be convex. For example, an object-side surface or an image-side surface of the fifth lens may be convex. At least one surface of the fifth lens may be concave. For example, the object-side surface or the image-side surface of the fifth lens may be concave, or both the object-side surface and the image-side surface of the fifth lens may be concave. The fifth lens may have an aspherical surface. For example, one surface or both surfaces of the fifth lens may be aspherical.

[0113] The sixth lens has a refractive power. For example, the sixth lens has a positive refractive power or a negative refractive power. At least one surface of the sixth lens may be convex. For example, an object-side surface or an image-side surface of the sixth lens may be convex, or both the object-side surface and the image-side surface of the sixth lens may be convex. At least one surface of the sixth lens may be concave. For example, the object-side surface or the image-side surface of the sixth lens may be concave, or both the object-side surface and the image-side surface of the sixth lens may be concave. At least one surface of the sixth

lens may have at least one inflection point. An inflection point is a point where a lens surface changes from convex to concave, or from concave to convex. A number of inflection points is counted from a center of the lens to an outer edge of the optical portion of the lens. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the sixth lens. Therefore, at least one surface of the sixth lens may have a paraxial region and a peripheral region having shapes that are different from each other. For example, a paraxial region of the object-side surface of the sixth lens may be concave, but a peripheral region thereof may be convex. The sixth lens may have an aspherical surface. For example, one surface or both surfaces of the sixth lens may be aspherical.

[0114] The seventh lens has a refractive power. For example, the seventh lens has a positive refractive power or a negative refractive power. One surface of the seventh lens may be convex. For example, an object-side surface of the seventh lens may be convex. At least one surface of the seventh lens may be concave. For example, an image-side surface of the seventh lens may be concave, or both the object-side surface and the image-side surface of the seventh lens may be concave. At least one surface of the seventh lens may have at least one inflection point. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the seventh lens. Therefore, at least one surface of the seventh lens may have a paraxial region and a peripheral region having shapes that different from each other. For example, a paraxial region of the object-side surface of the seventh lens may be concave, but a peripheral region thereof may be convex. The seventh lens may have an aspherical surface. For example, one surface or both surfaces of the seventh lens may be aspherical.

[0115] The lenses of the optical imaging system may be made of a light material having a high light transmittance. For example, the first to seventh lenses may be made of a plastic material. However, a material of the first to seventh lenses is not limited to the plastic material.

[0116] The aspherical surfaces of the first to seventh lenses may be represented by the following Equation 1:

$$Z = \frac{cY^2}{1 + \sqrt{1 - (1 + K)c^2 Y^2}} + AY^4 + BY^6 + CY^8 + DY^{10} + EY^{12} + FY^{14} + GY^{16} + HY^{18} + \sim \quad (1)$$

[0117] In Equation 1, c is a curvature of a lens surface and is equal to an inverse of a radius of curvature of the lens surface at an optical axis of the lens surface, K is a conic constant, Y is a distance from a certain point on an aspherical surface of the lens to an optical axis of the lens in a direction perpendicular to the optical axis, A to H are aspherical constants, and Z (or sag) is a distance between the certain point on the aspherical surface of the lens at the distance Y to the optical axis and a tangential plane perpendicular to the optical axis meeting the apex of the aspherical surface of the lens. Some of the examples disclosed in this application include an aspherical constant J . An additional term of JY^{20} may be added to the right side of Equation 1 to reflect the effect of the aspherical constant J .

[0118] The optical imaging system may satisfy one or more of the following Conditional Expressions 1 to 5:

$$0.1 < L1w/L7w < 0.4 \quad (\text{Conditional Expression 1})$$

$$0.5 < S6d/f < 1.4 \quad (\text{Conditional Expression 2})$$

$$0.4 < L1TR/L7TR < 0.8 \quad (\text{Conditional Expression 3})$$

$$0.5 < L1234TRavg/L7TR < 0.9 \quad (\text{Conditional Expression 4})$$

$$0.5 < L12345TRavg/L7TR < 0.9 \quad (\text{Conditional Expression 5})$$

[0119] In the above Conditional Expressions, $L1w$ is a weight of the first lens, and $L7w$ is a weight of the seventh lens.

[0120] $S6d$ is an inner diameter of the sixth spacer, and f is an overall focal length of the optical imaging system.

[0121] $L1TR$ is an overall outer diameter of the first lens, and $L7TR$ is an overall outer diameter of the seventh lens. The overall outer diameter of a lens is a diameter of the lens including both the optical portion of the lens and the rib of the lens.

[0122] $L1234TRavg$ is an average value of overall outer diameters of the first to fourth lenses, and $L12345TRavg$ is an average value of overall outer diameters of the first to fifth lenses.

[0123] Conditional Expressions 1 and 3 specify ranges of a weight ratio and an overall outer diameter ratio between the first lens and the seventh lens to facilitate a self-alignment between the lenses and an alignment by the lens barrel.

[0124] Conditional Expression 2 specifies a range of a ratio of the inner diameter of the sixth spacer to the overall focal length of the optical imaging system to minimize a flare phenomenon.

[0125] Conditional Expressions 4 and 5 specify overall outer diameter ratios between the lenses to facilitate aberration correction.

[0126] The optical imaging system may also satisfy one or more of the following Conditional Expressions 6 to 10:

$$0.1 < L1w/L7w < 0.3 \quad (\text{Conditional Expression 6})$$

$$0.5 < S6d/f < 1.2 \quad (\text{Conditional Expression 7})$$

$$0.4 < L1TR/L7TR < 0.7 \quad (\text{Conditional Expression 8})$$

$$0.5 < L1234TRavg/L7TR < 0.75 \quad (\text{Conditional Expression 9})$$

$$0.5 < L12345TRavg/L7TR < 0.76 \quad (\text{Conditional Expression 10})$$

[0127] Conditional Expressions 6 to 10 are the same as Conditional Expressions 1 to 5, except that Conditional Expressions 6 to 10 specify narrower ranges.

[0128] The optical imaging system may also satisfy one or more of the following Conditional Expressions 11 to 31:

$$0.01 < R1/R4 < 1.3 \quad (\text{Conditional Expression 11})$$

$$0.1 < R1/R5 < 0.7 \quad (\text{Conditional Expression 12})$$

$$0.05 < R1/R6 < 0.9 \quad (\text{Conditional Expression 13})$$

$$0.2 < R1/R11 < 1.2 \quad (\text{Conditional Expression 14})$$

$$0.8 < R1/R14 < 1.2 \quad (\text{Conditional Expression 15})$$

-continued

- $0.6 < (R11 + R14)/(2 * R1) < 3.0$ (Conditional Expression 16)
- $0.4 < D13/D57 < 1.2$ (Conditional Expression 17)
- $0.1 < (1/f1 + 1/f2 + 1/f3 + 1/f4 + 1/f5 + 1/f6 + 1/f7) * f < 0.8$ (Conditional Expression 18)
- $0.1 < (1/f1 + 1/f2 + 1/f3 + 1/f4 + 1/f5 + 1/f6 + 1/f7) * TTL < 1.0$ (Conditional Expression 19)
- $0.2 < TD1/D67 < 0.8$ (Conditional Expression 20)
- $0.1 < (R11 + R14)/(R5 + R6) < 1.0$ (Conditional Expression 21)
- $SD12 < SD34$ (Conditional Expression 22)
- $SD56 < SD67$ (Conditional Expression 23)
- $SD56 < SD34$ (Conditional Expression 24)
- $0.6 < TTL/(2 * (IMG HT)) < 0.9$ (Conditional Expression 25)
- $0.2 < \Sigma SD/\Sigma TD < 0.7$ (Conditional Expression 26)
- $0 < \min(f1:f3)/\max(f4:f7) < 0.4$ (Conditional Expression 27)
- $0.4 < (\Sigma TD)/TTL < 0.7$ (Conditional Expression 28)
- $0.7 < SL/TTL < 1.0$ (Conditional Expression 29)
- $0.81 < f12/f123 < 0.96$ (Conditional Expression 30)
- $0.6 < f12/f1234 < 0.84$ (Conditional Expression 31)

[0129] In the above Conditional Expressions, R1 is a radius of curvature of an object-side surface of the first lens, R4 is a radius of curvature of an image-side surface of the second lens, R5 is a radius of curvature of an object-side surface of the third lens, R6 is a radius of curvature of an image-side surface of the third lens, R11 is a radius of curvature of an object-side surface of the sixth lens, and R14 is a radius of curvature of an image-side surface of the seventh lens.

[0130] D13 is a distance along an optical axis of the optical imaging system from the object-side surface of the first lens to the image-side surface of the third lens, and D57 is a distance along the optical axis from an object-side surface of the fifth lens to the image-side surface of the seventh lens.

[0131] f1 is a focal length of the first lens, f2 is a focal length of the second lens, f3 is a focal length of the third lens, f4 is a focal length of the fourth lens, f5 is a focal length of the fifth lens, f6 is a focal length of the sixth lens, f7 is a focal length of the seventh lens, f is an overall focal length of the optical imaging system, and TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of an image sensor of the optical imaging system.

[0132] TD1 is a thickness along the optical axis of the first lens, and D67 is a distance along the optical axis from the object-side surface of the sixth lens to the image-side surface of the seventh lens.

[0133] SD12 is a distance along the optical axis from an image-side surface of the first lens to an object-side surface of the second lens, SD34 is a distance along the optical axis from the image-side surface of the third lens to an object-side surface of the fourth lens, SD56 is a distance along the optical axis from an image-side surface of the fifth lens to the object-side surface of the sixth lens, and SD67 is a

distance along the optical axis from an image-side surface of the sixth lens to an object-side surface of the seventh lens.

[0134] IMG HT is one-half of a diagonal length of the imaging plane of the image sensor.

[0135] Σ SD is a sum of air gaps along the optical axis between the lenses, and Σ TD is a sum of thicknesses along the optical axis of the lenses. An air gap is a distance along the optical axis between adjacent lenses.

[0136] $\min(f1:f3)$ is a minimum value of absolute values of the focal lengths of the first to third lenses, and $\max(f4:f7)$ is a maximum value of absolute values of the focal lengths of the fourth to seventh lenses.

[0137] SL is a distance along the optical axis from the stop to the imaging plane of the image sensor.

[0138] f12 is a composite focal length of the first and second lenses, f123 is a composite focal length of the first to third lenses, and f1234 is a composite focal length of the first to fourth lenses.

[0139] Conditional Expression 11 specifies a design range of the second lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the second lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 11, and it is difficult to expect a sufficient correction of astigmatic field curves for the second lens having a radius of curvature that is below the lower limit value of Conditional Expression 11.

[0140] Conditional Expressions 12 and 13 specify a design range of the third lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the third lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 12 or 13, and it is difficult to expect a sufficient correction of astigmatic field curves for the third lens having a radius of curvature that is below the lower limit value of Conditional Expression 12 or 13.

[0141] Conditional Expression 14 specifies a design range of the sixth lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the sixth lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 14, and the sixth lens having a radius of curvature that is below the lower limit value of Conditional Expression 14 is apt to cause a flare phenomenon.

[0142] Conditional Expression 15 specifies a design range of the seventh lens for minimizing the aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the seventh lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 15, and the seventh lens having a radius of curvature that is below the lower limit value of Conditional Expression 15 is apt to cause an imaging plane curvature.

[0143] Conditional Expression 16 specifies a ratio of a sum of radii of curvature of the sixth lens and the seventh lens to twice a radius of curvature of the first lens for correcting the longitudinal spherical aberration and achieving excellent optical performance.

[0144] Conditional Expression 17 specifies a ratio of the optical imaging system mountable in a compact terminal. For example, an optical imaging system having a ratio that exceeds the upper limit value of Conditional Expression 17 may cause a problem that the total length of the optical imaging system becomes long, and an optical imaging system having a ratio that is below the lower limit value of

Conditional Expression 17 may cause a problem that a lateral cross-section of the optical imaging system becomes large.

[0145] Conditional Expressions 18 and 19 specify a refractive power ratio of the first to seventh lenses for facilitating mass production of the optical imaging system. For example, an optical imaging system having a refractive power ratio that exceeds the upper limit value of Conditional Expression 18 or 19 or is below the lower limit value of Conditional Expression 18 or 19 is difficult to commercialize because the refractive power of one or more of the first to seventh lenses is too great.

[0146] Conditional Expression 20 specifies a thickness range of the first lens for implementing a compact optical imaging system. For example, the first lens having a thickness that exceeds the upper value of Conditional Expression 20 or is below the lower limit value of Conditional Expression 20 is too thick or too thin to be manufactured.

[0147] Conditional Expression 22 specifies a design condition of the first to fourth lenses for improving chromatic aberration. For example, a case in which a distance between the first lens and the second lens is shorter than a distance between the third lens and the fourth lens is advantageous for improving the chromatic aberration.

[0148] Conditional Expressions 25 to 28 specify design conditions for implementing a compact optical imaging system. For example, lenses that deviate from the numerical range of Conditional Expression 26 or 28 are difficult to form by injection molding and process.

[0149] Conditional Expressions 29 to 31 specify design conditions of the optical imaging system in consideration of a position of the stop. For example, an optical imaging system that does not satisfy one or more of Conditional Expressions 29 to 31 may have a longer overall length due to the refractive power of the lenses disposed behind the stop.

[0150] Next, various examples of the optical imaging system will be described. In the tables that appear in the following examples, S1 denotes an object-side surface of the first lens, S2 denotes an image-side surface of the first lens, S3 denotes an object-side surface of the second lens, S4 denotes an image-side surface of the second lens, S5 denotes an object-side surface of the third lens, S6 denotes an image-side surface of the third lens, S7 denotes an object-side surface of the fourth lens, S8 denotes an image-side surface of the fourth lens, S9 denotes an object-side surface of the fifth lens, S10 denotes an image-side surface of the fifth lens, S11 denotes an object-side surface of the sixth lens, S12 denotes an image-side surface of the sixth lens, S13 denotes an object-side surface of the seventh lens, S14 denotes an image-side surface of the seventh lens, S15 denotes an object-side surface of the filter, S16 denotes an image-side surface of the filter, and S17 denotes the imaging plane.

First Example

[0151] FIG. 1 is a view illustrating a first example of an optical imaging system, and FIG. 2 illustrates aberration curves of the optical imaging system of FIG. 1.

[0152] An optical imaging system 1 includes a first lens 1001, a second lens 2001, a third lens 3001, a fourth lens 4001, a fifth lens 5001, a sixth lens 6001, and a seventh lens 7001.

[0153] The first lens 1001 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2001 has a negative refractive power, and an object-side surface thereof

is convex and an image-side surface thereof is concave. The third lens 3001 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4001 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5001 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6001 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6001. The seventh lens 7001 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7001, and one inflection point is formed on the image-side surface of the seventh lens 7001.

[0154] The optical imaging system 1 further includes a stop, a filter 8001, and an image sensor 9001. The stop is disposed between the second lens 2001 and the third lens 3001 to adjust an amount of light incident on the image sensor 9001. The filter 8001 is disposed between the seventh lens 7001 and the image sensor 9001 to block infrared rays. The image sensor 9001 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 1, the stop is disposed at a distance of 0.671 mm from the object-side surface of the first lens 1001 toward the imaging plane of the optical imaging system 1. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 1 listed in Table 59 that appears later in this application.

[0155] Table 1 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 1, and Table 2 below shows aspherical surface coefficients of the lenses of FIG. 1.

TABLE 1

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First Lens	1.3225	0.5006	1.546	56.114	0.920
S2		4.3551	0.0200			0.881
S3	Second Lens	3.5653	0.1500	1.668	20.353	0.850
S4		2.1771	0.1624			0.780
(Stop)						
S5	Third Lens	3.1365	0.2284	1.547	56.114	0.775
S6		5.3019	0.2563			0.805
S7	Fourth Lens	-13.3504	0.2620	1.547	56.114	0.852
S8		-10.1423	0.2108			0.996
S9	Fifth Lens	-3.9034	0.2300	1.657	21.494	1.066
S10		-7.2623	0.1660			1.350
S11	Sixth Lens	3.0156	0.3939	1.621	21.494	1.561
S12		3.1904	0.2101			1.812
S13	Seventh Lens	1.6112	0.4621	1.537	55.711	2.420
S14		1.2425	0.1961			2.479
S15	Filter	Infinity	0.1100	1.519	64.197	2.878
S16		Infinity	0.6293			2.915
S17	Imaging Plane	Infinity	0.0120			3.268

TABLE 2

	K	A	B	C	D	E	F	G	H
S1	-0.0435	-0.0109	0.0629	-0.3496	0.8258	-1.1525	0.7549	-0.2137	0
S2	-13.409	-0.1278	0.1627	0.0433	-0.8903	1.7755	-1.5568	0.5056	0
S3	-3.4003	-0.1481	0.3107	-0.2639	0.1553	-0.0446	0.299	-0.2867	0
S4	0.21	-0.0544	0.1307	-0.1443	0.9025	-2.9213	4.8603	-2.7957	0
S5	3.2497	-0.0995	0.0696	-0.678	2.7904	-5.9532	7.488	-3.502	0
S6	0.0843	-0.0685	-0.0666	0.3302	-1.5683	4.6995	-6.2319	3.6509	0
S7	4E-08	-0.155	-0.0515	-0.2096	0.1445	0.5442	-0.6037	0.1711	0
S8	5E-09	-0.1321	0.0922	-0.6514	0.9947	-0.5706	0.1385	-0.0115	0
S9	-0.3203	-0.1327	0.4222	-1.1911	1.2725	-0.4351	-0.1439	0.0842	0
S10	10.535	-0.1849	0.2573	-0.4641	0.511	-0.2753	0.069	-0.0065	0
S11	-50	0.1562	-0.6443	0.8704	-0.7495	0.3744	-0.0961	0.0098	0
S12	-34.584	0.0511	-0.2768	0.3083	-0.2075	0.0819	-0.0171	0.0015	0
S13	-0.9404	-0.5384	0.2288	-0.0102	-0.0258	0.0106	-0.002	0.0002	-7E-06
S14	-0.996	-0.4669	0.3054	-0.1607	0.0625	-0.0162	0.0026	-0.0002	8E-06

Second Example

[0156] FIG. 3 is a view illustrating a second example of an optical imaging system, and FIG. 4 illustrates aberration curves of the optical imaging system of FIG. 3.

[0157] An optical imaging system 2 includes a first lens 1002, a second lens 2002, a third lens 3002, a fourth lens 4002, a fifth lens 5002, a sixth lens 6002, and a seventh lens 7002.

[0158] The first lens 1002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3002 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4002 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6002 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one

or both of the object-side surface and the image-side surface of the sixth lens 6002. The seventh lens 7002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7002, and one inflection point is formed on the image-side surface of the seventh lens 7002.

[0159] The optical imaging system 2 further includes a stop, a filter 8002, and an image sensor 9002. The stop is disposed between the second lens 2002 and the third lens 3002 to adjust an amount of light incident on the image sensor 9002. The filter 8002 is disposed between the seventh lens 7002 and the image sensor 9002 to block infrared rays. The image sensor 9002 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 3, the stop is disposed at a distance of 1.425 mm from the object-side surface of the first lens 1002 toward the imaging plane of the optical imaging system 2. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 2 listed in Table 59 that appears later in this application.

[0160] Table 3 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 3, and Table 4 below shows aspherical surface coefficients of the lenses of FIG. 3.

TABLE 3

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.3706	0.5431	1.546	56.114	1.572
S2	Lens	3.8377	0.1516			1.517
S3	Second	3.4329	0.7078	1.546	56.114	1.478
S4	Lens	-17.0251	0.0225			1.428
S5	Third Lens	5.1429	0.2247	1.679	19.236	1.300
(Stop)						
S6		2.5333	0.5888			1.230
S7	Fourth	-1446.167	0.3404	1.679	19.236	1.404
S8	Lens	-1446.167	0.2070			1.600
S9	Fifth	3.6434	0.3264	1.546	56.114	1.857
S10	Lens	3.8224	0.3171			2.199
S11	Sixth	3.8509	0.4406	1.679	19.236	2.415
S12	Lens	3.0494	0.1774			2.808
S13	Seventh	1.7430	0.6133	1.537	53.955	3.115
S14	Lens	1.5635	0.2466			3.314
S15	Filter	Infinity	0.1100	1.519	64.166	3.655
S16		Infinity	0.8047			3.688
S17	Imaging Plane	Infinity	0.0051			4.075

TABLE 4

	K	A	B	C	D	E	F	G	H	J
S1	-7.5196	0.0476	-0.039	0.0108	-0.0002	-0.006	0.0045	-0.0012	0.0001	0
S2	-19.661	-0.0106	-0.0481	0.0183	0.0105	-0.0109	0.0039	-0.0006	3E-05	0
S3	0.042	-0.0249	-0.0196	0.0094	0.0041	0.0108	-0.014	0.0056	-0.0008	0
S4	0	0.0098	-0.0507	0.0341	0.0229	-0.0518	0.0341	-0.0103	0.0012	0
S5	-5.6502	-0.0476	0.0152	-0.0398	0.11	-0.1327	0.082	-0.0252	0.0031	0
S6	0.5327	-0.067	0.0583	-0.0705	0.0922	-0.0854	0.0499	-0.0161	0.0024	0
S7	0	-0.0158	-0.0083	-0.0305	0.0756	-0.0736	0.035	-0.0077	0.0005	0
S8	0	-0.0099	-0.0427	0.0077	0.0285	-0.0272	0.01	-0.0013	0	0
S9	-44.395	0.1048	-0.1251	0.08	-0.0437	0.0187	-0.0058	0.001	-7E-05	0
S10	-4.0715	-0.0175	0.0211	-0.0368	0.0252	-0.01	0.0024	-0.0003	2E-05	0
S11	-1.1211	0.0034	-0.0742	0.0637	-0.0381	0.0134	-0.0026	0.0003	-1E-05	0
S12	0.0464	-0.092	0.0339	-0.0168	0.0044	-0.0005	1E-05	3E-06	-2E-07	0
S13	-0.795	-0.2987	0.11	-0.0259	0.0046	-0.0007	7E-05	-5E-06	2E-07	-5E-09
S14	-1.3233	-0.199	0.0846	-0.0285	0.0073	-0.0013	0.0002	-1E-05	5E-07	-9E-09

Third Example

[0161] FIG. 5 is a view illustrating a third example of an optical imaging system, and FIG. 6 illustrates aberration curves of the optical imaging system of FIG. 5.

[0162] An optical imaging system 3 includes a first lens 1003, a second lens 2003, a third lens 3003, a fourth lens 4003, a fifth lens 5003, a sixth lens 6003, and a seventh lens 7003.

[0163] The first lens 1003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3003 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4003 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6003 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one

or both of the object-side surface and the image-side surface of the sixth lens 6003. The seventh lens 7003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7003, and one inflection point is formed on the image-side surface of the seventh lens 7003.

[0164] The optical imaging system 3 further includes a stop, a filter 8003, and an image sensor 9003. The stop is disposed between the second lens 2003 and the third lens 3003 to adjust an amount of light incident on the image sensor 9003. The filter 8003 is disposed between the seventh lens 7003 and the image sensor 9003 to block infrared rays. The image sensor 9003 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 5, the stop is disposed at a distance of 1.259 mm from the object-side surface of the first lens 1003 toward the imaging plane of the optical imaging system 3. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 3 listed in Table 59 that appears later in this application.

[0165] Table 5 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 5, and Table 6 below shows aspherical surface coefficients of the lenses of FIG. 5.

TABLE 5

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1022	0.4835	1.546	56.114	1.408
S2	Lens	3.3563	0.1357			1.350
S3	Second	3.0907	0.6198	1.546	56.114	1.308
S4	Lens	-13.9876	0.0200			1.271
S5	Third Lens	4.8553	0.2000	1.679	19.236	1.157
(Stop)						
S6		2.3669	0.5599			1.095
S7	Fourth	-2272.129	0.3012	1.679	19.236	1.270
S8	Lens	-7278.426	0.1848			1.442
S9	Fifth	3.3546	0.2946	1.546	56.114	1.646
S10	Lens	3.5201	0.2604			1.947
S11	Sixth	3.4723	0.3932	1.679	19.236	2.150
S12	Lens	2.7354	0.1549			2.500
S13	Seventh	1.5570	0.5518	1.537	53.955	2.749
S14	Lens	1.3661	0.2501			2.950
S15	Filter	Infinity	0.1100	1.519	64.166	3.293
S16		Infinity	0.6646			3.328
S17	Imaging Plane	Infinity	0.0054			3.699

TABLE 6

	K	A	B	C	D	E	F	G	H	J
S1	-7.5279	0.0685	-0.0723	0.0313	-0.0131	-0.0097	0.0144	-0.0054	0.0007	0
S2	-19.893	-0.0114	-0.0921	0.0405	0.0318	-0.0345	0.0116	-0.001	-0.0002	0
S3	-0.0142	-0.0359	-0.0288	-0.0087	0.0581	0.0053	-0.0505	0.0291	-0.0054	0
S4	0	0.0225	-0.1301	0.1638	-0.0413	-0.1012	0.1103	-0.0452	0.0067	0
S5	-6.2325	-0.061	-0.0037	-0.0472	0.3094	-0.5229	0.4199	-0.1649	0.0257	0
S6	0.4782	-0.092	0.0962	-0.1588	0.2881	-0.3518	0.2616	-0.1062	0.0192	0
S7	0	-0.0151	-0.0532	0.0425	0.0094	-0.0356	0.0085	0.009	-0.0039	0
S8	0	-0.0101	-0.0934	0.0497	0.0399	-0.0661	0.0321	-0.0053	0	0
S9	-49.08	0.1451	-0.2207	0.1683	-0.1105	0.058	-0.0226	0.0051	-0.0005	0
S10	-5.4303	-0.0164	0.0172	-0.0595	0.0534	-0.0275	0.0084	-0.0014	1E-04	0
S11	-1.136	0.0251	-0.1801	0.1935	-0.1377	0.0586	-0.014	0.0017	-9E-05	0
S12	0.0272	-0.1034	0.0166	3E-05	-0.0063	0.0037	-0.0009	0.0001	-5E-06	0
S13	-0.8	-0.4195	0.2062	-0.0728	0.0211	-0.0048	0.0007	-8E-05	4E-06	-1E-07
S14	-1.3207	-0.2931	0.1671	-0.0741	0.0239	-0.0053	0.0008	-7E-05	4E-06	-8E-08

Fourth Example

[0166] FIG. 7 is a view illustrating a fourth example of an optical imaging system, and FIG. 8 illustrates aberration curves of the optical imaging system of FIG. 7.

[0167] An optical imaging system 4 includes a first lens 1004, a second lens 2004, a third lens 3004, a fourth lens 4004, a fifth lens 5004, a sixth lens 6004, and a seventh lens 7004.

[0168] The first lens 1004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3004 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4004 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6004 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one

or both of the object-side surface and the image-side surface of the sixth lens 6004. The seventh lens 7004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7004, and one inflection point is formed on the image-side surface of the seventh lens 7004.

[0169] The optical imaging system 4 further includes a stop, a filter 8004, and an image sensor 9004. The stop is disposed between the second lens 2004 and the third lens 3004 to adjust an amount of light incident on the image sensor 9004. The filter 8004 is disposed between the seventh lens 7004 and the image sensor 9004 to block infrared rays. The image sensor 9004 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 7, the stop is disposed at a distance of 1.169 mm from the object-side surface of the first lens 1004 toward the imaging plane of the optical imaging system 4. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 4 listed in Table 59 that appears later in this application.

[0170] Table 7 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 7, and Table 8 below shows aspherical surface coefficients of the lenses of FIG. 7.

TABLE 7

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.9512	0.4488	1.546	56.114	1.307
S2	Lens	3.1152	0.1260			1.253
S3	Second	2.8686	0.5753	1.546	56.114	1.214
S4	Lens	-12.9825	0.0186			1.180
S5	Third Lens	4.5064	0.1856	1.679	19.236	1.074
(Stop)						
S6		2.1969	0.5197			1.016
S7	Fourth	-2108.865	0.2796	1.679	19.236	1.179
S8	Lens	-6755.436	0.1715			1.338
S9	Fifth	3.1135	0.2734	1.546	56.114	1.528
S10	Lens	3.2672	0.2417			1.808
S11	Sixth	3.2228	0.3650	1.679	19.236	1.996
S12	Lens	2.5388	0.1438			2.320
S13	Seventh	1.4451	0.5122	1.537	53.955	2.500
S14	Lens	1.2680	0.2501			2.738
S15	Filter	Infinity	0.1100	1.519	64.166	2.940
S16		Infinity	0.5924			2.971
S17	Imaging Plane	Infinity	0.0054			3.251

TABLE 8

	K	A	B	C	D	E	F	G	H	J
S1	-7.5279	0.0857	-0.105	0.0528	-0.0256	-0.0221	0.0379	-0.0166	0.0023	0
S2	-19.893	-0.0142	-0.1337	0.0682	0.0621	-0.0783	0.0306	-0.0031	-0.0006	0
S3	-0.0142	-0.0449	-0.0418	-0.0147	0.1136	0.012	-0.1333	0.0892	-0.0193	0
S4	0	0.0281	-0.189	0.276	-0.0808	-0.2297	0.2908	-0.1382	0.024	0
S5	-6.2325	-0.0763	-0.0054	-0.0795	0.6054	-1.1875	1.107	-0.5047	0.0912	0
S6	0.4782	-0.115	0.1396	-0.2676	0.5637	-0.7991	0.6898	-0.325	0.0682	0
S7	0	-0.0188	-0.0772	0.0717	0.0184	-0.081	0.0225	0.0277	-0.0139	0
S8	0	-0.0127	-0.1356	0.0837	0.0781	-0.1502	0.0847	-0.0163	0	0
S9	-49.08	0.1815	-0.3205	0.2837	-0.2161	0.1317	-0.0595	0.0158	-0.0017	0
S10	-5.4303	-0.0205	0.025	-0.1003	0.1046	-0.0624	0.0222	-0.0043	0.0003	0
S11	-1.136	0.0314	-0.2615	0.3261	-0.2695	0.133	-0.0369	0.0053	-0.0003	0
S12	0.0272	-0.1293	0.0241	5E-05	-0.0123	0.0085	-0.0024	0.0003	-2E-05	0
S13	-0.8	-0.5247	0.2994	-0.1227	0.0414	-0.0108	0.002	-0.0002	2E-05	-4E-07
S14	-1.3207	-0.3666	0.2425	-0.1248	0.0468	-0.0121	0.002	-0.0002	1E-05	-3E-07

Fifth Example

[0171] FIG. 9 is a view illustrating a fifth example of an optical imaging system, and FIG. 10 illustrates aberration curves of the optical imaging system of FIG. 9.

[0172] An optical imaging system 5 includes a first lens 1005, a second lens 2005, a third lens 3005, a fourth lens 4005, a fifth lens 5005, a sixth lens 6005, and a seventh lens 7005.

[0173] The first lens 1005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6005. The seventh lens 7005 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7005.

[0174] The optical imaging system 5 further includes a stop, a filter 8005, and an image sensor 9005. The stop is disposed between the first lens 1005 and the second lens 2005 to adjust an amount of light incident on the image

sensor 9005. The filter 8005 is disposed between the seventh lens 7005 and the image sensor 9005 to block infrared rays. The image sensor 9005 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 9, the stop is disposed at a distance of 0.383 mm from the object-side surface of the first lens 1005 toward the imaging plane of the optical imaging system 5. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 5 listed in Table 59 that appears later in this application.

[0175] Table 9 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 9, and Table 10 below shows aspherical surface coefficients of the lenses of FIG. 9.

TABLE 9

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1824	0.3329	1.546	56.114	1.380
S2	Lens	1.9439	0.0500			1.369
S3 (Stop)	Second	1.6857	0.7322	1.546	56.114	1.335
S4	Lens	28.3727	0.0500			1.264
S5	Third	7.1536	0.2200	1.679	19.236	1.185
S6	Lens	2.9223	0.4264			1.050
S7	Fourth	46.9146	0.3121	1.646	23.528	1.112
S8	Lens	17.5860	0.2616			1.268
S9	Fifth	2.2655	0.2700	1.646	23.528	1.774
S10	Lens	2.3143	0.3731			1.839
S11	Sixth	8.5186	0.6078	1.546	56.114	2.160
S12	Lens	-1.9871	0.3782			2.308
S13	Seventh	-4.7165	0.3600	1.546	56.114	2.780
S14	Lens	1.8919	0.1457			2.998
S15	Filter	Infinity	0.1100	1.519	64.166	3.353
S16		Infinity	0.6600			3.385
S17	Imaging Plane	Infinity	0.0100			3.712

TABLE 10

	K	A	B	C	D	E	F	G	H	J
S1	-3.5715	0.0005	0.0011	-0.0181	0.0025	0.0107	-0.0084	0.0026	-0.0003	0
S2	-9.1496	-0.0513	-0.0055	0.0116	0.0161	-0.0207	0.0078	-0.001	0	0
S3	-2.5622	-0.0879	0.1115	-0.1204	0.1625	-0.1325	0.0578	-0.0118	0.0006	0
S4	-90	-0.078	0.2103	-0.4384	0.6397	-0.6153	0.3736	-0.1288	0.0189	0
S5	0	-0.1133	0.2975	-0.5447	0.7496	-0.7199	0.4525	-0.1642	0.0257	0
S6	4.6946	-0.0705	0.1434	-0.2144	0.1998	-0.0956	-0.0142	0.0399	-0.0137	0
S7	0	-0.0972	0.1221	-0.3303	0.5457	-0.6222	0.4555	-0.1995	0.0405	0
S8	0	-0.1596	0.2027	-0.3281	0.3412	-0.2472	0.1212	-0.0385	0.0064	0

TABLE 10-continued

	K	A	B	C	D	E	F	G	H	J
S9	-18.27	-0.0564	-0.0069	0.0518	-0.0566	0.0228	-0.0011	-0.0019	0.0004	0
S10	-15.127	-0.0603	-0.0145	0.0594	-0.0601	0.0318	-0.0096	0.0015	-1E-04	0
S11	0	0.0027	-0.0398	0.025	-0.0137	0.005	-0.001	1E-04	-4E-06	0
S12	-1.1693	0.1224	-0.1006	0.0535	-0.0195	0.005	-0.0008	8E-05	-3E-06	0
S13	-4.4446	-0.097	-0.0137	0.0358	-0.0141	0.0028	-0.0003	2E-05	-5E-07	0
S14	-8.7431	-0.0906	0.0342	-0.009	0.0017	-0.0002	2E-05	-1E-06	3E-08	0

Sixth Example

[0176] FIG. 11 is a view illustrating a sixth example of an optical imaging system, and FIG. 12 illustrates aberration curves of the optical imaging system of FIG. 11.

[0177] An optical imaging system 6 includes a first lens 1006, a second lens 2006, a third lens 3006, a fourth lens 4006, a fifth lens 5006, a sixth lens 6006, and a seventh lens 7006.

[0178] The first lens 1006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6006. The seventh lens 7006 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7006.

[0179] The optical imaging system 6 further includes a stop, a filter 8006, and an image sensor 9006. The stop is disposed between the first lens 1006 and the second lens 2006 to adjust an amount of light incident on the image

sensor 9006. The filter 8006 is disposed between the seventh lens 7006 and the image sensor 9006 to block infrared rays. The image sensor 9006 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 11, the stop is disposed at a distance of 0.352 mm from the object-side surface of the first lens 1006 toward the imaging plane of the optical imaging system 6. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 6 listed in Table 59 that appears later in this application.

[0180] Table 11 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 11, and Table 12 below shows aspherical surface coefficients of the lenses of FIG. 11.

TABLE 11

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.0068	0.3061	1.546	56.114	1.269
S2	Lens	1.7875	0.0460			1.259
S3 (Stop)	Second	1.5501	0.6732	1.546	56.114	1.228
S4	Lens	26.0899	0.0460			1.162
S5	Third	6.5780	0.2023	1.679	19.236	1.090
S6	Lens	2.6872	0.3921			0.966
S7	Fourth	43.1399	0.2870	1.646	23.528	1.017
S8	Lens	16.1710	0.2406			1.159
S9	Fifth	2.0832	0.2483	1.646	23.528	1.449
S10	Lens	2.1281	0.3430			1.675
S11	Sixth	7.8332	0.5589	1.546	56.114	1.986
S12	Lens	-1.8272	0.3478			2.083
S13	Seventh	-4.3370	0.3310	1.546	56.114	2.556
S14	Lens	1.7397	0.1457			2.669
S15	Filter	Infinity	0.1100	1.519	64.166	2.953
S16		Infinity	0.5893			2.983
S17	Imaging Plane	Infinity	0.0100			3.258

TABLE 12

	K	A	B	C	D	E	F	G	H	J
S1	-3.5715	0.0006	0.0017	-0.0326	0.0053	0.027	-0.0251	0.0092	-0.0013	0
S2	-9.1496	-0.0659	-0.0083	0.0208	0.0344	-0.0521	0.0231	-0.0036	0	0
S3	-2.5622	-0.113	0.1695	-0.2165	0.3458	-0.3334	0.1719	-0.0415	0.0024	0
S4	-90	-0.1003	0.3198	-0.7886	1.361	-1.5481	1.1116	-0.4533	0.0785	0
S5	0	-0.1458	0.4526	-0.9799	1.5947	-1.8113	1.3465	-0.5778	0.107	0
S6	4.6946	-0.0906	0.2181	-0.3857	0.425	-0.2404	-0.0421	0.1403	-0.0569	0
S7	0	-0.125	0.1857	-0.5941	1.161	-1.5655	1.3554	-0.7021	0.1686	0
S8	0	-0.2053	0.3083	-0.5902	0.7259	-0.622	0.3607	-0.1355	0.0267	0
S9	-18.27	-0.0725	-0.0104	0.0931	-0.1203	0.0574	-0.0031	-0.0068	0.0017	0
S10	-15.127	-0.0776	-0.022	0.1069	-0.1279	0.0799	-0.0284	0.0054	-0.0004	0
S11	0	0.0035	-0.0606	0.045	-0.0292	0.0125	-0.0029	0.0003	-2E-05	0
S12	-1.1693	0.1574	-0.153	0.0962	-0.0415	0.0126	-0.0025	0.0003	-1E-05	0
S13	-4.4446	-0.1248	-0.0209	0.0644	-0.0301	0.007	-0.0009	6E-05	-2E-06	0
S14	-8.7431	-0.1165	0.052	-0.0161	0.0035	-0.0006	7E-05	-4E-06	1E-07	0

Seventh Example

[0181] FIG. 13 is a view illustrating a seventh example of an optical imaging system, and FIG. 14 illustrates aberration curves of the optical imaging system of FIG. 13.

[0182] An optical imaging system 7 includes a first lens 1007, a second lens 2007, a third lens 3007, a fourth lens 4007, a fifth lens 5007, a sixth lens 6007, and a seventh lens 7007.

[0183] The first lens 1007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2007 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3007 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5007 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6007. The seventh lens 7007 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7007.

FIG. 13, and Table 14 below shows aspherical surface coefficients of the lenses of FIG. 13.

TABLE 13

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.73233	0.73124	1.546	56.114	1.250
S2 (Stop)	Lens	12.53699	0.07002	1.181		
S3	Second	5.58930	0.20000	1.667	20.353	1.147
S4	Lens	2.57397	0.39715	1.100		
S5	Third	8.06552	0.38474	1.546	56.114	1.128
S6	Lens	7.83668	0.19259	1.247		
S7	Fourth	6.68716	0.24423	1.546	56.114	1.276
S8	Lens	30.32847	0.27130	1.374		
S9	Fifth	-3.28742	0.24968	1.667	20.353	1.481
S10	Lens	-4.51593	0.13884	1.734		
S11	Sixth	5.67988	0.51987	1.546	56.114	2.150
S12	Lens	-1.89003	0.31663	2.318		
S13	Seventh	-3.93255	0.30000	1.546	56.114	2.640
S14	Lens	1.74183	0.19371	2.747		
S15	Filter	Infinity	0.11000	1.518	64.166	3.146
S16		Infinity	0.77000			3.177
S17	Imaging Plane	Infinity	0.01000			3.536

TABLE 14

	K	A	B	C	D	E	F	G	H	J
S1	-0.7464	0.0139	0.0344	-0.0749	0.1029	-0.0706	0.0173	0.0042	-0.0023	0
S2	36.669	-0.0823	0.195	-0.3067	0.3634	-0.323	0.1902	-0.0632	0.0086	0
S3	-1.3559	-0.1603	0.3305	-0.4059	0.3324	-0.1787	0.0673	-0.0166	0.0018	0
S4	-0.4109	-0.0907	0.1444	0.1155	-0.7969	1.5009	-1.4406	0.7219	-0.147	0
S5	0	-0.0739	0.0463	-0.1203	0.1165	-0.0578	-0.0089	0.0233	-0.0057	0
S6	0	-0.0932	0.0034	0.0521	-0.1827	0.2457	-0.2173	0.1126	-0.0241	0
S7	25.148	-0.1235	-0.1887	0.3763	-0.554	0.6731	-0.5796	0.2782	-0.0538	0
S8	-99	-9E-05	-0.3274	0.3588	-0.3195	0.3451	-0.2608	0.0995	-0.0144	0
S9	-70.894	0.0205	0.0483	-0.5284	0.7583	-0.4915	0.1636	-0.0271	0.0018	0
S10	2.2832	0.1759	-0.3448	0.2283	-0.0716	0.011	-0.0007	-4E-06	1E-06	0
S11	-99	0.1188	-0.2169	0.1675	-0.0871	0.0276	-0.0049	0.0005	-2E-05	0
S12	-3.3067	0.1644	-0.1849	0.1159	-0.049	0.0138	-0.0024	0.0002	-9E-06	0
S13	-2.4772	-0.1026	-0.0482	0.074	-0.0308	0.0067	-0.0008	6E-05	-2E-06	0
S14	-1.1028	-0.2935	0.2033	-0.1127	0.0457	-0.0129	0.0024	-0.0003	2E-05	-5E-07

[0184] The optical imaging system 7 further includes a stop, a filter 8007, and an image sensor 9007. The stop is disposed between the first lens 1007 and the second lens 2007 to adjust an amount of light incident on the image sensor 9007. The filter 8007 is disposed between the seventh lens 7007 and the image sensor 9007 to block infrared rays. The image sensor 9007 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 13, the stop is disposed at a distance of 0.731 mm from the object-side surface of the first lens 1007 toward the imaging plane of the optical imaging system 7. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 7 listed in Table 59 that appears later in this application.

[0185] Table 13 below shows physical properties of the lenses and other elements of the optical imaging system of

Eighth Example

[0186] FIG. 15 is a view illustrating an eighth example of an optical imaging system, and FIG. 16 illustrates aberration curves of the optical imaging system of FIG. 15.

[0187] An optical imaging system 8 includes a first lens 1008, a second lens 2008, a third lens 3008, a fourth lens 4008, a fifth lens 5008, a sixth lens 6008, and a seventh lens 7008.

[0188] The first lens 1008 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2008 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3008 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4008 has a

positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5008** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6008** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6008**. The seventh lens **7008** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the

TABLE 15-continued

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S12	Lens	-1.85438	0.30690			2.100
S13	Seventh	-4.06715	0.30000	1.546	56.114	2.300
S14	Lens	1.68940	0.22233			2.631
S15	Filter	Infinity	0.11000	1.518	64.166	2.912
S16		Infinity	0.77152			2.942
S17	Imaging Plane	Infinity	0.00847			3.281

TABLE 16

	K	A	B	C	D	E	F	G	H	J
S1	-0.785	0.0169	0.0171	-0.0159	-0.0114	0.0583	-0.067	0.0338	-0.0066	0
S2	45.58	-0.0601	0.1355	-0.2175	0.2636	-0.2331	0.1319	-0.0411	0.0051	0
S3	1.584	-0.1444	0.2883	-0.3594	0.3009	-0.1422	0.021	0.0121	-0.0046	0
S4	-0.5948	-0.0935	0.1656	-0.0201	-0.372	0.7699	-0.7293	0.3536	-0.0683	0
S5	0	-0.0677	0.0256	-0.0658	0.0582	-0.0383	0.013	0.0013	-0.0005	0
S6	0	-0.0847	-0.0499	0.1996	-0.3976	0.4333	-0.3115	0.1361	-0.026	0
S7	25.588	-0.1098	-0.2727	0.6279	-0.9136	0.9353	-0.6674	0.2828	-0.0509	0
S8	-99	0.0297	-0.4755	0.7447	-0.8912	0.8375	-0.5032	0.1625	-0.0211	0
S9	-66.103	-0.0453	0.1301	-0.541	0.7178	-0.4567	0.1508	-0.0248	0.0016	0
S10	1.6574	0.126	-0.2272	0.1439	-0.0478	0.0097	-0.0013	0.0001	-5E-06	0
S11	-99	0.1115	-0.1711	0.1182	-0.0561	0.0162	-0.0026	0.0002	-8E-06	0
S12	-3.3351	0.1192	-0.1147	0.0615	-0.0222	0.0053	-0.0008	6E-05	-2E-06	0
S13	-2.7152	-0.0748	-0.0688	0.08	-0.032	0.0069	-0.0009	6E-05	-2E-06	0
S14	-1.1007	-0.2576	0.1553	-0.0741	0.0262	-0.0065	0.0011	-0.0001	7E-06	-2E-07

object-side surface and the image-side surface of the seventh lens **7008**.

[0189] The optical imaging system **8** further includes a stop, a filter **8008**, and an image sensor **9008**. The stop is disposed between the first lens **1008** and the second lens **2008** to adjust an amount of light incident on the image sensor **9008**. The filter **8008** is disposed between the seventh lens **7008** and the image sensor **9008** to block infrared rays. The image sensor **9008** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 15, the stop is disposed at a distance of 0.696 mm from the object-side surface of the first lens **1008** toward the imaging plane of the optical imaging system **8**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 8 listed in Table 59 that appears later in this application.

[0190] Table 15 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 15, and Table 16 below shows aspherical surface coefficients of the lenses of FIG. 15.

TABLE 15

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.76093	0.69658	1.546	56.114	1.285
S2 (Stop)	Lens	12.78634	0.09190			1.237
S3	Second	5.69283	0.19245	1.667	20.353	1.198
S4	Lens	2.65937	0.39218			1.100
S5	Third	8.20428	0.41830	1.546	56.114	1.143
S6	Lens	7.91545	0.16876			1.265
S7	Fourth	6.75201	0.26903	1.546	56.114	1.284
S8	Lens	55.74455	0.23097			1.367
S9	Fifth	-3.08690	0.42919	1.667	20.353	1.418
S10	Lens	-4.62560	0.09404			1.699
S11	Sixth	5.18773	0.58281	1.546	56.114	2.150

Ninth Example

[0191] FIG. 17 is a view illustrating a ninth example of an optical imaging system, and FIG. 18 illustrates aberration curves of the optical imaging system of FIG. 17.

[0192] An optical imaging system **9** includes a first lens **1009**, a second lens **2009**, a third lens **3009**, a fourth lens **4009**, a fifth lens **5009**, a sixth lens **6009**, and a seventh lens **7009**.

[0193] The first lens **1009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6009**. The seventh lens **7009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7009**, and one inflection point is formed on the image-side surface of the seventh lens **7009**.

[0194] The optical imaging system **9** further includes a stop, a filter **8009**, and an image sensor **9009**. The stop is disposed between the second lens **2009** and the third lens

3009 to adjust an amount of light incident on the image sensor **9009**. The filter **8009** is disposed between the seventh lens **7009** and the image sensor **9009** to block infrared rays. The image sensor **9009** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **17**, the stop is disposed at a distance of 1.158 mm from the object-side surface of the first lens **1009** toward the imaging plane of the optical imaging system **9**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 9 listed in Table 59 that appears later in this application.

[0195] Table 17 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **17**, and Table 18 below shows aspherical surface coefficients of the lenses of FIG. **17**.

TABLE 17

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.141	0.481	1.546	56.114	1.450
S2	Lens	3.251	0.110			1.350
S3	Second	3.253	0.542	1.546	56.114	1.285
S4	Lens	-15.773	0.025			1.232
S5 (Stop)	Third	8.425	0.230	1.679	19.236	1.157
S6	Lens	3.514	0.625			1.095
S7	Fourth	25.986	0.296	1.679	19.236	1.265
S8	Lens	15.894	0.230			1.452
S9	Fifth	3.048	0.400	1.546	56.114	1.675
S10	Lens	3.616	0.290			2.092
S11	Sixth	3.762	0.400	1.679	19.236	2.153
S12	Lens	2.792	0.204			2.476
S13	Seventh	1.614	0.510	1.537	53.955	2.938
S14	Lens	1.326	0.196			3.102
S15	Filter	Infinity	0.110	1.518	64.197	3.420
S16	Infinity	Infinity	0.639			3.450
S17	Imaging Plane	Infinity	0.011			3.730

TABLE 18

	K	A	B	C	D	E	F	G	H	J
S1	-8.038	0.0707	-0.0797	0.0334	0.0072	-0.0491	0.0465	-0.0186	0.0032	-0.0002
S2	-20.594	-0.0019	-0.1494	0.2041	-0.2922	0.3755	-0.3085	0.1486	-0.0387	0.0042
S3	-0.0908	-0.0339	-0.0641	0.1368	-0.2821	0.4921	-0.4815	0.2605	-0.0746	0.0088
S4	-0.4822	-0.0436	0.1761	-0.3256	0.1999	0.1916	-0.4291	0.3203	-0.1141	0.0162
S5	-1.1841	-0.1073	0.2544	-0.4683	0.4991	-0.2863	0.0565	0.0325	-0.0229	0.0044
S6	0.8733	-0.0693	0.0357	0.2048	-0.8833	1.7328	-1.9742	1.3464	-0.5106	0.083
S7	-0.4999	-0.0314	0.0135	-0.2894	0.9716	-1.7181	1.7923	-1.1152	0.3837	-0.0563
S8	-1E-06	-0.0273	-0.1177	0.212	-0.2544	0.2157	-0.1264	0.0469	-0.0093	0.0007
S9	-41.843	0.1624	-0.3487	0.4016	-0.3105	0.1396	-0.027	-0.0038	0.0026	-0.0003
S10	-5.1424	0.0397	-0.1364	0.1569	-0.1229	0.0633	-0.0212	0.0044	-0.0005	3E-05
S11	-2.1666	0.0056	-0.1809	0.1985	-0.1438	0.0641	-0.0173	0.0028	-0.0002	9E-06
S12	-0.0207	-0.1043	0.0239	-0.0063	-0.0007	0.0007	-3E-06	-4E-05	7E-06	-4E-07
S13	-0.7948	-0.4128	0.1863	-0.0516	0.0101	-0.0015	0.0002	-1E-05	6E-07	-1E-08
S14	-1.3226	-0.3105	0.1713	-0.0712	0.0213	-0.0043	0.0006	-5E-05	2E-06	-5E-08

Tenth Example

[0196] FIG. **19** is a view illustrating a tenth example of an optical imaging system, and FIG. **20** illustrates aberration curves of the optical imaging system of FIG. **19**.

[0197] An optical imaging system **10** includes a first lens **1010**, a second lens **2010**, a third lens **3010**, a fourth lens

4010, a fifth lens **5010**, a sixth lens **6010**, and a seventh lens **7010**.

[0198] The first lens **1010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6010**. The seventh lens **7010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7010**, and one inflection point is formed on the image-side surface of the seventh lens **7010**.

[0199] The optical imaging system **10** further includes a stop, a filter **8010**, and an image sensor **9010**. The stop is disposed between the second lens **2010** and the third lens **3010** to adjust an amount of light incident on the image sensor **9010**. The filter **8010** is disposed between the seventh lens **7010** and the image sensor **9010** to block infrared rays. The image sensor **9010** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **19**, the stop is disposed at a distance of 1.179 mm from the object-side surface of the first lens **1010** toward the imaging

plane of the optical imaging system **10**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 10 listed in Table 59 that appears later in this application.

[0200] Table 19 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **19**, and Table 20 below shows aspherical surface coefficients of the lenses of FIG. **19**.

TABLE 19

Surface No.	Element	Radius of Curvature	Thick-ness/ Distance	Index of Re- fraction	Abbe Number	Effective Aperture Radius
S1	First	2.3878602	0.35	1.546	56.114	1.470
S2	Lens	2.62132398	0.10425			1.439
S3	Second	2.53822644	0.7	1.546	56.114	1.405
S4	Lens	36.207255	0.025			1.322
S5 (Stop)	Third	7.79607111	0.23	1.679	19.236	1.287
S6	Lens	3.49120755	0.3961335			1.325
S7	Fourth	5.43312406	0.5053863	1.546	56.114	1.461
S8	Lens	9.39050621	0.4436731			1.563
S9	Fifth	4.67888662	0.4842217	1.546	56.114	1.772
S10	Lens	9.57328318	0.4757916			2.209
S11	Sixth	7.18942036	0.4753531	1.679	19.236	2.238
S12	Lens	3.83971065	0.2154454			2.557
S13	Seventh	1.81373827	0.5598542	1.546	56.114	3.026
S14	Lens	1.38723655	0.2300778			3.262
S15	Filter	Infinity	0.11	1.518	64.197	3.724
S16		Infinity	0.6349996			3.763
S17	Imaging Plane	Infinity	0.015			4.160

TABLE 20

	K	A	B	C	D	E	F	G	H	J
S1	-0.9977	-0.0046	-0.0139	-0.024	0.0712	-0.0925	0.0658	-0.0257	0.0051	-0.0004
S2	-10.008	0.0879	-0.2624	0.4378	-0.5581	0.4751	-0.2476	0.0753	-0.0122	0.0008
S3	-0.5121	-0.0082	0.0297	-0.2049	0.402	-0.4509	0.3219	-0.1392	0.0325	-0.0031
S4	42.587	-0.0077	-0.0816	0.5107	-1.169	1.3206	-0.8193	0.2839	-0.0514	0.0038
S5	14.891	-0.0724	0.0409	0.4245	-1.2505	1.5731	-1.0799	0.4229	-0.0892	0.0079
S6	1.8054	-0.0886	0.1906	-0.2777	0.2837	-0.2167	0.1131	-0.0327	0.0033	0.0002
S7	-10.152	0.028	-0.4038	1.2639	-2.2626	2.4845	-1.6998	0.7029	-0.16	0.0153
S8	1.7534	-0.0643	0.1117	-0.3935	0.7438	-0.8306	0.565	-0.2306	0.0519	-0.005
S9	-44.62	0.0891	-0.1333	0.115	-0.0807	0.043	-0.017	0.0044	-0.0006	4E-05
S10	-4.9001	0.0976	-0.137	0.0999	-0.0465	0.0132	-0.0022	0.0002	-1E-05	2E-07
S11	-13.159	0.0954	-0.1535	0.1202	-0.0703	0.0264	-0.006	0.0008	-6E-05	2E-06
S12	0.7792	-0.0211	-0.0288	0.0188	-0.0093	0.003	-0.0006	6E-05	-4E-06	9E-08
S13	-0.8786	-0.3003	0.0918	-0.0015	-0.0073	0.0023	-0.0004	3E-05	-1E-06	3E-08
S14	-1.3086	-0.2504	0.1203	-0.0427	0.0114	-0.0021	0.0003	-2E-05	9E-07	-2E-08

Eleventh Example

[0201] FIG. 21 is a view illustrating an eleventh example of an optical imaging system, and FIG. 22 illustrates aberration curves of the optical imaging system of FIG. 21.

[0202] An optical imaging system 11 includes a first lens 1011, a second lens 2011, a third lens 3011, a fourth lens 4011, a fifth lens 5011, a sixth lens 6011, and a seventh lens 7011.

[0203] The first lens 1011 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2011 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3011 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4011 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5011 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6011 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface

of the sixth lens 6011. The seventh lens 7011 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7011, and one inflection point is formed on the image-side surface of the seventh lens 7011.

[0204] The optical imaging system 11 further includes a stop, a filter 8011, and an image sensor 9011. The stop is disposed between the second lens 2011 and the third lens 3011 to adjust an amount of light incident on the image sensor 9011. The filter 8011 is disposed between the seventh lens 7011 and the image sensor 9011 to block infrared rays. The image sensor 9011 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 21, the stop is disposed at a distance of 1.205 mm from the object-side surface of the first lens 1011 toward the imaging plane of the optical imaging system 11. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 11 listed in Table 59 that appears later in this application.

[0205] Table 21 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 21, and Table 22 below shows aspherical surface coefficients of the lenses of FIG. 21.

TABLE 21

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.34218008	0.35	1.546	56.114	1.570
S2	Lens	2.46082267	0.130273			1.582
S3	Second	2.30906902	0.7	1.546	56.114	1.578
S4	Lens	38.8335574	0.025			1.553
S5 (Stop)	Third	7.6526629	0.2529802	1.679	19.236	1.474
S6	Lens	3.32721743	0.5047159			1.325
S7	Fourth	5.67437537	0.3466492	1.679	19.236	1.461
S8	Lens	6.58110754	0.40646			1.563
S9	Fifth	4.79255093	0.35	1.546	56.114	1.772
S10	Lens	10.114446	0.5278215			2.209
S11	Sixth	6.16226916	0.43	1.679	19.236	2.238
S12	Lens	3.92995117	0.2624604			2.557
S13	Seventh	1.96658005	0.5643968	1.546	56.114	3.026
S14	Lens	1.49045699	0.2268426			3.262
S15	Filter	Infinity	0.11	1.518	64.197	3.641
S16		Infinity	0.7973487			3.681
S17	Imaging Plane	Infinity	0.015			4.171

TABLE 22

	K	A	B	C	D	E	F	G	H	J
S1	-1	-0.0146	0.01	-0.0522	0.0643	-0.0468	0.0217	-0.0061	0.0009	-6E-05
S2	-9.9316	0.0529	-0.0805	0.0187	0.0015	0.0183	-0.0206	0.0089	-0.0018	0.0001
S3	-0.3035	-0.0312	0.1094	-0.2703	0.3068	-0.1783	0.0505	-0.0026	-0.0019	0.0003
S4	42.587	-0.14	0.5905	-1.3419	1.845	-1.6056	0.8858	-0.299	0.0561	-0.0045
S5	14.878	-0.1698	0.5798	-1.2212	1.6455	-1.4398	0.807	-0.2778	0.0533	-0.0044
S6	2.4782	-0.0974	0.3302	-0.8477	1.4713	-1.6606	1.1916	-0.5238	0.1288	-0.0136
S7	-10.89	-0.0126	-0.1352	0.3555	-0.5088	0.426	-0.2043	0.0479	-0.0019	-0.0008
S8	2.4559	-0.0438	-0.0219	-0.0159	0.1351	-0.2288	0.1952	-0.0929	0.0235	-0.0025
S9	-44.519	0.0902	-0.1408	0.1256	-0.0754	0.0253	-0.0032	-0.0008	0.0003	-3E-05
S10	2.7864	0.0708	-0.1198	0.1033	-0.057	0.0193	-0.004	0.0005	-3E-05	9E-07
S11	-17.823	0.0653	-0.1084	0.075	-0.0394	0.0131	-0.0026	0.0003	-2E-05	5E-07
S12	0.5903	-0.0273	-0.0096	0.0039	-0.0025	0.001	-0.0002	2E-05	-1E-06	3E-08
S13	-0.8439	-0.2851	0.1057	-0.0237	0.0034	-0.0003	5E-07	2E-06	-1E-07	4E-09
S14	-1.3672	-0.238	0.119	-0.0473	0.0138	-0.0027	0.0003	-3E-05	1E-06	-2E-08

Twelfth Example

[0206] FIG. 23 is a view illustrating a twelfth example of an optical imaging system, and FIG. 24 illustrates aberration curves of the optical imaging system of FIG. 23.

[0207] An optical imaging system 12 includes a first lens 1012, a second lens 2012, a third lens 3012, a fourth lens 4012, a fifth lens 5012, a sixth lens 6012, and a seventh lens 7012.

[0208] The first lens 1012 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2012 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3012 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4012 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5012 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6012 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of

the sixth lens 6012. The seventh lens 7012 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7012.

[0209] The optical imaging system 12 further includes a stop, a filter 8012, and an image sensor 9012. The stop is disposed in front of the first lens 1012 to adjust an amount of light incident on the image sensor 9012. The filter 8012 is disposed between the seventh lens 7012 and the image sensor 9012 to block infrared rays. The image sensor 9012 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 23, the stop is disposed at a distance of 0.250 mm from the object-side surface of the first lens 1012 toward the imaging plane of the optical imaging system 12. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 12 listed in Table 59 that appears later in this application.

[0210] Table 23 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 23, and Table 24 below shows aspherical surface coefficients of the lenses of FIG. 23.

TABLE 23

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S1	First Lens	1.7211	0.6349	1.544	56.114	1.100
(Stop)						
S2		11.4571	0.1212			1.071
S3	Second Lens	119.1721	0.2033	1.661	20.353	1.057
S4		4.4758	0.0843			1.043
S5	Third Lens	4.5258	0.3109	1.544	56.114	1.051
S6		20.6082	0.2158			1.015
S7	Fourth Lens	13.2152	0.2369	1.544	56.114	1.019
S8		16.2733	0.2103			1.070
S9	Fifth Lens	-6.5732	0.4119	1.651	21.494	1.076
S10		-10.4553	0.3710			1.320
S11	Sixth Lens	3.4779	0.6318	1.544	56.114	1.556
S12		3.1994	0.2672			2.337
S13	Seventh Lens	2.8804	0.5060	1.544	56.114	2.489
S14		1.7054	0.1384			2.666
S15	Filter	Infinity	0.2100			3.102
S16		Infinity	0.5794			3.177
S17	Imaging Plane	Infinity	0.0106			3.529

both of the object-side surface and the image-side surface of the sixth lens **6013**. The seventh lens **7013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7013**, and one inflection point is formed on the image-side surface of the seventh lens **7013**.

[0214] The optical imaging system **13** further includes a stop, a filter **8013**, and an image sensor **9013**. The stop is disposed in front of the first lens **1013** to adjust an amount of light incident on the image sensor **9013**. The filter **8013** is disposed between the seventh lens **7013** and the image sensor **9013** to block infrared rays. The image sensor **9013** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **25**, the stop is disposed at a distance of 0.200 mm from the object-side surface of the first lens **1013** toward the imaging plane of the optical imaging system **13**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example **13** listed in Table **59** that appears later in this application.

TABLE 24

	K	A	B	C	D	E	F	G	H
S1	0.0432	-0.0088	0.0131	-0.0627	0.1199	-0.1345	0.077	-0.018	-0.0004
S2	-26.097	-0.0562	0.051	-0.0514	0.0595	-0.0683	0.0462	-0.0139	-7E-05
S3	-99	-0.1283	0.1953	-0.2779	0.5135	-0.8812	0.9662	-0.5723	0.1395
S4	-16.567	-0.0971	0.1552	-0.3608	0.985	-2.059	2.5647	-1.6683	0.4378
S5	-1.6774	-0.0377	0.065	-0.4515	1.687	-3.5163	4.2391	-2.6607	0.6752
S6	57.913	-0.0559	0.0533	-0.341	1.3373	-2.8539	3.4811	-2.2114	0.5781
S7	-66.305	-0.1749	-0.0635	0.0963	-0.2061	0.5819	-0.9	0.6874	-0.1979
S8	19.549	-0.1228	-0.0686	0.0207	0.1647	-0.2695	0.1725	-0.0616	0.0161
S9	29.709	-0.0709	0.0826	-0.3062	0.6009	-0.6459	0.3344	-0.0761	0
S10	-31.338	-0.1255	0.1076	-0.1494	0.1908	-0.1423	0.0506	-0.0065	0
S11	-46.453	0.0038	-0.1455	0.1534	-0.126	0.0705	-0.0225	0.0029	0
S12	-31.504	0.0093	-0.0326	0.0149	-0.0033	0.0003	-1E-05	-7E-07	0
S13	-0.5233	-0.2947	0.1709	-0.0627	0.0154	-0.0025	0.0003	-1E-05	3E-07
S14	-0.8257	-0.2584	0.1353	-0.0565	0.0166	-0.0032	0.0004	-3E-05	7E-07

Thirteenth Example

[0211] FIG. **25** is a view illustrating a thirteenth example of an optical imaging system, and FIG. **26** illustrates aberration curves of the optical imaging system of FIG. **25**.

[0212] An optical imaging system **13** includes a first lens **1013**, a second lens **2013**, a third lens **3013**, a fourth lens **4013**, a fifth lens **5013**, a sixth lens **6013**, and a seventh lens **7013**.

[0213] The first lens **1013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5013** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or

[0215] Table **25** below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **25**, and Table **26** below shows aspherical surface coefficients of the lenses of FIG. **25**.

TABLE 25

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S1	First Lens	2.0468	0.6797	1.544	56.114	1.100
(Stop)						
S2		15.4382	0.1358			1.127
S3	Second Lens	5.7788	0.2300	1.661	20.353	1.142
S4		2.6751	0.2609			1.153
S5	Third Lens	4.0817	0.2952	1.544	56.114	1.246
S6		7.3687	0.3995			1.258
S7	Fourth Lens	7.0190	0.3141	1.544	56.114	1.340
S8		7.7267	0.2822			1.388
S9	Fifth Lens	-9.0899	0.3555	1.661	20.353	1.405
S10		-9.9293	0.1000			1.576
S11	Sixth Lens	3.0002	0.6701	1.544	56.114	1.700
S12		3.7492	0.3200			2.316
S13	Seventh Lens	2.9879	0.6382	1.535	55.712	2.566
S14		1.6763	0.1830			2.822
S15	Filter	Infinity	0.2100			2.988
S16		Infinity	0.5699			3.045

TABLE 25-continued

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S17	Imaging Plane	Infinity	0.0084			3.298

2014 to adjust an amount of light incident on the image sensor **9014**. The filter **8014** is disposed between the seventh lens **7014** and the image sensor **9014** to block infrared rays. The image sensor **9014** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 27, the stop is disposed at a distance of 0.768 mm from the object-side surface of the first lens **1014** toward the imaging plane of the optical imaging system **14**. This distance is

TABLE 26

	K	A	B	C	D	E	F	G	H
S1	0.1134	-0.0041	0.0058	-0.0209	0.0313	-0.0267	0.0115	-0.0022	4E-05
S2	-31.971	-0.0343	0.0186	0.018	-0.0672	0.0895	-0.0703	0.0307	-0.0057
S3	-61.479	-0.0628	0.0273	0.0375	-0.0593	0.0197	0.0233	-0.0229	0.0061
S4	-5.9901	-0.0361	0.0057	0.1204	-0.3118	0.4572	-0.3962	0.1898	-0.0382
S5	0.4276	-0.01	-0.0036	-0.0725	0.164	-0.1983	0.1346	-0.041	0.0037
S6	-16.707	-0.0183	-0.0111	0.0191	-0.1026	0.1931	-0.1846	0.0944	-0.0194
S7	-17.469	-0.087	0.0448	-0.1275	0.1878	-0.176	0.0891	-0.0173	0.0002
S8	-45.166	-0.0908	0.0709	-0.1666	0.2236	-0.2004	0.1091	-0.0324	0.0042
S9	9.4915	-0.0528	0.0726	-0.1573	0.1702	-0.1138	0.0418	-0.0065	0
S10	27.883	-0.0964	0.0215	-0.01	0.0057	-0.0007	-0.0003	7E-05	0
S11	-18.4	0.0175	-0.105	0.0871	-0.0593	0.0265	-0.0067	0.0007	0
S12	-27.836	0.0419	-0.0472	0.0168	-0.0037	0.0005	-4E-05	1E-06	0
S13	-0.521	-0.2204	0.0759	-0.0132	0.0012	-4E-05	-3E-06	3E-07	-7E-09
S14	-0.8591	-0.2015	0.0852	-0.0311	0.0079	-0.0012	0.0001	-6E-06	1E-07

Fourteenth Example

[0216] FIG. 27 is a view illustrating a fourteenth example of an optical imaging system, and FIG. 28 illustrates aberration curves of the optical imaging system of FIG. 27.

[0217] An optical imaging system **14** includes a first lens **1014**, a second lens **2014**, a third lens **3014**, a fourth lens **4014**, a fifth lens **5014**, a sixth lens **6014**, and a seventh lens **7014**.

[0218] The first lens **1014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2014** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5014** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6014**. The seventh lens **7014** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7014**.

[0219] The optical imaging system **14** further includes a stop, a filter **8014**, and an image sensor **9014**. The stop is disposed between the first lens **1014** and the second lens

equal to TTL-SL and can be calculated from the values of TTL and SL for Example 14 listed in Table 59 that appears later in this application.

[0220] Table 27 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 27, and Table 28 below shows aspherical surface coefficients of the lenses of FIG. 27.

TABLE 27

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S1	First	1.9548	0.7678	1.544	56.114	1.300
S2	Lens	7.8859	0.1005			1.209
(Stop)						
S3	Second	4.8288	0.2366	1.661	20.353	1.213
S4	Lens	2.8662	0.4493			1.265
S5	Third	8.8655	0.4845	1.544	56.114	1.268
S6	Lens	16.6746	0.2752			1.403
S7	Fourth	10.6715	0.3698	1.544	56.114	1.456
S8	Lens	22.4472	0.2801			1.642
S9	Fifth	-4.3816	0.2711	1.661	20.353	1.769
S10	Lens	-4.3828	0.1050			2.019
S11	Sixth	7.9522	0.5677	1.544	56.114	2.357
S12	Lens	-3.0368	0.4648			2.647
S13	Seventh	-7.5079	0.3200	1.544	56.114	3.123
S14	Lens	1.7962	0.1891			3.381
S15	Filter	Infinity	0.1100			3.605
S16		Infinity	0.6749			3.635
S17	Imaging Plane	Infinity	-0.0163			3.930

TABLE 28

	K	A	B	C	D	E	F	G	H	J
S1	-0.8127	0.0142	0.0092	-0.0157	0.0206	-0.0137	0.0037	0.0003	-0.0003	0
S2	5.6538	-0.0472	0.0448	-0.0321	0.0158	-0.0059	0.001	0.0004	-0.0002	0
S3	-10.668	-0.0824	0.0792	-0.0266	-0.0158	0.0274	-0.0153	0.0039	-0.0004	0
S4	-0.1737	-0.0508	0.0303	0.1129	-0.3063	0.4131	-0.3101	0.1243	-0.0205	0
S5	0	-0.0377	0.0156	-0.0597	0.0773	-0.0624	0.0268	-0.0045	3E-05	0
S6	0	-0.0706	0.0482	-0.0575	-0.0009	0.0419	-0.0392	0.0166	-0.0028	0
S7	46.114	-0.1374	0.0451	0.0051	-0.0298	0.0052	0.0076	-0.0027	0.0001	0
S8	99	-0.1096	-0.0451	0.1394	-0.1519	0.0948	-0.0333	0.006	-0.0004	0
S9	-99	-0.0865	0.1152	-0.1605	0.1182	-0.0466	0.0099	-0.0011	5E-05	0
S10	-0.2245	0.0593	-0.0542	0.0004	0.0119	-0.0044	0.0007	-5E-05	1E-06	0
S11	-99	0.1031	-0.1094	0.0579	-0.0216	0.005	-0.0007	4E-05	-1E-06	0
S12	-4.7232	0.1521	-0.1221	0.0592	-0.0202	0.0046	-0.0007	5E-05	-2E-06	0
S13	-1.1986	-0.0323	-0.0724	0.0507	-0.0141	0.0021	-0.0002	8E-06	-2E-07	0
S14	-1.2644	-0.1675	0.0662	-0.0204	0.0047	-0.0007	8E-05	-5E-06	2E-07	-2E-09

Fifteenth Example

[0221] FIG. 29 is a view illustrating a fifteenth example of an optical imaging system, and FIG. 30 illustrates aberration curves of the optical imaging system of FIG. 29.

[0222] An optical imaging system 15 includes a first lens 1015, a second lens 2015, a third lens 3015, a fourth lens 4015, a fifth lens 5015, a sixth lens 6015, and a seventh lens 7015.

[0223] The first lens 1015 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2015 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3015 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4015 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens 5015 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6015 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6015. The seventh lens 7015 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7015.

[0224] The optical imaging system 15 further includes a stop, a filter 8015, and an image sensor 9015. The stop is

disposed between the first lens 1015 and the second lens 2015 to adjust an amount of light incident on the image sensor 9015. The filter 8015 is disposed between the seventh lens 7015 and the image sensor 9015 to block infrared rays. The image sensor 9015 forms an imaging plane on which an image of a subject is formed.

[0225] Table 29 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 29, and Table 30 below shows aspherical surface coefficients of the lenses of FIG. 29.

TABLE 29

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.7773	0.6238	1.544	56.114	1.217
S2	Lens	6.4566	0.1000			1.158
(Stop)						
S3	Second	4.4103	0.2363	1.661	20.353	1.157
S4	Lens	2.6584	0.4138			1.184
S5	Third	6.5879	0.4640	1.544	56.114	1.177
S6	Lens	10.5233	0.1777			1.282
S7	Fourth	13.4749	0.3627	1.544	56.114	1.306
S8	Lens	-20.2300	0.2325			1.444
S9	Fifth	-3.1831	0.2000	1.661	20.353	1.456
S10	Lens	-4.2151	0.1000			1.625
S11	Sixth	6.7646	0.6089	1.544	56.114	2.207
S12	Lens	-2.8792	0.4211			2.145
S13	Seventh	-6.9958	0.3200	1.544	56.114	2.280
S14	Lens	1.6934	0.1485			3.165
S15	Filter	Infinity	0.1100			2.850
S16		Infinity	0.7007			2.888
S17	Imaging Plane	Infinity	-0.0200			3.276

TABLE 30

	K	A	B	C	D	E	F	G	H	J
S1	-0.5383	0.0108	0.0209	-0.0477	0.0729	-0.06	0.0243	-0.0027	-0.0007	0
S2	5.8135	-0.0459	0.0189	0.0248	-0.0559	0.0486	-0.026	0.0094	-0.0019	0
S3	-10.011	-0.085	0.066	0.02	-0.0808	0.0756	-0.0332	0.0069	-0.0006	0
S4	-0.1875	-0.0544	0.0068	0.26	-0.6655	0.9329	-0.7519	0.3313	-0.061	0
S5	0	-0.0569	0.0063	-0.0275	-0.0046	0.0401	-0.0485	0.0264	-0.0053	0
S6	0	-0.0775	-0.0976	0.271	-0.5329	0.5567	-0.3323	0.1128	-0.0176	0
S7	47.015	-0.0863	-0.1024	0.2298	-0.2721	0.1091	0.0392	-0.0378	0.0065	0
S8	-99	-0.0603	-0.0348	0.057	-0.0468	0.0241	-0.007	0.001	-6E-05	0
S9	-99	-0.2672	0.6153	-0.9745	0.9138	-0.5236	0.1786	-0.0332	0.0026	0
S10	-0.0701	0.0268	-0.0377	-0.0253	0.035	-0.0133	0.0024	-0.0002	7E-06	0
S11	-97.721	0.1556	-0.2109	0.1424	-0.0678	0.02	-0.0033	0.0003	-1E-05	0

TABLE 30-continued

	K	A	B	C	D	E	F	G	H	J
S12	-1.5998	0.2298	-0.1811	0.0905	-0.0342	0.0088	-0.0014	0.0001	-4E-06	0
S13	4.8341	-0.1142	-0.0024	0.0306	-0.013	0.0027	-0.0003	2E-05	-5E-07	0
S14	-1.0993	-0.2618	0.1449	-0.0599	0.0171	-0.0032	0.0004	-3E-05	1E-06	-2E-08

Sixteenth Example

[0226] FIG. 31 is a view illustrating a sixteenth example of an optical imaging system, and FIG. 32 illustrates aberration curves of the optical imaging system of FIG. 31.

[0227] An optical imaging system 16 includes a first lens 1016, a second lens 2016, a third lens 3016, a fourth lens 4016, a fifth lens 5016, a sixth lens 6016, and a seventh lens 7016.

[0228] The first lens 1016 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2016 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3016 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4016 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5016 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6016 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6016. The seventh lens 7016 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens 7016, and one inflection point is formed on the image-side surface of the seventh lens 7016.

[0229] The optical imaging system 16 further includes a stop, a filter 8016, and an image sensor 9016. The stop is disposed between the first lens 1016 and the second lens 2016 to adjust an amount of light incident on the image sensor 9016. The filter 8016 is disposed between the seventh

lens 7016 and the image sensor 9016 to block infrared rays. The image sensor 9016 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 31, the stop is disposed at a distance of 0.641 mm from the object-side surface of the first lens 1016 toward the imaging plane of the optical imaging system 16. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 16 listed in Table 59 that appears later in this application.

[0230] Table 31 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 31, and Table 32 below shows aspherical surface coefficients of the lenses of FIG. 31.

TABLE 31

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.7977	0.6409	1.544	56.114	1.270
S2	Lens	3.7422	0.1191			1.211
(Stop)						
S3	Second	3.0573	0.2200	1.661	20.353	1.190
S4	Lens	2.7951	0.3931			1.130
S5	Third	10.6215	0.4640	1.544	56.114	1.153
S6	Lens	9.0266	0.1000			1.289
S7	Fourth	7.9876	0.3621	1.544	56.114	1.328
S8	Lens	138.7678	0.2334			1.454
S9	Fifth	-4.1765	0.2198	1.661	20.353	1.518
S10	Lens	-4.1394	0.1000			1.656
S11	Sixth	4.6134	0.6089	1.544	56.114	2.000
S12	Lens	-3.5921	0.4726			2.038
S13	Seventh	-7.0016	0.3200	1.544	56.114	2.049
S14	Lens	1.6938	0.1107			2.685
S15	Filter	Infinity	0.2100			2.942
S16		Infinity	0.5300			3.008
S17	Imaging Plane	Infinity	0.0200			3.292

TABLE 32

	K	A	B	C	D	E	F	G	H	J
S1	-0.812	0.0136	0.0311	-0.0769	0.1226	-0.1099	0.0531	-0.0116	0.0005	0
S2	-6.6917	-0.0631	0.0174	0.0714	-0.1648	0.1763	-0.1086	0.0376	-0.0059	0
S3	-14.579	-0.0707	0.0068	0.1319	-0.2129	0.173	-0.0715	0.0127	-0.0005	0
S4	-0.188	-0.0614	-0.0138	0.3338	-0.7392	0.9251	-0.6781	0.276	-0.0477	0
S5	0	-0.0572	0.0435	-0.1733	0.2724	-0.2421	0.0931	-0.0042	-0.0038	0
S6	0	-0.1356	-0.0309	0.2183	-0.5547	0.6931	-0.486	0.1856	-0.0304	0
S7	30.023	-0.2107	0.0007	0.1568	-0.2854	0.2586	-0.1154	0.0236	-0.0019	0
S8	-99	-0.1858	-0.0192	0.2616	-0.4111	0.3392	-0.1538	0.0357	-0.0033	0
S9	-98.995	-0.2935	0.5043	-0.5157	0.2657	-0.0658	0.0056	0.0005	-8E-05	0
S10	-0.0701	-0.0775	0.2223	-0.2703	0.1529	-0.0452	0.0073	-0.0006	2E-05	0
S11	-97.878	0.1479	-0.1956	0.1288	-0.0598	0.0172	-0.0028	0.0002	-8E-06	0
S12	1.4166	0.1234	-0.1416	0.087	-0.0341	0.0088	-0.0014	0.0001	-4E-06	0
S13	9.5503	-0.2864	0.1096	0.0149	-0.0214	0.0064	-0.0009	6E-05	-2E-06	0
S14	-1.2786	-0.3076	0.1777	-0.0626	0.0143	-0.0022	0.0002	-1E-05	5E-07	-7E-09

Seventeenth Example

[0231] FIG. 33 is a view illustrating a seventeenth example of an optical imaging system, and FIG. 34 illustrates aberration curves of the optical imaging system of FIG. 33.

[0232] An optical imaging system 17 includes a first lens 1017, a second lens 2017, a third lens 3017, a fourth lens 4017, a fifth lens 5017, a sixth lens 6017, and a seventh lens 7017.

[0233] The first lens 1017 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2017 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3017 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4017 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5017 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6017 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6017. The seventh lens 7017 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7017, and one inflection point is formed on the image-side surface of the seventh lens 7017.

[0234] The optical imaging system 17 further includes a stop, a filter 8017, and an image sensor 9017. The stop is

[0235] Table 33 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 33, and Table 34 below shows aspherical surface coefficients of the lenses of FIG. 33.

TABLE 33

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.7485	0.6674	1.547	56.114	1.300
S2	Lens	7.7065	0.0200			1.229
S3	Second	1.8213	0.1562	1.660	20.400	1.115
S4	Lens	1.4314	0.4979			1.000
(Stop)						
S5	Third	4.2378	0.1600	1.660	20.400	0.993
S6	Lens	3.6200	0.1668			1.033
S7	Fourth	9.2634	1.0427	1.547	56.114	1.114
S8	Lens	28.8419	0.2336			1.480
S9	Fifth	13.9672	0.2922	1.650	21.494	1.522
S10	Lens	11.4259	0.0356			1.730
S11	Sixth	5.8246	0.4995	1.650	21.494	1.773
S12	Lens	6.8164	0.1833			2.172
S13	Seventh	1.8633	0.5958	1.537	55.711	3.150
S14	Lens	1.4257	0.2454			3.009
S15	Filter	Infinity	0.1100			3.214
S16		Infinity	0.4585			3.253
S17	Imaging Plane	Infinity	0.0150			3.531

TABLE 34

	K	A	B	C	D	E	F	G	H	J
S1	-0.0374	0.0041	0.0102	-0.0274	0.0363	-0.023	0.0071	-0.0007	0	0
S2	29.499	-0.0579	0.1753	-0.2761	0.2564	-0.1451	0.0461	-0.0064	0	0
S3	-1.6808	-0.0951	0.1875	-0.1212	-0.0118	0.0698	-0.0436	0.0084	0	0
S4	-0.1875	-0.0734	-0.0279	0.4656	-0.9504	1.0182	-0.5704	0.132	0	0
S5	1.3424	-0.1042	0.1034	-0.1017	0.1369	-0.1182	0.0626	-0.014	0	0
S6	6.3397	-0.1322	0.0914	-0.0544	0.0396	-0.0158	0.0024	-4E-05	0	0
S7	21.918	-0.0361	0.0211	-0.0154	0.0179	-0.009	0.0019	-0.0001	0	0
S8	25.736	-0.0408	0.011	-0.0055	0.0031	-0.0008	1E-04	-4E-06	0	0
S9	1.6857	-0.0998	0.1128	-0.1386	0.0983	-0.0424	0.0105	-0.0012	0	0
S10	-99	-0.0846	0.0566	-0.0552	0.0277	-0.0068	0.0008	-3E-05	0	0
S11	0	-0.0205	-0.0335	0.0064	0	0	0	0	0	0
S12	0	-0.0019	-0.0483	0.031	-0.0123	0.0029	-0.0004	2E-05	0	0
S13	-0.9427	-0.2417	0.0607	-0.0015	-0.0024	0.0006	-7E-05	4E-06	-8E-08	0
S14	-1.0048	-0.2102	0.0796	-0.0236	0.0052	-0.0008	7E-05	-3E-06	7E-08	0

disposed between the second lens 2017 and the third lens 3017 to adjust an amount of light incident on the image sensor 9017. The filter 8017 is disposed between the seventh lens 7017 and the image sensor 9017 to block infrared rays. The image sensor 9017 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 33, the stop is disposed at a distance of 0.9137 mm from the object-side surface of the first lens 1017 toward the imaging plane of the optical imaging system 17. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 17 listed in Table 59 that appears later in this application.

Eighteenth Example

[0236] FIG. 35 is a view illustrating an eighteenth example of an optical imaging system, and FIG. 36 illustrates aberration curves of the optical imaging system of FIG. 35.

[0237] An optical imaging system 18 includes a first lens 1018, a second lens 2018, a third lens 3018, a fourth lens 4018, a fifth lens 5018, a sixth lens 6018, and a seventh lens 7018.

[0238] The first lens 1018 has a positive refractive power, and an object-side surface thereof is convex and an image-

side surface thereof is concave. The second lens **2018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens **5018** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. The sixth lens **6018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6018**. The seventh lens **7018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7018**, and one inflection point is formed on the image-side surface of the seventh lens **7018**.

TABLE 35

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.8575	0.6450	1.547	56.114	1.410
S2	Lens	7.5625	0.0250			1.373
S3	Second	2.3869	0.2400	1.660	20.400	1.275
S4	Lens	1.7513	0.4645			1.130
(Stop)						
S5	Third	4.0476	0.1300	1.660	20.400	1.092
S6	Lens	3.9656	0.2812			1.124
S7	Fourth	11.4640	0.6806	1.547	56.114	1.340
S8	Lens	-165.9873	0.3664			1.596
S9	Fifth	-83.3972	0.2354	1.650	21.494	1.696
S10	Lens	312.1902	0.1713			1.903
S11	Sixth	9.6436	0.8638	1.650	21.494	2.074
S12	Lens	6.8472	0.1129			2.622
S13	Seventh	2.1549	0.5985	1.537	55.711	3.411
S14	Lens	1.5639	0.2730			3.397
S15	Filter	Infinity	0.1100			3.542
S16		Infinity	0.5510			3.577
S17	Imaging Plane	Infinity	0.0150			3.954

TABLE 36

	K	A	B	C	D	E	F	G	H	J
S1	-0.0815	0.0055	0.0014	-0.0029	0.0061	-0.0051	0.002	-0.0003	0	0
S2	25.622	-0.0509	0.133	-0.2101	0.1935	-0.1078	0.0334	-0.0045	0	0
S3	-1.6604	-0.0775	0.1997	-0.3026	0.2917	-0.1691	0.0543	-0.0074	0	0
S4	0.011	-0.0397	0.0365	0.0313	-0.0936	0.1109	-0.0595	0.0137	0	0
S5	0.49	-0.0743	0.0501	-0.0466	0.0606	-0.0386	0.013	-0.0015	0	0
S6	7.0482	-0.0885	0.0384	-0.0259	0.0308	-0.0159	0.0036	-0.0003	0	0
S7	21.918	-0.0229	0.0042	-0.0033	0.0044	-0.0002	-0.0006	0.0001	0	0
S8	25.736	-0.0244	0.007	-0.0107	0.0065	-0.0016	0.0002	-7E-06	0	0
S9	1.6857	-0.0527	0.078	-0.0871	0.0455	-0.0137	0.0025	-0.0002	0	0
S10	76.281	-0.0552	0.0643	-0.0583	0.0243	-0.005	0.0005	-2E-05	0	0
S11	-52.836	0.0115	-0.0347	0.0203	-0.0087	0.0021	-0.0002	1E-05	0	0
S12	0	-0.0341	0.0013	0	0	0	0	0	0	0
S13	-0.9427	-0.1816	0.0377	-0.0008	-0.001	0.0002	-2E-05	9E-07	-2E-08	0
S14	-1.0048	-0.1579	0.0494	-0.0121	0.0022	-0.0003	2E-05	-8E-07	1E-08	0

[0239] The optical imaging system **18** further includes a stop, a filter **8018**, and an image sensor **9018**. The stop is disposed between the second lens **2018** and the third lens **3018** to adjust an amount of light incident on the image sensor **9018**. The filter **8018** is disposed between the seventh lens **7018** and the image sensor **9018** to block infrared rays. The image sensor **9018** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 35, the stop is disposed at a distance of 0.8665 mm from the object-side surface of the first lens **1018** toward the imaging plane of the optical imaging system **18**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 18 listed in Table 59 that appears later in this application.

[0240] Table 35 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 35, and Table 36 below shows aspherical surface coefficients of the lenses of FIG. 35.

Nineteenth Example

[0241] FIG. 37 is a view illustrating a nineteenth example of an optical imaging system, and FIG. 38 illustrates aberration curves of the optical imaging system of FIG. 37.

[0242] An optical imaging system **19** includes a first lens **1019**, a second lens **2019**, a third lens **3019**, a fourth lens **4019**, a fifth lens **5019**, a sixth lens **6019**, and a seventh lens **7019**.

[0243] The first lens **1019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5019** has a negative refractive power, and an

object-side surface thereof is concave and an image-side surface thereof is concave. The sixth lens **6019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6019**. The seventh lens **7019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7019**, and one inflection point is formed on the image-side surface of the seventh lens **7019**. [0244] The optical imaging system **19** further includes a stop, a filter **8019**, and an image sensor **9019**. The stop is disposed between the second lens **2019** and the third lens **3019** to adjust an amount of light incident on the image

TABLE 37-continued

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S8	Lens	98.7612	0.3664			1.609
S9	Fifth	-99.1146	0.2354	1.650	21.494	1.696
S10	Lens	195.9458	0.0598			1.903
S11	Sixth	8.0269	0.8638	1.650	21.494	2.017
S12	Lens	7.0025	0.1107			2.570
S13	Seventh	2.1549	0.6513	1.537	55.711	3.411
S14	Lens	1.5639	0.2946			3.295
S15	Filter	Infinity	0.1100			3.521
S16		Infinity	0.5174			3.558
S17	Imaging Plane	Infinity	0.015			3.941

TABLE 38

	K	A	B	C	D	E	F	G	H	J
S1	-0.0815	0.0055	0.0014	-0.0029	0.0061	-0.0051	0.002	-0.0003	0	0
S2	25.622	-0.0509	0.133	-0.2101	0.1935	-0.1078	0.0334	-0.0045	0	0
S3	-1.7152	-0.0796	0.199	-0.2926	0.2754	-0.1567	0.0498	-0.0068	0	0
S4	0.011	-0.0397	0.0365	0.0313	-0.0936	0.1109	-0.0595	0.0137	0	0
S5	0.49	-0.0743	0.0501	-0.0466	0.0606	-0.0386	0.013	-0.0015	0	0
S6	7.0482	-0.0885	0.0384	-0.0259	0.0308	-0.0159	0.0036	-0.0003	0	0
S7	21.918	-0.0229	0.0042	-0.0033	0.0044	-0.0002	-0.0006	0.0001	0	0
S8	25.736	-0.0244	0.007	-0.0107	0.0065	-0.0016	0.0002	-7E-06	0	0
S9	1.6857	-0.0527	0.078	-0.0871	0.0455	-0.0137	0.0025	-0.0002	0	0
S10	76.281	-0.0552	0.0643	-0.0583	0.0243	-0.005	0.0005	-2E-05	0	0
S11	-52.836	0.0115	-0.0347	0.0203	-0.0087	0.0021	-0.0002	1E-05	0	0
S12	0	-0.034	0.0045	-0.0014	0.0003	-2E-05	6E-07	0	0	0
S13	-0.9427	-0.1816	0.0377	-0.0008	-0.001	0.0002	-2E-05	9E-07	-2E-08	0
S14	-1.0048	-0.1579	0.0494	-0.0121	0.0022	-0.0003	2E-05	-8E-07	1E-08	0

sensor **9019**. The filter **8019** is disposed between the seventh lens **7019** and the image sensor **9019** to block infrared rays. The image sensor **9019** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 37, the stop is disposed at a distance of 0.8997 mm from the object-side surface of the first lens **1019** toward the imaging plane of the optical imaging system **19**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 19 listed in Table 59 that appears later in this application.

[0245] Table 37 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 37, and Table 38 below shows aspherical surface coefficients of the lenses of FIG. 37.

TABLE 37

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refrac- tion	Abbe Number	Effective Aperture Radius
S1	First	1.8876	0.6427	1.547	56.114	1.410
S2	Lens	7.5583	0.0250			1.369
S3	Second	2.5872	0.2400	1.660	20.400	1.272
S4	Lens	1.9097	0.4388			1.130
(Stop)						
S5	Third	3.9821	0.1399	1.660	20.400	1.092
S6	Lens	3.8971	0.2690			1.132
S7	Fourth	10.2645	0.7282	1.547	56.114	1.352

Twentieth Example

[0246] FIG. 39 is a view illustrating a twentieth example of an optical imaging system, and FIG. 40 illustrates aberration curves of the optical imaging system of FIG. 39.

[0247] An optical imaging system **20** includes a first lens **1020**, a second lens **2020**, a third lens **3020**, a fourth lens **4020**, a fifth lens **5020**, a sixth lens **6020**, and a seventh lens **7020**.

[0248] The first lens **1020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the

sixth lens **6020**. The seventh lens **7020** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7020**.

[0249] The optical imaging system **20** further includes a stop, a filter **8020**, and an image sensor **9020**. The stop is disposed between the second lens **2020** and the third lens **3020** to adjust an amount of light incident on the image sensor **9020**. The filter **8020** is disposed between the seventh lens **7020** and the image sensor **9020** to block infrared rays. The image sensor **9020** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **39**, the stop is disposed at a distance of 1.002 mm from the object-side surface of the first lens **1020** toward the imaging plane of the optical imaging system **20**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 20 listed in Table 59 that appears later in this application.

[0250] Table 39 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **39**, and Table 40 below shows aspherical surface coefficients of the lenses of FIG. **39**.

TABLE 39

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.8047	0.5769	1.544	56.114	1.270
S2	Lens	5.0109	0.0406			1.230
S3	Second	4.8095	0.3545	1.544	56.114	1.204
S4	Lens	14.1878	0.0300			1.158
S5 (Stop)	Third	3.6592	0.2000	1.661	20.350	1.087
S6	Lens	2.1487	0.4249			1.050
S7	Fourth	21.5791	0.3654	1.544	56.114	1.050
S8	Lens	9.6990	0.0619			1.187
S9	Fifth	6.2306	0.2825	1.639	21.525	1.212
S10	Lens	8.4970	0.3479			1.367
S11	Sixth	10.1847	0.5847	1.544	56.114	1.650
S12	Lens	-1.5171	0.3562			1.934
S13	Seventh	-2.7118	0.3000	1.544	56.114	2.303
S14	Lens	2.0636	0.1646			2.650
S15	Filter	Infinity	0.2100	1.518	64.197	
S16		Infinity	0.6300			
S17	Imaging Plane	Infinity	0.0099			

TABLE 40

	K	A	B	C	D	E	F	G	H
S1	-1.5984	0.022	0.0011	-0.0095	0.0071	-0.0076	0.0028	-0.0002	0
S2	0	-0.0267	-0.08	0.1204	-0.1085	0.0777	-0.0361	0.0074	0
S3	0	0.0185	-0.0944	0.1151	-0.0877	0.0713	-0.0433	0.0104	0
S4	93.032	-0.0833	0.3002	-0.6564	0.7873	-0.5697	0.2292	-0.0392	0
S5	-11.518	-0.2115	0.4874	-0.8074	0.9509	-0.7204	0.3239	-0.0644	0
S6	-4.4222	-0.0999	0.1985	-0.0999	-0.0975	0.2773	-0.2246	0.0743	0
S7	0	-0.0315	-0.1501	0.4497	-1.0958	1.4445	-1.0093	0.2957	0
S8	0	-0.1532	-0.084	0.3675	-0.5986	0.475	-0.1986	0.0366	0
S9	-76.367	-0.2472	-0.1038	0.5308	-0.6528	0.4225	-0.1503	0.0226	0
S10	0	-0.1927	-0.1015	0.3168	-0.3163	0.1912	-0.0703	0.0115	0
S11	0	0.0245	-0.0539	-0.0674	0.1082	-0.0625	0.0168	-0.0017	0
S12	-1.5099	0.2023	-0.1451	0.0004	0.0431	-0.0194	0.0035	-0.0002	0
S13	-6.0002	0.009	-0.1914	0.1596	-0.0593	0.0123	-0.0015	1E-04	-3E-06
S14	-0.8696	-0.1901	0.0765	-0.0229	0.0049	-0.0008	9E-05	-6E-06	2E-07

Twenty-First Example

[0251] FIG. **41** is a view illustrating a twenty-first example of an optical imaging system, and FIG. **42** illustrates aberration curves of the optical imaging system of FIG. **41**.

[0252] An optical imaging system **21** includes a first lens **1021**, a second lens **2021**, a third lens **3021**, a fourth lens **4021**, a fifth lens **5021**, a sixth lens **6021**, and a seventh lens **7021**.

[0253] The first lens **1021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both

of the object-side surface and the image-side surface of the sixth lens **6021**. The seventh lens **7021** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7021**, and two inflection points are formed on the image-side surface of the seventh lens **7021**.

[0254] The optical imaging system **21** further includes a stop, a filter **8021**, and an image sensor **9021**. The stop is disposed between the first lens **1021** and the second lens **2021** to adjust an amount of light incident on the image sensor **9021**. The filter **8021** is disposed between the seventh lens **7021** and the image sensor **9021** to block infrared rays. The image sensor **9021** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **41**, the stop is disposed at a distance of 0.374 mm from the object-side surface of the first lens **1021** toward the imaging plane of the optical imaging system **21**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 21 listed in Table 59 that appears later in this application.

[0255] Table 41 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **41**, and Table 42 below shows aspherical surface coefficients of the lenses of FIG. **41**.

TABLE 41

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1873	0.3243	1.546	56.114	1.450
S2	Lens	1.8391	0.0497			1.441
S3 (Stop)	Second	1.6361	0.7740	1.546	56.114	1.415
S4	Lens	30.6063	0.0300			1.354
S5	Third	7.2628	0.2100	1.678	19.236	1.270
S6	Lens	2.9652	0.4149			1.120
S7	Fourth	14.3312	0.3269	1.645	23.528	1.182
S8	Lens	12.1292	0.2502			1.337
S9	Fifth	2.1804	0.2500	1.645	23.528	1.580
S10	Lens	2.1733	0.3831			1.892
S11	Sixth	8.6678	0.6610	1.546	56.114	2.429
S12	Lens	-1.9375	0.3110			2.544
S13	Seventh	-7.6533	0.3650	1.546	56.114	2.916
S14	Lens	1.6261	0.2200			3.075
S15	Filter	Infinity	0.1100	1.518	64.166	3.378
S16		Infinity	0.6351			3.414
S17	Imaging Plane	Infinity	0.0049			3.763

TABLE 42

	K	A	B	C	D	E	F	G	H
S1	-3.7488	0.0012	-0.0066	-0.0004	-0.0198	0.0252	-0.0132	0.0034	-0.0004
S2	-7.1577	-0.061	-0.0104	0.0163	0.0115	-0.0163	0.0063	-0.0009	0
S3	-2.6408	-0.0742	0.0698	-0.0582	0.0727	-0.0412	0.0034	0.0048	-0.0013
S4	-99	-0.0752	0.197	-0.3925	0.5174	-0.4377	0.2286	-0.0663	0.008
S5	0	-0.1076	0.2644	-0.4642	0.6109	-0.5485	0.3128	-0.0997	0.0134
S6	4.364	-0.0584	0.0882	-0.068	-0.0405	0.1629	-0.1817	0.0962	-0.0201
S7	0	-0.0603	0.0743	-0.2389	0.4197	-0.4882	0.353	-0.1472	0.0274
S8	0	-0.1174	0.165	-0.2983	0.348	-0.2864	0.1556	-0.0507	0.0077
S9	-15.429	-0.0562	0.0005	0.0397	-0.0576	0.0355	-0.0117	0.0015	3E-05
S10	-9.1654	-0.1003	0.0623	-0.0379	0.0141	-0.0032	5E-05	0.0002	-3E-05
S11	0	-0.001	-0.0216	0.0157	-0.0111	0.0043	-0.0009	8E-05	-3E-06
S12	-1.7327	0.1074	-0.0935	0.0649	-0.0289	0.0078	-0.0012	0.0001	-4E-06
S13	0.6082	-0.1509	0.0462	0.0036	-0.0043	0.001	-0.0001	6E-06	-2E-07
S14	-8.5925	-0.0951	0.041	-0.0124	0.0026	-0.0004	4E-05	-2E-06	4E-08

Twenty-Second Example

[0256] FIG. **43** is a view illustrating a twenty-second example of an optical imaging system, and FIG. **44** illustrates aberration curves of the optical imaging system of FIG. **43**.

[0257] An optical imaging system **22** includes a first lens **1022**, a second lens **2022**, a third lens **3022**, a fourth lens **4022**, a fifth lens **5022**, a sixth lens **6022**, and a seventh lens **7022**.

[0258] The first lens **1022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2022** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4022** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5022** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition,

at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6022**. The seventh lens **7022** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7022**, and two inflection points are formed on the image-side surface of the seventh lens **7022**.

[0259] The optical imaging system **22** further includes a stop, a filter **8022**, and an image sensor **9022**. The stop is disposed between the first lens **1022** and the second lens **2022** to adjust an amount of light incident on the image sensor **9022**. The filter **8022** is disposed between the seventh lens **7022** and the image sensor **9022** to block infrared rays. The image sensor **9022** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **43**, the stop is disposed at a distance of 0.920 mm from the object-side surface of the first lens **1022** toward the imaging plane of the optical imaging system **22**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 22 listed in Table 59 that appears later in this application.

[0260] Table 43 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **43**, and Table 44 below shows aspherical surface coefficients of the lenses of FIG. **43**.

Twenty-Third Example

[0261] FIG. **45** is a view illustrating a twenty-third example of an optical imaging system, and FIG. **46** illustrates aberration curves of the optical imaging system of FIG. **45**.

[0262] An optical imaging system **23** includes a first lens **1023**, a second lens **2023**, a third lens **3023**, a fourth lens **4023**, a fifth lens **5023**, a sixth lens **6023**, and a seventh lens **7023**.

[0263] The first lens **1023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both

TABLE 43

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.9701	0.9200	1.546	56.114	1.435
S2 (Stop)	Lens	6.1422	0.0762			1.327
S3	Second	5.3743	0.2000	1.677	19.238	1.311
S4	Lens	3.5472	0.3480			1.231
S5	Third	10.0771	0.3764	1.546	56.114	1.271
S6	Lens	25.5187	0.1640			1.351
S7	Fourth	5.8924	0.2000	1.667	20.377	1.359
S8	Lens	4.6147	0.2547			1.460
S9	Fifth	5.0940	0.2295	1.619	25.960	1.756
S10	Lens	4.3859	0.3402			1.654
S11	Sixth	4.9999	0.7714	1.546	56.114	2.420
S12	Lens	-1.8739	0.3896			2.467
S13	Seventh	-2.1172	0.3000	1.546	56.114	3.169
S14	Lens	2.8301	0.1800			3.066
S15	Filter	Infinity	0.2100	1.518		3.715
S16		Infinity	0.6368			3.801
S17	Imaging Plane	Infinity	0.0032			4.254

TABLE 44

	K	A	B	C	D	E	F	G	H	J
S1	-1.1385	0.0141	0.023	-0.0501	0.0713	-0.0603	0.0298	-0.0079	0.0009	0
S2	12.673	-0.0899	0.0792	-0.0381	-0.0163	0.0343	-0.0229	0.0077	-0.0011	0
S3	9.9647	-0.1473	0.1118	0.0661	-0.2646	0.2998	-0.1775	0.0556	-0.0072	0
S4	-0.5888	-0.076	0.0676	0.0602	-0.1804	0.1698	-0.0679	0.0057	0.0025	0
S5	0	-0.0278	0.0424	-0.1578	0.2776	-0.3017	0.1871	-0.0609	0.0081	0
S6	-99	-0.0505	0.0344	-0.0587	0.0428	0.0016	-0.0357	0.0253	-0.0056	0
S7	0	-0.138	0.0096	0.0579	-0.2108	0.3235	-0.2566	0.1009	-0.0155	0
S8	0	-0.1363	0.1001	-0.1765	0.2075	-0.1546	0.071	-0.0193	0.0025	0
S9	0	-0.2113	0.2288	-0.2271	0.1631	-0.0851	0.0308	-0.0071	0.0008	0
S10	-62.082	-0.1439	0.0555	-0.0007	-0.029	0.0245	-0.009	0.0016	-0.0001	0
S11	-21.515	0.0047	-0.0144	0.0029	-0.0019	0.0006	-8E-05	1E-06	2E-07	0
S12	-3.7544	0.1035	-0.0491	0.0125	-0.0024	0.0003	-2E-05	-3E-07	9E-08	0
S13	-11.142	-0.0315	-0.0345	0.0239	-0.0062	0.0009	-7E-05	3E-06	-5E-08	0
S14	-1.2542	-0.091	0.025	-0.0054	0.0009	-0.0001	1E-05	-1E-06	6E-08	-1E-09

of the object-side surface and the image-side surface of the sixth lens **6023**. The seventh lens **7023** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7023**, and two inflection points are formed on the image-side surface of the seventh lens **7023**.

[0264] The optical imaging system **23** further includes a stop, a filter **8023**, and an image sensor **9023**. The stop is disposed between the first lens **1023** and the second lens **2023** to adjust an amount of light incident on the image sensor **9023**. The filter **8023** is disposed between the seventh lens **7023** and the image sensor **9023** to block infrared rays. The image sensor **9023** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **45**, the stop is disposed at a distance of 1.060 mm from the object-side surface of the first lens **1023** toward the imaging plane of the optical imaging system **23**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 23 listed in Table 59 that appears later in this application.

[0265] Table 45 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **45**, and Table 46 below shows aspherical surface coefficients of the lenses of FIG. **45**.

TABLE 45

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.0122	1.0601	1.546	56.114	1.510
S2 (Stop)	Lens	5.8868	0.0711			1.375
S3	Second	5.4922	0.2000	1.677	19.238	1.353
S4	Lens	3.7029	0.2969			1.240
S5	Third	9.9760	0.3703	1.546	56.114	1.270
S6	Lens	29.4320	0.1826			1.356
S7	Fourth	8.6841	0.2108	1.667	20.377	1.355
S8	Lens	6.0112	0.2603			1.479
S9	Fifth	5.7445	0.2195	1.619	25.960	1.750
S10	Lens	5.0434	0.3482			1.736
S11	Sixth	4.8476	0.9335	1.546	56.114	2.400
S12	Lens	-1.7967	0.3268			2.438
S13	Seventh	-1.9512	0.3000	1.546	56.114	2.692
S14	Lens	2.8062	0.1700			3.158
S15	Filter	Infinity	0.2100	1.518	64.197	3.744
S16		Infinity	0.6441			3.826
S17	Imaging Plane	Infinity	-0.0041			4.252

TABLE 46

	K	A	B	C	D	E	F	G	H	J
S1	-1.1332	0.0154	0.0091	-0.0108	0.01	-0.0053	0.0015	-0.0002	-1E-05	0
S2	12.901	-0.0771	0.0315	0.036	-0.0911	0.0858	-0.0454	0.0132	-0.0016	0
S3	9.6379	-0.1255	0.0603	0.0962	-0.2261	0.222	-0.1214	0.036	-0.0045	0
S4	-0.8719	-0.061	0.0524	0.0027	0.0041	-0.069	0.0931	-0.0498	0.0101	0
S5	0	-0.0121	0.0151	-0.088	0.1573	-0.1697	0.1016	-0.0315	0.004	0
S6	-99	-0.0237	-0.0073	0.0365	-0.1182	0.1732	-0.1421	0.0603	-0.0104	0
S7	0	-0.104	-0.0751	0.2741	-0.5665	0.6635	-0.4435	0.1557	-0.0222	0
S8	0	-0.1088	0.0332	-0.0258	-0.0057	0.0249	-0.0175	0.0048	-0.0003	0
S9	0	-0.1824	0.192	-0.1693	0.1022	-0.0423	0.0117	-0.002	0.0002	0
S10	-96.971	-0.1318	0.0661	-0.0223	-0.0048	0.0087	-0.0034	0.0006	-4E-05	0
S11	-34.065	-0.0075	-0.0001	-0.0106	0.0081	-0.0035	0.0008	-1E-04	4E-06	0
S12	-2.7443	0.1131	-0.0726	0.0291	-0.0078	0.0012	-9E-05	1E-06	1E-07	0
S13	-10.221	-0.0393	-0.0198	0.0148	-0.0036	0.0005	-3E-05	1E-06	-2E-08	0
S14	-1.2844	-0.0938	0.0319	-0.0092	0.002	-0.0003	4E-05	-3E-06	1E-07	-3E-09

Twenty-Fourth Example

[0266] FIG. **47** is a view illustrating a twenty-fourth example of an optical imaging system, and FIG. **48** illustrates aberration curves of the optical imaging system of FIG. **47**.

[0267] An optical imaging system **24** includes a first lens **1024**, a second lens **2024**, a third lens **3024**, a fourth lens **4024**, a fifth lens **5024**, a sixth lens **6024**, and a seventh lens **7024**.

[0268] The first lens **1024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3024** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5024** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addi-

tion, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6024**. The seventh lens **7024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7024**, and one inflection point is formed on the image-side surface of the seventh lens **7024**. [0269] The optical imaging system **24** further includes a stop, a filter **8024**, and an image sensor **9024**. The stop is disposed between the second lens **2024** and the third lens **3024** to adjust an amount of light incident on the image sensor **9024**. The filter **8024** is disposed between the seventh lens **7024** and the image sensor **9024** to block infrared rays. The image sensor **9024** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 47, the stop is disposed at a distance of 1.082 mm from the object-side surface of the first lens **1024** toward the imaging plane of the optical imaging system **24**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 24 listed in Table 59 that appears later in this application. [0270] Table 47 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 47, and Table 48 below shows aspherical surface coefficients of the lenses of FIG. 47.

TABLE 47

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.3369	0.4321	1.546	56.114	1.365
S2	Lens	2.8574	0.0250			1.352
S3	Second	2.5422	0.6000	1.546	56.114	1.326
S4	Lens	36.4170	0.0250			1.254
S5 (Stop)	Third	8.1937	0.2300	1.679	19.236	1.217
S6	Lens	3.3336	0.3222			1.227
S7	Fourth	6.3427	0.5711	1.546	56.114	1.322
S8	Lens	11.2370	0.4049			1.372
S9	Fifth	18.9615	0.5067	1.546	56.114	1.590
S10	Lens	6.6837	0.0732			1.931
S11	Sixth	2.3548	0.6194	1.546	56.114	2.023
S12	Lens	2.5651	0.1492			2.456
S13	Seventh	1.4247	0.5400	1.546	56.114	2.710
S14	Lens	1.2822	0.3444			2.982
S15	Filter	Infinity	0.2100	1.518	64.197	3.258
S16		Infinity	0.6347			3.334
S17	Imaging Plane	Infinity	0.0150			3.734

TABLE 48

	K	A	B	C	D	E	F	G	H	J
S1	-0.9157	-0.0242	0.0483	-0.0925	0.0385	0.0577	-0.0925	0.0579	-0.0178	0.0022
S2	-12.376	0.0627	-0.1415	-0.3392	0.8991	-0.7358	0.1834	0.0755	-0.0533	0.0088
S3	-0.8319	0.031	-0.03	-0.6522	1.4923	-1.3976	0.6352	-0.1105	-0.0112	0.0048
S4	-7.367	-0.1852	1.7179	-6.8471	14.821	-19.261	15.464	-7.5184	2.0307	-0.2341
S5	12.337	-0.2536	1.7489	-6.6898	14.646	-19.491	16.071	-8.0307	2.2327	-0.2657
S6	1.1454	-0.0901	0.2168	-0.6218	1.4502	-2.2709	2.2634	-1.3948	0.4895	-0.0747
S7	-12.034	0.0424	-0.6838	2.5289	-5.5859	7.6559	-6.5535	3.3828	-0.9545	0.1124
S8	5.8592	-0.0168	-0.1532	0.4479	-0.9325	1.2364	-1.0356	0.5306	-0.1517	0.0187
S9	-43.521	0.0196	0.0447	-0.1445	0.1741	-0.1293	0.0589	-0.0164	0.0026	-0.0002
S10	-9.9703	-0.0233	-0.0527	0.0821	-0.0601	0.0246	-0.0062	0.001	-9E-05	4E-06
S11	-16.199	0.1383	-0.3024	0.3056	-0.2185	0.1017	-0.0304	0.0057	-0.0006	3E-05
S12	0.0118	-0.0979	0.0662	-0.0617	0.0337	-0.0119	0.0028	-0.0004	3E-05	-1E-06
S13	-0.8414	-0.3646	0.1533	-0.0353	0.0033	0.0004	-0.0001	2E-05	-8E-07	1E-08
S14	-1.4251	-0.2584	0.1351	-0.0538	0.0161	-0.0034	0.0005	-4E-05	2E-06	-4E-08

Twenty-Fifth Example

[0271] FIG. 49 is a view illustrating a twenty-fifth example of an optical imaging system, and FIG. 50 illustrates aberration curves of the optical imaging system of FIG. 49.

[0272] An optical imaging system **25** includes a first lens **1025**, a second lens **2025**, a third lens **3025**, a fourth lens **4025**, a fifth lens **5025**, a sixth lens **6025**, and a seventh lens **7025**.

[0273] The first lens **1025** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2025** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3025** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4025** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5025** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6025** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition,

tion, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6025**. The seventh lens **7025** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7025**. [0274] The optical imaging system **25** further includes a stop, a filter **8025**, and an image sensor **9025**. The stop is disposed between the second lens **2025** and the third lens **3025** to adjust an amount of light incident on the image sensor **9025**. The filter **8025** is disposed between the seventh lens **7025** and the image sensor **9025** to block infrared rays. The image sensor **9025** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **49**, the stop is disposed at a distance of 1.201 mm from the object-side surface of the first lens **1025** toward the imaging plane of the optical imaging system **25**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 25 listed in Table 59 that appears later in this application. [0275] Table 49 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **49**, and Table 50 below shows aspherical surface coefficients of the lenses of FIG. **49**.

TABLE 49

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.2889	0.4895	1.546	56.114	1.564
S2	Lens	2.8751	0.1225			1.556
S3	Second	3.1931	0.5641	1.546	56.114	1.519
S4	Lens	102.3291	0.0250			1.495
S5 (Stop)	Third	9.0291	0.2300	1.679	19.236	1.430
S6	Lens	4.0323	0.4394			1.411
S7	Fourth	6.6204	0.3813	1.546	56.114	1.543
S8	Lens	14.3245	0.5330			1.563
S9	Fifth	5.4175	0.4127	1.679	19.236	1.840
S10	Lens	3.5247	0.2029			2.415
S11	Sixth	2.3899	0.5978	1.546	56.114	2.201
S12	Lens	4.4770	0.3962			2.763
S13	Seventh	2.3256	0.5184	1.546	56.114	3.015
S14	Lens	1.4122	0.2273			3.288
S15	Filter	Infinity	0.2100	1.518	64.197	3.711
S16		Infinity	0.6350			3.786
S17	Imaging Plane	Infinity	0.0150			4.203

TABLE 50

	K	A	B	C	D	E	F	G	H	J
S1	-1	-0.0109	0.0161	-0.0521	0.0675	-0.0541	0.0251	-0.0062	0.0007	-2E-05
S2	-12.313	0.0249	-0.0812	0.0686	-0.0854	0.0923	-0.0564	0.019	-0.0034	0.0002
S3	-1.1961	-0.0151	-0.0414	0.0709	-0.1526	0.202	-0.1389	0.052	-0.0102	0.0008
S4	-7.0515	-0.0439	0.2205	-0.5763	0.8204	-0.7024	0.3734	-0.1213	0.0221	-0.0017
S5	9.4925	-0.0841	0.2664	-0.6308	0.9198	-0.8507	0.5017	-0.1833	0.0381	-0.0035
S6	1.6278	-0.0537	0.0672	-0.0789	0.0603	-0.0261	0.0045	0.001	-0.0003	-4E-05
S7	-4.8767	-0.0251	-0.0455	0.1569	-0.312	0.3626	-0.2555	0.1067	-0.024	0.0022
S8	5.8592	-0.0325	-0.0105	0.0226	-0.033	0.0214	-0.0047	-0.0015	0.001	-0.0001
S9	-43.521	-0.009	-0.005	0.0283	-0.0424	0.0317	-0.0144	0.004	-0.0006	4E-05
S10	-16.247	-0.0574	0.03	-0.0024	-0.0083	0.0056	-0.0019	0.0004	-4E-05	2E-06
S11	-12.323	0.0445	-0.0879	0.0791	-0.052	0.0213	-0.0055	0.0009	-8E-05	3E-06
S12	-0.1058	-0.0342	0.019	-0.0122	0.0033	-0.0005	6E-05	-9E-06	8E-07	-3E-08
S13	-0.7464	-0.2683	0.0838	-0.0065	-0.0032	0.0012	-0.0002	2E-05	-8E-07	1E-08
S14	-1.4016	-0.2382	0.1163	-0.0418	0.0106	-0.0018	0.0002	-1E-05	5E-07	-8E-09

Twenty-Sixth Example

[0276] FIG. **51** is a view illustrating a twenty-sixth example of an optical imaging system, and FIG. **52** illustrates aberration curves of the optical imaging system of FIG. **51**.

[0277] An optical imaging system **26** includes a first lens **1026**, a second lens **2026**, a third lens **3026**, a fourth lens **4026**, a fifth lens **5026**, a sixth lens **6026**, and a seventh lens **7026**.

[0278] The first lens **1026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2026** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3026** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5026** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6026** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In

addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6026**. The seventh lens **7026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7026**. [0279] The optical imaging system **26** further includes a stop, a filter **8026**, and an image sensor **9026**. The stop is disposed between the second lens **2026** and the third lens **3026** to adjust an amount of light incident on the image sensor **9026**. The filter **8026** is disposed between the seventh lens **7026** and the image sensor **9026** to block infrared rays. The image sensor **9026** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **51**, the stop is disposed at a distance of 0.963 mm from the object-side surface of the first lens **1026** toward the imaging plane of the optical imaging system **26**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 26 listed in Table 59 that appears later in this application. [0280] Table 51 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **51**, and Table 52 below shows aspherical surface coefficients of the lenses of FIG. **51**.

TABLE 51

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.7493	0.7080	1.546	56.114	1.280
S2	Lens	7.7627	0.0250			1.225
S3	Second	3.6883	0.2300	1.667	20.353	1.160
S4 (Stop)	Lens	2.4524	0.3551			1.033
S5	Third	39.9140	0.2300	1.667	20.353	1.053
S6	Lens	22.4233	0.0250			1.090
S7	Fourth	6.6877	0.3582	1.546	56.114	1.130
S8	Lens	17.1426	0.3932			1.201
S9	Fifth	10.0343	0.3525	1.656	21.525	1.329
S10	Lens	6.5555	0.2520			1.664
S11	Sixth	-324.8644	0.6107	1.656	21.525	1.841
S12	Lens	12.2860	0.0342			2.288
S13	Seventh	1.9518	0.8257	1.536	55.656	2.578
S14	Lens	1.7567	0.2187			2.963
S15	Filter	Infinity	0.2100	1.518	64.197	3.258
S16		Infinity	0.6350			3.334
S17	Imaging Plane	Infinity	0.0150			3.729

TABLE 52

	K	A	B	C	D	E	F	G	H	J
S1	-0.2398	5E-05	0.0225	-0.0553	0.0791	-0.0725	0.0408	-0.0137	0.0019	0
S2	6.0424	-0.0363	0.0343	0.0144	-0.1124	0.1667	-0.1307	0.054	-0.0092	0
S3	-1.7137	-0.0472	0.041	0.0264	-0.116	0.1895	-0.1701	0.0827	-0.0161	0
S4	-0.2358	-0.0167	-0.01	0.0564	-0.0195	-0.1069	0.2279	-0.1897	0.0625	0
S5	-0.0716	-0.0169	-0.0047	-0.1892	0.6295	-1.0256	0.9612	-0.4977	0.1127	0
S6	-1.1573	0.0199	-0.1372	0.1444	-0.0555	0.1408	-0.2746	0.2067	-0.0539	0
S7	-28.459	0.0213	-0.1017	0.0611	0.0456	0.018	-0.1503	0.1307	-0.0346	0
S8	-2.3038	-0.0386	0.0394	-0.1206	0.2443	-0.4112	0.4746	-0.3301	0.1229	-0.0182
S9	-3.3254	-0.1025	0.044	-0.1067	0.238	-0.3262	0.2409	-0.0929	0.0146	0
S10	-25.215	-0.0274	-0.1331	0.1909	-0.1562	0.0771	-0.0231	0.0041	-0.0003	0
S11	23.202	0.1679	-0.2882	0.2414	-0.1422	0.0533	-0.0119	0.0015	-8E-05	0
S12	-49.948	0.0068	-0.0175	0.0027	0.0001	-0.0001	4E-05	-6E-06	4E-07	0
S13	-1.9292	-0.2614	0.126	-0.0405	0.0094	-0.0015	0.0002	-9E-06	2E-07	0
S14	-0.8288	-0.1737	0.0652	-0.0206	0.0046	-0.0007	6E-05	-3E-06	7E-08	0

Twenty-Seventh Example

[0281] FIG. **53** is a view illustrating a twenty-seventh example of an optical imaging system, and FIG. **54** illustrates aberration curves of the optical imaging system of FIG. **53**.

[0282] An optical imaging system **27** includes a first lens **1027**, a second lens **2027**, a third lens **3027**, a fourth lens **4027**, a fifth lens **5027**, a sixth lens **6027**, and a seventh lens **7027**.

[0283] The first lens **1027** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2027** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3027** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4027** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5027** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6027** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. In addition,

tion, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6027**. The seventh lens **7027** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens **7027**, and one inflection point is formed on the image-side surface of the seventh lens **7027**. [0284] The optical imaging system **27** further includes a stop, a filter **8027**, and an image sensor **9027**. The stop is disposed between the second lens **2027** and the third lens **3027** to adjust an amount of light incident on the image sensor **9027**. The filter **8027** is disposed between the seventh lens **7027** and the image sensor **9027** to block infrared rays. The image sensor **9027** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **53**, the stop is disposed at a distance of 0.872 mm from the object-side surface of the first lens **1027** toward the imaging plane of the optical imaging system **27**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 27 listed in Table 59 that appears later in this application.

[0285] Table 53 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **53**, and Table 54 below shows aspherical surface coefficients of the lenses of FIG. **53**.

TABLE 53

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First Lens	1.7603	0.6172	1.546	56.114	1.100
S2		14.1233	0.0250			1.040
S3	Second Lens	5.8341	0.2300	1.667	20.353	1.011
S4 (Stop)		3.1227	0.3733			0.919
S5	Third Lens	-49.9417	0.3799	1.546	56.114	0.995
S6		-15.1870	0.1809			1.096
S7	Fourth Lens	23.3680	0.3032	1.667	20.353	1.124
S8		12.2098	0.3354			1.309
S9	Fifth Lens	-4.3948	0.4729	1.546	56.114	1.471
S10		-1.5983	0.0250			1.698
S11	Sixth Lens	-6.0815	0.5447	1.546	56.114	1.822
S12		-3.0145	0.2724			2.192
S13	Seventh Lens	-6.1494	0.4224	1.546	56.114	2.462
S14		1.6367	0.1933			2.880
S15	Filter	Infinity	0.2100	1.518	64.197	3.223
S16		Infinity	0.6445			3.300
S17	Imaging Plane	Infinity	0.0099			3.728

TABLE 54

	K	A	B	C	D	E	F	G	H	J
S1	-1.0054	0.0225	0.0222	-0.0696	0.1604	-0.2238	0.1806	-0.0791	0.0141	0
S2	-1.5097	-0.1275	0.3975	-0.6982	0.6801	-0.322	0.0288	0.029	-0.0076	0
S3	6.0294	-0.163	0.4504	-0.8514	1.0525	-0.8203	0.4235	-0.138	0.0213	0
S4	-0.8846	-0.0449	0.0393	0.1574	-0.6934	1.3171	-1.3069	0.6799	-0.143	0
S5	0	-0.0513	-0.0193	-0.016	0.0043	0.0034	-0.0155	0.0319	-0.0128	0
S6	0	-0.1089	-0.0569	0.3576	-0.9255	1.1947	-0.8604	0.3322	-0.0547	0
S7	-7.5	-0.2139	-0.0107	0.1788	-0.1827	-0.1159	0.3046	-0.1897	0.0405	0
S8	-43.341	-0.1402	-0.061	0.2777	-0.4123	0.3523	-0.1857	0.0564	-0.0071	0
S9	-35.081	-0.0602	0.0736	-0.1046	0.1084	-0.0726	0.0255	-0.0041	0.0002	0
S10	-1.5734	0.1621	-0.2197	0.1896	-0.107	0.0396	-0.0091	0.0011	-6E-05	0
S11	0.5153	0.2137	-0.3167	0.2399	-0.1217	0.0384	-0.0069	0.0007	-3E-05	0
S12	-1.1466	0.1967	-0.2565	0.1542	-0.0532	0.0115	-0.0015	0.0001	-4E-06	0
S13	-0.9056	-0.0077	-0.2094	0.1883	-0.0749	0.0167	-0.0022	0.0002	-5E-06	0
S14	-1.2797	-0.2192	0.1006	-0.0338	0.0088	-0.0018	0.0003	-2E-05	1E-06	-3E-08

Twenty-Eighth Example

[0286] FIG. **55** is a view illustrating a twenty-eighth example of an optical imaging system, and FIG. **56** illustrates aberration curves of the optical imaging system of FIG. **55**.

[0287] An optical imaging system **28** includes a first lens **1028**, a second lens **2028**, a third lens **3028**, a fourth lens **4028**, a fifth lens **5028**, a sixth lens **6028**, and a seventh lens **7028**.

[0288] The first lens **1028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2028** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3028** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5028** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition,

tion, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6028**. The seventh lens **7028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7028**, and one inflection point is formed on the image-side surface of the seventh lens **7028**. [0289] The optical imaging system **28** further includes a stop, a filter **8028**, and an image sensor **9028**. The stop is disposed between the second lens **2028** and the third lens **3028** to adjust an amount of light incident on the image sensor **9028**. The filter **8028** is disposed between the seventh lens **7028** and the image sensor **9028** to block infrared rays. The image sensor **9028** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **55**, the stop is disposed at a distance of 0.866 mm from the object-side surface of the first lens **1028** toward the imaging plane of the optical imaging system **28**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 28 listed in Table 59 that appears later in this application.

[0290] Table 55 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **55**, and Table 56 below shows aspherical surface coefficients of the lenses of FIG. **55**.

TABLE 55

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.8830	0.5872	1.546	56.114	1.050
S2	Lens	18.0733	0.0492			0.962
S3	Second	4.5995	0.2300	1.667	20.353	0.934
S4 (Stop)	Lens	2.5464	0.3929			0.837
S5	Third	-21.7546	0.2745	1.546	56.114	1.100
S6	Lens	-13.5144	0.0611			1.106
S7	Fourth	25.3349	0.2655	1.546	56.114	1.200
S8	Lens	25.3360	0.3710			1.285
S9	Fifth	9.4682	0.3930	1.656	21.525	1.500
S10	Lens	5.1029	0.3790			1.754
S11	Sixth	6.4162	0.8885	1.546	56.114	2.041
S12	Lens	6.3521	0.0460			2.631
S13	Seventh	1.9665	0.8854	1.536	55.656	3.050
S14	Lens	1.7699	0.3098			3.456
S15	Filter	Infinity	0.2100	1.518	64.197	3.768
S16		Infinity	0.6537			3.829
S17	Imaging Plane	Infinity	-0.0037			4.129

TABLE 56

	K	A	B	C	D	E	F	G	H	J
S1	-0.1525	0.0035	0.0054	-0.0238	0.0587	-0.0925	0.0808	-0.0376	0.0069	0
S2	-36.188	-0.0554	0.191	-0.4954	0.9092	-1.1194	0.849	-0.3546	0.0617	0
S3	-0.1164	-0.0883	0.2264	-0.5273	0.9947	-1.274	1.0104	-0.4343	0.076	0
S4	0.3326	-0.0462	0.097	-0.2316	0.5455	-0.848	0.7854	-0.3759	0.0708	0
S5	51.758	-0.0119	-0.0911	0.3617	-0.9067	1.3845	-1.3014	0.6835	-0.1493	0
S6	42.164	0.0924	-0.5269	1.3558	-2.2584	2.5093	-1.8107	0.7611	-0.139	0
S7	-4.7579	0.1336	-0.5938	1.261	-1.8115	1.7924	-1.1666	0.4427	-0.0728	0
S8	-3.4393	0.0471	-0.1842	0.2886	-0.3575	0.3273	-0.1971	0.067	-0.0093	0
S9	-8.5449	-0.0502	-0.0588	0.1599	-0.2027	0.1398	-0.0542	0.0105	-0.0007	0
S10	-18.064	-0.044	-0.0734	0.1425	-0.1303	0.0691	-0.0217	0.0038	-0.0003	0
S11	-4.6497	0.0633	-0.1193	0.0882	-0.0426	0.0135	-0.0028	0.0004	-2E-05	0
S12	-50	0.034	-0.0497	0.0246	-0.0072	0.0013	-0.0001	7E-06	-2E-07	0
S13	-2.4291	-0.1201	0.0167	0.0022	-0.0009	0.0001	-6E-06	1E-07	9E-10	0
S14	-1.0032	-0.1111	0.0248	-0.0032	-0.0001	0.0001	-2E-05	2E-06	-8E-08	1E-09

Twenty-Ninth Example

[0291] FIG. **57** is a view illustrating a twenty-ninth example of an optical imaging system, and FIG. **58** illustrates aberration curves of the optical imaging system of FIG. **57**.

[0292] An optical imaging system **29** includes a first lens **1029**, a second lens **2029**, a third lens **3029**, a fourth lens **4029**, a fifth lens **5029**, a sixth lens **6029**, and a seventh lens **7029**.

[0293] The first lens **1029** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2029** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3029** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4029** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5029** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6029** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition,

tion, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6029**. The seventh lens **7029** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7029**.

[0294] The optical imaging system **29** further includes a stop, a filter **8029**, and an image sensor **9029**. The stop is disposed between the second lens **2029** and the third lens **3029** to adjust an amount of light incident on the image sensor **9029**. The filter **8029** is disposed between the seventh lens **7029** and the image sensor **9029** to block infrared rays. The image sensor **9029** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **57**, the stop is disposed at a distance of 0.904 mm from the object-side surface of the first lens **1029** toward the imaging plane of the optical imaging system **29**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 29 listed in Table 59 that appears later in this application.

[0295] Table 57 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **57**, and Table 58 below shows aspherical coefficients of the lenses of FIG. **57**.

[0296] Table 59 below shows an overall focal length f of the optical imaging system, an overall length TTL of the optical imaging system (a distance from the object-side surface of the first lens to the imaging plane), a distance SL from the stop to the imaging plane, an f-number (F No.) of the optical imaging system (the overall focal length f of the optical imaging system divided by the diameter of an entrance pupil of the optical imaging system, where both f and the diameter of the entrance pupil are expressed in mm), an image height (IMG HT) on the imaging plane (one-half of a diagonal length of the imaging plane), and a field of view (FOV) of the optical imaging system for each of Examples 1-29 described herein. The values of f , TTL, SL, and IMG HT are expressed in mm. The values of F No. are dimensionless values. The values of FOV are expressed in degrees.

TABLE 59

Example	f	TTL	SL	F No.	IMG HT	FOV
1	3.671	4.200	3.529	1.99	3.26	81.90
2	4.780	5.827	4.402	1.57	4.05	79.02
3	4.256	5.190	3.931	1.58	3.68	80.22
4	3.950	4.819	3.650	1.58	3.25	77.47

TABLE 57

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.8987	0.6486	1.546	56.114	1.260
S2	Lens	7.3568	0.0250			1.216
S3	Second	3.8789	0.2300	1.667	20.353	1.161
S4 (Stop)	Lens	2.7620	0.3408			1.053
S5	Third	-50.1242	0.2819	1.546	56.114	1.120
S6	Lens	-14.9889	0.0597			1.158
S7	Fourth	12.0498	0.2698	1.546	56.114	1.220
S8	Lens	12.5657	0.2919			1.320
S9	Fifth	9.5926	0.3500	1.667	20.353	1.520
S10	Lens	5.2748	0.3344			1.762
S11	Sixth	6.8735	0.8484	1.546	56.114	2.052
S12	Lens	7.4933	0.0591			2.641
S13	Seventh	2.0337	0.8836	1.536	55.656	3.070
S14	Lens	1.8436	0.3048			3.425
S15	Filter	Infinity	0.2100	1.518	64.197	3.764
S16		Infinity	0.6441			3.825
S17	Imaging Plane	Infinity	0.0150			4.134

TABLE 58

	K	A	B	C	D	E	F	G	H	J
S1	-0.1061	-0.0082	0.0469	-0.0925	0.0811	-0.0129	-0.032	0.0224	-0.0047	0
S2	-36.188	-0.0502	0.1624	-0.4029	0.6931	-0.7643	0.5021	-0.1789	0.0264	0
S3	0.0036	-0.0795	0.2057	-0.548	1.0742	-1.291	0.9097	-0.3412	0.052	0
S4	0.4038	-0.0325	0.0884	-0.3009	0.7004	-0.9194	0.6738	-0.2424	0.0308	0
S5	51.758	0.0055	-0.1746	0.5018	-0.9395	1.1442	-0.9144	0.4407	-0.0937	0
S6	42.164	0.0953	-0.4992	1.0397	-1.2284	0.8169	-0.2802	0.0384	4E-06	0
S7	-4.7579	0.1185	-0.4938	0.8554	-0.8643	0.5167	-0.185	0.0417	-0.0054	0
S8	-3.4393	0.0492	-0.194	0.3147	-0.3773	0.3249	-0.1878	0.063	-0.0088	0
S9	-8.5449	-0.0638	0.0289	-0.0884	0.1649	-0.171	0.0983	-0.0306	0.0041	0
S10	-18.064	-0.0543	-0.0172	0.0321	-0.0179	0.004	5E-06	-0.0001	8E-06	0
S11	-4.6497	0.0535	-0.0909	0.0613	-0.0311	0.011	-0.0026	0.0004	-2E-05	0
S12	-50	0.0103	-0.0176	0.0057	-0.0015	0.0003	-4E-05	2E-06	-6E-08	0
S13	-2.606	-0.1177	0.0192	-0.0004	-1E-04	-1E-05	4E-06	-4E-07	9E-09	0
S14	-1.0102	-0.0979	0.0187	-0.0024	0.0001	2E-05	-6E-06	6E-07	-3E-08	6E-10

TABLE 59-continued

Example	f	TTL	SL	F No.	IMG HT	FOV
5	4.350	5.300	4.917	1.58	3.38	79.58
6	4.000	4.877	4.525	1.58	3.25	76.86
7	4.280	5.100	4.369	1.71	3.54	77.84
8	4.320	5.285	4.589	1.68	3.26	72.96
9	4.401	5.300	4.142	1.69	3.73	79.31
10	4.831	5.955	4.776	1.66	4.13	79.74
11	5.086	6.000	4.795	1.59	4.13	76.86
12	4.447	5.144	4.894	2.07	3.53	75.63
13	4.657	5.652	5.452	2.12	3.26	69.07
14	4.700	5.650	4.882	1.81	3.93	78.82
15	4.400	5.200		1.81	3.26	72.55
16	3.994	5.125	4.484	1.57	3.26	77.38
17	4.460	5.450	4.5363	1.72	3.53	76.77
18	4.700	5.720	4.8535	1.67	3.93	78.23
19	4.620	5.700	4.8003	1.64	3.93	79.17
20	4.020	4.940	3.938	1.58	3.23	76.00

TABLE 59-continued

Example	f	TTL	SL	F No.	IMG HT	FOV
21	4.333	5.320	4.946	1.50	3.75	80.30
22	4.589	5.600	4.680	1.60	4.25	84.74
23	4.737	5.800	4.740	1.57	4.25	82.21
24	4.451	5.703	4.621	1.63	3.73	78.60
25	4.825	6.000	4.799	1.54	4.20	80.78
26	4.592	5.478	4.515	1.79	3.73	76.90
27	4.302	5.240	4.368	1.95	3.73	80.46
28	4.966	5.993	5.127	2.36	4.13	78.45
29	4.667	5.797	4.893	1.85	4.13	81.80

[0297] Table 60 below shows in mm a focal length f1 of the first lens, a focal length f2 of the second lens, a focal length f3 of the third lens, a focal length f4 of the fourth lens, a focal length f5 of the fifth lens, a focal length f6 of the sixth lens, and a focal length f7 of the seventh lens for each of Examples 1-29 described herein.

TABLE 60

Example	f1	f2	f3	f4	f5	f6	f7
1	3.284	-8.740	13.548	75.065	-13.184	44.153	-17.987
2	10.035	5.292	-7.613	22346862	86.584	-27.728	146.074
3	9.060	4.692	-7.025	-4861.622	80.126	-24.191	1985.391
4	8.409	4.355	-6.520	-4512.292	74.369	-22.452	1842.731
5	-64.233	3.248	-7.428	-43.722	52.425	3.010	-2.424
6	-59.064	2.987	-6.830	-40.204	48.207	2.768	-2.229
7	3.596	-7.349	-1245.238	15.657	-19.723	2.662	-2.171
8	3.659	-7.678	-842.989	14.045	-15.662	2.578	-2.147
9	9.952	4.985	-9.042	-60.959	28.461	-19.130	-36.205
10	32.070	4.958	-9.510	22.572	16.180	-12.867	-20.135
11	43.522	4.462	-8.873	52.492	16.288	-17.315	-19.385
12	3.626	-6.978	10.551	125.381	-28.155	-367.720	-9.031
13	4.246	-7.696	16.245	121.347	-231.607	20.905	-8.575
14	4.553	-11.109	33.932	36.853	268.352	4.100	-2.623
15	4.290	-10.606	30.978	14.871	-21.133	3.784	-2.465
16	5.677	-73.551	-122.716	15.510	207.375	3.799	-2.466
17	3.980	-11.900	-41.500	24.500	-99.990	-800.060	-21.550
18	4.330	-11.590	-800.000	19.640	-99.990	-166.350	-16.430
19	4.420	-12.710	-799.990	20.900	-99.990	-206.860	-17.260
20	4.858	13.152	-8.241	-32.625	34.583	2.462	-2.100
21	-31.530	3.137	-7.545	-130.033	80.886	2.966	-2.423
22	4.929	-16.125	30.244	-34.061	-58.155	2.600	-2.172
23	5.107	-17.583	27.458	-30.256	-75.811	2.526	-2.062
24	18.149	4.970	-8.433	25.591	-19.167	25.748	69.101
25	15.861	6.019	-10.927	22.137	-16.283	8.518	-8.229
26	3.971	-11.857	-77.132	19.846	-30.042	-18.041	68.790
27	3.620	-10.428	39.821	-38.762	4.342	10.303	-2.323
28	3.802	-8.955	64.595	12384.769	-17.503	299.093	57.797
29	4.499	-15.674	39.058	453.779	-18.160	102.612	59.134

[0298] Table 61 below shows in mm a thickness (L1edgeT) of an edge of the first lens, a thickness (L2edgeT) of the edge of the second lens, a thickness (L3edgeT) of the edge of the third lens, a thickness (L4edgeT) of the edge of the fourth lens, a thickness (L5edgeT) of the edge of the fifth lens, a thickness (L6edgeT) of the edge of the sixth lens, and a thickness (L7edgeT) of the edge of the seventh lens for each of Examples 1-29 described herein.

TABLE 61

Example	L1edgeT	L2edgeT	L3edgeT	L4edgeT	L5edgeT	L6edgeT	L7edgeT
1	0.201	0.210	0.200	0.203	0.246	0.233	0.320
2	0.282	0.313	0.416	0	0.325	0.371	0.382
3	0.251	0.280	0.359	0.218	0.293	0.356	0.364
4	0.233	0.259	0.333	0.203	0.272	0.330	0.376
5	0.220	0.270	0.348	0.224	0.259	0.269	0.437
6	0.203	0.249	0.320	0.208	0.237	0.247	0.473
7	0.222	0.377	0.235	0.240	0.189	0.260	0.323
8	0.179	0.351	0.253	0.254	0.383	0.263	0.645
9	0.257	0.255	0.340	0.276	0.365	0.307	0.278
10	0.248	0.333	0.383	0.361	0.582	0.355	0.641
11	0.254	0.125	0.425	0.277	0.380	0.461	0.328
12	0.269	0.308	0.190	0.230	0.410	0.714	0.300
13	0.372	0.402	0.183	0.213	0.284	0.711	0.752
14	0.323	0.418	0.258	0.328	0.292	0.401	0.493
15	0.205	0.407	0.201	0.333	0.278	0.348	0.815
16	0.218	0.347	0.211	0.259	0.277	0.251	0.950
17	0.170	0.220	0.180	0.900	0.230	0.370	0.740
18	0.100	0.330	0.140	0.490	0.230	0.560	0.830
19	0.110	0.320	0.150	0.540	0.230	0.570	0.910
20	0.212	0.210	0.351	0.213	0.236	0.357	0.445
21	0.220	0.248	0.350	0.237	0.252	0.242	0.540
22	0.373	0.329	0.222	0.285	0.186	0.221	0.456
23	0.435	0.330	0.215	0.331	0.143	0.268	0.556
24	0.248	0.303	0.393	0.456	0.352	0.835	0.513
25	0.247	0.240	0.413	0.254	0.352	0.632	0.553
26	0.250	0.275	0.277	0.250	0.337	0.479	0.782
27	0.252	0.293	0.238	0.374	0.258	0.415	0.686
28	0.293	0.298	0.252	0.251	0.409	0.715	0.678
29	0.246	0.280	0.254	0.273	0.356	0.630	0.692

[0299] Table 62 below shows in mm a sag value (L5S1 sag) of the object-side surface of the fifth lens, a sag value (L5S2 sag) of the image-side surface of the fifth lens, a thickness (Yc71P1) of the seventh lens at a first inflection point on the object-side surface of the seventh lens, a thickness (Yc71P2) of the seventh lens at a second inflection point on the object-side surface of the seventh lens, a thickness (Yc72P1) of the seventh lens at a first inflection point on the image-side surface of the seventh lens, and a thickness (Yc72P2) of the seventh lens at a second inflection point on the image-side surface of the seventh lens for each of Examples 1-29 described herein.

TABLE 62

Example	L5S1 sag	L5S2 sag	Yc71P1	Yc71P2	Yc72P1	Yc72P2
1	-0.342	-0.314	0.508	0.449	0.585	—
2	0.197	0.199	0.678	1	0.795	—
3	0.153	0.181	0.610	0.712	0.719	—
4	0.200	0.202	0.568	0.670	0.667	—
5	0.115	0.139	0.930	—	0.811	—
6	0.116	0.128	0.831	—	0.741	—
7	-0.466	-0.526	2.933	—	4.142	—
8	-0.417	-0.463	3.001	—	4.432	—
9	0.210	0.245	0.569	0.641	0.670	—
10	0.129	0.052	0.628	0.923	0.790	—
11	0.070	0.060	0.633	0.519	0.738	—
12	-0.261	-0.263	0.473	—	0.631	—

TABLE 62-continued

Example	L5S1 sag	L5S2 sag	Yc71P1	Yc71P2	Yc72P1	Yc72P2
13	-0.408	-0.480	0.746	0.904	0.888	—
14	-0.499	-0.478	0.806	—	0.771	—
15	-0.485	-0.407	0.890	—	0.920	—
16	-0.479	-0.422	—	—	0.781	—

TABLE 62-continued

Example	L5S1 sag	L5S2 sag	Yc71P1	Yc71P2	Yc72P1	Yc72P2
17	-0.280	-0.330	0.880	2.410	1.440	—
18	-0.340	-0.360	0.910	2.820	1.580	—
19	-0.350	-0.360	0.950	2.800	1.600	—
20	-0.301	-0.528	0.849	—	0.718	—
21	0.202	0.201	0.967	—	0.535	0.904
22	0.334	0.378	0.719	—	0.402	0.845
23	0.354	0.431	1.004	—	0.511	0.913
24	0.210	0.372	0.610	0.706	0.722	—
25	0.199	0.269	0.603	—	0.797	—
26	0.270	0.286	0.889	—	1.015	—
27	0.276	0.509	—	—	0.968	—
28	0.092	0.103	0.955	1.103	1.128	—
29	0.179	0.173	0.964	1.114	1.130	—

[0300] Table 63 below shows an inner diameter of each of the first through seventh spacers for each of Examples 1-29 described herein. S1d is an inner diameter of the first spacer SP1, S2d is an inner diameter of the second spacer SP2, S3d is an inner diameter of the third spacer SP3, S4d is an inner diameter of the fourth spacer SP4, S5d is an inner diameter of the fifth spacer SP5, S6d is an inner diameter of the sixth spacer SP6, and S7d is an inner diameter of the seventh spacer SP7.

TABLE 63

Example	S1d	S2d	S3d	S4d	S5d	S6d	S7d
1	1.730	1.550	1.650	2.070	2.910	4.240	—
2	1.500	1.340	1.320	1.720	2.310	3.030	—
3	1.330	1.220	1.200	1.580	2.050	2.690	—
4	1.240	1.150	1.030	1.480	1.900	2.460	—
5	1.340	1.230	1.030	1.500	1.980	2.660	—
6	1.260	1.150	0.950	1.270	1.790	2.380	—
7	2.310	2.160	2.540	2.940	4.060	4.840	5.120
8	2.420	2.200	2.560	2.820	3.600	4.490	—
9	2.580	2.400	2.490	2.970	4.160	4.890	5.510
10	2.800	2.650	2.730	3.540	3.420	4.440	5.740
11	3.170	3.000	2.790	3.080	4.180	5.490	—
12	2.120	2.100	2.040	2.120	2.810	4.640	—
13	2.290	2.480	2.630	2.820	3.480	5.210	—
14	2.430	2.480	2.890	3.380	4.570	6.180	—
15	2.320	2.360	2.560	2.930	3.700	4.350	—
16	2.410	2.300	2.660	3.030	3.760	—	—
17	2.304	2.000	2.314	3.060	3.548	5.304	—

TABLE 63-continued

Example	S1d	S2d	S3d	S4d	S5d	S6d	S7d
21	2.880	2.630	2.290	2.930	4.380	5.510	—
22	2.660	2.490	2.720	3.150	4.380	5.810	—
23	2.660	2.470	—	3.130	3.780	5.130	—
24	2.680	2.510	2.540	3.000	3.960	5.280	—
25	3.070	2.920	2.900	3.320	4.400	5.750	5.930
26	2.390	2.090	2.240	2.650	3.620	4.780	5.080
27	2.060	1.890	2.150	2.700	3.610	4.560	4.840
28	1.890	1.840	2.330	2.730	3.730	5.430	6.030
29	2.390	2.150	2.400	2.820	3.940	5.680	6.020

[0301] Table 64 below shows in mm³ a volume of each of the first through seventh lenses for each of Examples 1-29 described herein. L1v is a volume of the first lens, L2v is a volume of the second lens, L3v is a volume of the third lens, L4v is a volume of the fourth lens, L5v is a volume of the fifth lens, L6v is a volume of the sixth lens, and L7v is a volume of the seventh lens.

TABLE 64

Example	L1v	L2v	L3v	L4v	L5v	L6v	L7v
1	2.361	2.136	2.508	3.117	4.947	7.771	13.630
2	8.018	9.563	9.605	8.413	12.033	16.720	28.027
3	7.068	7.912	8.188	6.550	7.990	12.999	20.487
4	6.344	6.949	7.760	6.208	6.896	10.336	16.560
5	5.725	8.018	8.377	7.959	10.343	11.103	27.151
6	3.948	5.668	5.721	5.266	6.019	10.492	20.321
7	5.234	5.060	5.146	4.140	5.986	8.138	19.681
8	5.387	4.797	5.748	5.234	8.644	9.466	23.258
9	5.639	4.858	6.675	7.163	11.037	11.936	27.122
10	5.199	7.998	6.680	7.988	20.416	20.093	36.079
11	6.750	7.286	8.850	8.075	13.416	20.371	25.358
12	3.812	4.671	4.055	5.063	11.284	25.762	16.565
13	5.416	6.513	5.011	6.233	8.575	27.331	33.722
14	6.276	6.532	7.753	9.564	12.572	16.077	29.974
15	4.235	5.537	5.593	7.547	9.420	8.999	27.326
16	4.653	4.657	6.231	6.713	10.267	11.740	33.537
17	5.042	4.656	3.428	20.255	7.215	14.931	30.446
18	5.187	6.691	4.372	16.797	9.254	30.554	39.300
19	5.213	6.669	4.416	18.623	8.042	29.510	40.895
20	3.768	3.460	4.028	5.007	6.979	11.351	18.888
21	5.617	7.960	6.846	7.224	12.525	12.815	28.597
22	9.538	6.253	6.836	7.614	9.636	19.918	32.859
23	11.129	6.012	5.802	9.643	7.925	20.060	37.307
24	5.111	5.865	6.312	10.093	12.273	29.379	26.367
25	6.801	6.939	8.141	8.791	14.589	27.072	34.703
26	5.147	4.509	4.470	4.812	8.939	18.212	35.936
27	3.810	3.975	3.927	6.189	7.516	13.035	31.859
28	4.752	4.366	6.456	5.072	9.867	36.871	47.470
29	5.627	4.949	5.142	5.079	9.362	31.583	47.908

TABLE 63-continued

Example	S1d	S2d	S3d	S4d	S5d	S6d	S7d
18	2.630	2.220	2.360	3.320	4.080	5.810	—
19	2.620	2.210	2.380	3.360	4.010	5.700	—
20	2.420	2.230	2.070	2.410	3.080	4.230	—

[0302] Table 65 below shows in mg a weight of each of the first through seventh lenses for each of Examples 1-29 described herein. L1w is a weight of the first lens, L2w is a weight of the second lens, L3w is a weight of the third lens, L4w is a weight of the fourth lens, L5w is a weight of the fifth lens, L6w is a weight of the sixth lens, and L7w is a weight of the seventh lens.

TABLE 65

Example	L1w	L2w	L3w	L4w	L5w	L6w	L7w
1	2.455	2.628	2.608	3.242	6.184	9.714	13.767
2	8.339	9.945	12.007	10.516	12.514	20.900	28.307
3	7.351	8.229	10.235	8.188	8.310	16.249	20.692
4	6.598	7.227	9.700	7.760	7.172	12.921	16.725
5	5.954	8.339	10.472	9.710	12.619	11.547	28.237
6	4.106	5.894	7.151	6.424	7.343	10.912	21.134
7	5.444	6.223	5.351	4.306	7.362	8.463	20.468
8	5.602	5.901	5.978	5.443	10.632	9.844	24.188
9	5.865	5.052	8.344	8.953	11.478	14.920	27.393
10	5.407	8.318	8.351	8.307	21.233	25.116	37.522
11	7.020	7.578	11.062	10.093	13.952	25.464	26.372
12	3.964	5.746	4.217	5.266	14.106	26.792	17.227
13	5.633	8.011	5.212	6.483	10.547	28.424	34.059
14	6.527	8.034	8.063	9.947	15.463	16.720	31.173
15	4.404	6.810	5.817	7.849	11.587	9.359	28.419
16	4.839	5.728	6.480	6.982	12.629	12.210	34.879
17	5.244	5.726	4.216	21.065	9.018	18.664	30.751
18	5.394	8.230	5.377	17.469	11.568	38.192	39.693
19	5.422	8.203	5.432	19.368	10.052	36.887	41.304
20	3.919	3.598	4.954	5.207	8.724	11.805	19.643
21	5.842	8.279	8.558	9.030	15.657	13.327	29.741
22	9.919	7.816	7.110	9.365	11.756	20.715	34.174
23	11.574	7.515	6.034	11.861	9.669	20.863	38.799
24	5.315	6.100	7.891	10.497	12.764	30.554	27.422
25	7.073	7.217	10.176	9.142	18.237	28.155	36.091
26	5.352	5.546	5.497	5.005	11.173	22.765	36.295
27	3.962	4.889	4.084	7.612	7.817	13.556	33.133
28	4.942	5.370	6.714	5.275	12.334	38.345	47.945
29	5.852	6.087	5.348	5.282	11.516	32.847	48.387

[0303] Table 66 below shows in mm an overall outer diameter (including a rib) of each of the first to seventh lenses for each of Examples 1-29 described herein. L1TR is

of the fourth lens, L5TR is an overall outer diameter of the fifth lens, L6TR is an overall outer diameter of the sixth lens, and L7TR is an overall outer diameter of the seventh lens.

TABLE 66

Example	L1TR	L2TR	L3TR	L4TR	L5TR	L6TR	L7TR
1	3.380	3.580	1.940	4.180	4.880	5.500	5.700
2	2.440	2.540	2.690	2.900	3.190	3.440	3.620
3	2.280	2.400	2.530	2.630	2.780	3.150	3.250
4	2.290	2.400	2.540	2.630	2.780	2.910	3.040
5	2.460	2.580	2.690	2.800	3.170	3.310	3.470
6	2.060	2.190	2.300	2.360	2.520	2.830	3.080
7	4.220	4.420	4.540	4.720	5.400	5.740	6.300
8	4.280	4.480	4.610	4.870	5.040	5.750	6.060
9	4.210	4.300	4.440	4.840	5.470	6.120	6.900
10	4.470	4.560	4.700	5.030	6.660	7.180	7.430
11	4.730	4.820	4.960	5.290	5.880	6.810	7.060
12	3.510	3.810	4.390	4.980	5.850	6.150	6.250
13	4.050	4.290	4.670	5.260	5.820	6.690	7.080
14	4.270	4.460	5.040	5.630	6.500	6.900	7.100
15	3.930	4.130	4.710	6.170	5.300	6.570	6.670
16	4.030	4.230	4.810	5.400	6.270	6.670	6.770
17	4.202	4.386	4.788	5.292	6.180	6.682	6.858
18	4.490	4.690	5.260	5.850	6.730	7.130	7.330
19	4.490	4.690	5.260	5.850	6.730	7.120	7.320
20	3.830	4.030	4.230	4.830	5.320	5.720	5.920
21	4.630	4.830	5.030	5.830	6.320	6.720	6.920
22	4.830	5.130	5.430	6.230	6.720	7.120	7.320
23	4.830	5.030	5.230	6.030	6.520	6.920	7.120
24	4.260	4.350	4.490	4.890	5.520	6.750	7.260
25	4.710	4.800	4.930	5.370	6.220	7.250	7.680
26	4.100	4.190	4.320	4.720	5.350	6.170	7.030
27	3.730	3.820	3.960	4.390	4.960	6.000	6.860
28	3.970	4.060	4.190	4.630	5.200	7.150	8.020
29	4.390	4.480	4.610	5.040	5.610	7.090	7.950

an overall outer diameter of the first lens, L2TR is an overall outer diameter of the second lens, L3TR is an overall outer diameter of the third lens, L4TR is an overall outer diameter

[0304] Table 67 below shows in mm a thickness of a flat portion of the rib of each of the first to seventh lenses for each of Examples 1-29 described herein. L1Tr is a thickness

of a flat portion of the rib of the first lens, L2rt is a thickness of a flat portion of the rib of the second lens, L3rt is a thickness of a flat portion of the rib of the third lens, L4rt is a thickness of a flat portion of the rib of the fourth lens, L5rt is a thickness of a flat portion of the rib of the fifth lens, L6rt is a thickness of a flat portion of the rib of the sixth lens, and L7rt is a thickness of a flat portion of the rib of the seventh lens.

TABLE 67

Example	L1rt	L2rt	L3rt	L4rt	L5rt	L6rt	L7rt
1	0.245	0.260	0.255	0.240	0.305	0.275	0.405
2	0.600	0.580	0.560	0.470	0.340	0.400	0.470
3	0.600	0.540	0.540	0.440	0.250	0.380	0.420
4	0.540	0.500	0.520	0.420	0.210	0.390	0.400
5	0.390	0.440	0.470	0.360	0.420	0.380	0.470
6	0.460	0.440	0.460	0.410	0.310	0.380	0.460
7	0.435	0.430	0.360	0.215	0.320	0.330	0.405
8	0.440	0.400	0.390	0.360	0.380	0.470	0.690
9	0.550	0.380	0.580	0.410	0.500	0.320	0.530
10	0.490	0.530	0.520	0.370	0.620	0.520	0.690
11	0.600	0.440	0.600	0.460	0.580	0.600	0.440
12	0.482	0.395	0.316	0.328	0.422	0.885	0.409
13	0.634	0.643	0.375	0.356	0.312	0.828	0.984
14	0.508	0.554	0.444	0.473	0.410	0.438	0.522
15	0.431	0.556	0.361	0.429	0.380	0.380	0.667
16	0.431	0.457	0.361	0.364	0.380	0.334	0.729
17	0.439	0.446	0.280	0.866	0.225	0.359	0.763
18	0.431	0.493	0.255	0.735	0.252	0.675	0.886
19	0.431	0.493	0.255	0.790	0.198	0.637	0.886
20	0.390	0.330	0.300	0.260	0.425	0.550	0.534
21	0.540	0.480	0.460	0.250	0.555	0.395	0.688
22	0.570	0.400	0.310	0.220	0.355	0.570	0.625
23	0.620	0.400	0.270	0.430	0.300	0.520	0.760
24	0.500	0.410	0.510	0.540	0.520	0.970	0.550
25	0.530	0.410	0.560	0.570	0.480	0.600	0.620
26	0.460	0.400	0.390	0.260	0.430	0.540	0.830
27	0.400	0.420	0.370	0.500	0.320	0.460	0.720
28	0.470	0.410	0.450	0.410	0.470	0.930	0.700
29	0.440	0.390	0.400	0.400	0.380	0.740	0.720

[0305] Table 68 below shows, for each of Examples 1-29 described herein, dimensionless values of each of the ratio L1w/L7w in Conditional Expressions 1 and 6, the ratio S6d/f in Conditional Expressions 2 and 7, the ratio L1TR/L7TR in Conditional Expressions 3 and 8, the ratio L1234TRavg/L7TR in Conditional Expressions 4 and 9, and the ratio L12345TRavg/L7TR in Conditional Expressions 5 and 10. The dimensionless value of each of these ratios is obtained by dividing two values expressed in a same unit of measurement.

TABLE 68

Example	L1w/L7w	S6d/f	L1TR/L7TR	L1234TRavg/L7TR	L12345TRavg/L7TR
1	0.178	1.155	0.5930	0.574	0.630
2	0.295	0.634	0.6740	0.7300	0.7602
3	0.355	0.632	0.7015	0.7569	0.7766
4	0.394	0.623	0.7533	0.8109	0.8316
5	0.211	0.611	0.7089	0.7586	0.7896
6	0.194	0.595	0.6688	0.7232	0.7422
7	0.266	1.131	0.6698	0.7103	0.7397
8	0.232	1.039	0.7063	0.7525	0.7683
9	0.214	1.111	0.6101	0.6446	0.6742
10	0.144	0.919	0.6016	0.6312	0.6843
11	0.266	1.079	0.6700	0.7011	0.7275
12	0.230	1.043	0.5616	0.6676	0.7213
13	0.165	1.119	0.5720	0.6451	0.6805
14	0.209	1.315	0.6014	0.6831	0.7296

TABLE 68-continued

Example	L1w/L7w	S6d/f	L1TR/L7TR	L1234TRavg/L7TR	L12345TRavg/L7TR
15	0.155	0.989	0.5892	0.7099	0.7268
16	0.139		0.5953	0.6821	0.7309
17	0.171	1.189	0.6127	0.6805	0.7246
18	0.136	1.236	0.6126	0.6920	0.7372
19	0.131	1.234	0.6134	0.6930	0.7383
20	0.200	1.052	0.6470	0.7145	0.7514
21	0.196	1.272	0.6691	0.7341	0.7699
22	0.290	1.266	0.6598	0.7384	0.7743
23	0.298	1.083	0.6784	0.7416	0.7764
24	0.194	1.186	0.5868	0.6195	0.6477
25	0.196	1.192	0.6133	0.6449	0.6779
26	0.147	1.041	0.5832	0.6163	0.6452
27	0.120	1.060	0.5437	0.5794	0.6082
28	0.103	1.093	0.4950	0.5252	0.5499
29	0.121	1.217	0.5522	0.5824	0.6070

[0306] FIGS. 59 and 60 are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.

[0307] The examples of an optical imaging system 100 described in this application may include a self-alignment structure as illustrated in FIGS. 59 and 60.

[0308] In one example illustrated in FIG. 59, the optical imaging system 100 includes a self-alignment structure in which optical axes of four consecutive lenses 1000, 2000, 3000, and 4000 are aligned with an optical axis of the optical imaging system 100 by coupling the four lenses 1000, 2000, 3000, and 4000 to one another.

[0309] The first lens 1000 disposed closest to object side of the optical imaging system 100 is disposed in contact with an inner surface of a lens barrel 200 to align the optical axis of the first lens 1000 with the optical axis of the optical imaging system 100, the second lens 2000 is coupled to the first lens 1000 to align the optical axis of the second lens 2000 with the optical axis of the optical imaging system 100, the third lens 3000 is coupled to the second lens 2000 to align the optical axis of the third lens 3000 with the optical axis of the optical imaging system 100, and the fourth lens 4000 is coupled to the third lens 3000 to align the optical axis of the fourth lens 4000 with the optical axis of the optical imaging system 100. The second lens 2000 to the fourth lens 4000 may not be disposed in contact with the inner surface of the lens barrel 200.

[0310] Although FIG. 59 illustrates that the first lens 1000 to the fourth lens 4000 are coupled to one another, the four consecutive lenses that are coupled to one another may be changed to the second lens 2000 to a fifth lens 5000, the third lens 3000 to a sixth lens 6000, or the fourth lens 4000 to a seventh lens 7000.

[0311] In another example illustrated in FIG. 60, the optical imaging system 100 includes a self-alignment structure in which optical axes of five consecutive lenses 1000, 2000, 3000, 4000, and 5000 are aligned with an optical axis of the optical imaging system 100 by coupling the five lenses 1000, 2000, 3000, 4000, and 5000 to one another.

[0312] The first lens 1000 disposed closest to an object side of the optical imaging system 100 is disposed in contact with an inner surface of the lens barrel 200 to align an optical axis of the first lens 1000 with the optical axis of the optical imaging system 100, the second lens 2000 is coupled to the first lens 1000 to align the optical axis of the second lens 2000 with the optical axis of the optical imaging system 100,

the third lens **3000** is coupled to the second lens **2000** to align the optical axis of the third lens **3000** with the optical axis of the optical imaging system **100**, the fourth lens **4000** is coupled to the third lens **3000** to align the optical axis of the fourth lens **4000** with the optical axis of the optical imaging system **100**, and the fifth lens **5000** is coupled to the fourth lens **4000** to align the optical axis of the fifth lens **5000** with the optical axis of the optical imaging system **100**. The second lens **2000** to the fifth lens **5000** may not be disposed in contact with the inner surface of the lens barrel **200**.

[0313] Although FIG. **60** illustrates that the first lens **1000** to the fifth lens **5000** are coupled to one another, the five consecutive lenses that are coupled to one another may be changed to the second lens **2000** to a sixth lens **6000**, or the third lens **3000** to a seventh lens **7000**.

[0314] FIG. **61** is a cross-sectional view illustrating an example of a seventh lens.

[0315] FIG. **61** illustrates the overall outer diameter (L7TR) of the seventh lens, the thickness (L7rt) of the flat portion of the rib of the seventh lens, the thickness (L7edgeT) of the edge of the seventh lens, the thickness (Yc71P1) of the seventh lens at the first inflection point on the object-side surface of the seventh lens, the thickness (Yc71P2) of the seventh lens at the second inflection point on the object-side surface of the seventh lens, and the thickness (Yc72P1) of the seventh lens at the first inflection point on the image-side surface of the seventh lens. Although not illustrated in FIG. **61**, the seventh lens may also have a second inflection point on the image-side surface of the seventh lens, and a thickness of the seventh lens at this inflection point is Yc72P2 as listed in Table 62.

[0316] FIG. **62** is a cross-sectional view illustrating an example of a shape of a rib of a lens.

[0317] The examples of the optical imaging system **100** described in this application may include a structure for preventing a flare phenomenon and reflection.

[0318] For example, the ribs of the first to seventh lenses **1000**, **2000**, **3000**, **4000**, **5000**, **6000**, and **7000** of the optical imaging system may be partially surface-treated to make the surface of the rib rough as shown in FIG. **62**. Methods of surface treatment may include chemical etching, physical grinding, or any other surface treatment method capable of increasing a roughness of a surface.

[0319] A surface-treated area EA may be formed in an entire area from an edge of the optical portion of the lens through which light actually passes to an outer end of the rib. However, as illustrated in FIG. **62**, non-treated areas NEA including step portions E11, E21, and E22 may not be surface-treated, or may be surface-treated to have a roughness less than a roughness of the surface-treated area EA. The step portions E11, E21, and E22 are portions where the thickness of the rib abruptly changes. A width G1 of a first non-treated area NEA formed on an object-side surface of the lens and including a first step portion E11 may be different from a width G2 of a second non-treated area NEA formed on an image-side surface of the lens and including a second step portion E21 and a third step portion E22. In the example illustrated in FIG. **62**, G1 is greater than G2.

[0320] The width G1 of the first non-treated area NEA includes the first step portion E11, the second step portion E21, and the third step portion E22 when viewed in an optical axis direction, and the width G2 of the second non-treated area NEA includes the second step portion E21

and the third step portion E22 but not the first step portion E11 when viewed in the optical axis direction. A distance G4 from the outer end of the rib to the second step portion E21 is smaller than a distance G3 from the outer end of the rib to the first step portion E11. Also, a distance G5 from the outer end of the rib to the third step portion E22 is smaller than the distance G3 from the outer end of the rib to the first step portion E11.

[0321] The positions at which the non-treated areas NEA and the step portions E11, E21, and E22 are formed as described above and shown in FIG. **62** may be advantageous for measuring a concentricity of the lens.

[0322] The examples described above enable the optical imaging system to be miniaturized and aberrations to be easily corrected.

[0323] While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

What is claimed is:

1. An optical imaging system comprising:
 - a first lens comprising a refractive power;
 - a second lens comprising a refractive power;
 - a third lens comprising negative refractive power and a convex object-side surface;
 - a fourth lens comprising a refractive power;
 - a fifth lens comprising a refractive power;
 - a sixth lens comprising a refractive power; and
 - a seventh lens comprising a refractive power,
 wherein the first to seventh lens are sequentially disposed from an object side toward an imaging plane,
 - wherein an absolute value of a radius of curvature of an image-side surface of the second lens is greater than an absolute value of a radius of curvature of an object-side surface of the first lens,
 - wherein an absolute value of a radius of curvature of an object-side surface of the fourth lens is greater than an absolute value of a radius of curvature of an object-side surface of the third lens, and
 - wherein an absolute value of a radius of curvature of an object-side surface of the third lens is greater than an absolute value of a radius of curvature of an image-side surface of the fifth lens.
2. The optical imaging system of claim 1, wherein the first lens has a convex object-side surface.
3. The optical imaging system of claim 1, wherein the second lens has a convex object-side surface.
4. The optical imaging system of claim 1, wherein the fourth lens has a convex object-side surface.

5. The optical imaging system of claim 1, wherein the fifth lens has a convex object-side surface.

6. The optical imaging system of claim 1, wherein the sixth lens has a convex object-side surface.

7. The optical imaging system of claim 1, wherein the seventh lens has a convex object-side surface.

8. An optical imaging system comprising:
a first lens comprising a refractive power;
a second lens comprising a refractive power;
a third lens comprising a convex object-side surface;
a fourth lens comprising a refractive power;
a fifth lens comprising positive refractive power;
a sixth lens comprising a refractive power; and
a seventh lens comprising a refractive power,
wherein the first to seventh lens are sequentially disposed
from an object side toward an imaging plane,
wherein an absolute value of a radius of curvature of an
image-side surface of the second lens is greater than an
absolute value of a radius of curvature of an object-side
surface of the first lens,

wherein an absolute value of a radius of curvature of an
object-side surface of the fourth lens is greater than an
absolute value of a radius of curvature of an object-side
surface of the third lens, and

wherein an absolute value of a radius of curvature of an
object-side surface of the third lens is greater than an
absolute value of a radius of curvature of an image-side
surface of the fifth lens.

9. The optical imaging system of claim 8, wherein the first
lens has a convex object-side surface.

10. The optical imaging system of claim 8, wherein the
second lens has a convex object-side surface.

11. The optical imaging system of claim 8, wherein the
fourth lens has a concave object-side surface.

12. The optical imaging system of claim 8, wherein the
fifth lens has a concave object-side surface.

13. The optical imaging system of claim 8, wherein the
sixth lens has a convex object-side surface.

14. The optical imaging system of claim 8, wherein the
seventh lens has a convex object-side surface.

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